Mathematical Methods I MTH1021
Lecture Notes Weeks 1 - 11

Dr. Dermot Green
School of Mathematics and Physics
Queen's University Belfast
Room 01.16 Old Physics Building

November 7, 2022

Contents

1 Introduction

Analysis is a branch of mathematics that studies continuous change. It includes the theories of differentiation, integration, measure, limits, infinite series and analytic functions. These theories are normally studied systematically in the context of real and complex functions. **Calculus** describes the mathematics of motion and change. Where there is motion or growth, where variable forces are at work producing acceleration, calculus is the right mathematics to apply. Both analysis and calculus are taught in every institution of higher learning in the world, because together they are essential to basic human thought in engineering, science and many social sciences. Today, calculus and it's extensions in mathematical analysis are far reaching, solving many problems in a wide range of fields that bring understanding about the universe and the world around us. For example:

1. Economists - to forecast global trends.
2. Oceanographers - to formulate theories about ocean currents.
3. Meteorologists - to describe the flow of air in the upper atmosphere.
4. Biologists - to forecast population size and to describe the way predators like foxes interact with their prey.
5. Medical Researchers - to design ultrasound and x-ray equipment for scanning the internal organs of the body.
6. Space Scientists - to design rockets and explore distant planets.
7. Psychologists - to understand optical illusions in visual perception.
8. Physicists - to design inertial navigations systems and to study the nature of time and the universe.
9. Electrical Engineers - to design stroboscopic flash equipment and to solve the differential equations that describe current flow in computers.
10. Stock Market Analysts - to predict prices and assess interest rate risk.

The list is practically endless, almost every professional field today uses analysis and calculus in some way. Are analysis and calculus difficult subjects? No, not really, but it sometimes seems that way to students, especially at the beginning because of the the new ideas and techniques involved. Success in mastering calculus depends on having a very solid basis in algebra, geometry and trigonometry to build upon. Working through examples is the best way to test and deepen your understanding of the theories involved. It is not unlike practising a musical instrument - everyone will improve with more practise. In this module you will be provided with numerous exercises. Some are "drill" exercises to help you develop your skills in calculation. More important, however, are the problems that develop reasoning skills and your ability to apply the techniques you have learned to concrete situations.

2 Numbers and functions

2.1 Numbers

Natural numbers

are those that are used for counting: 1, 2, 3, ... The set of natural numbers (sometimes called positive integers) is denoted by $\mathbb{N}$, i.e., $\mathbb{N} = \{1,\, 2,\, 3, \ldots\}$.

There is no largest natural number; if n is a natural number (which can be written in shorthand as $n \in \mathbb{N}$, meaning, n is in $\mathbb{N}$, or n belongs to $\mathbb{N}$), then $n+1$ is also a natural number ($n+1 \in \mathbb{N}$).

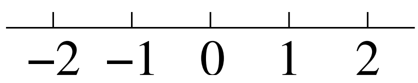
Natural numbers can be added, e.g., $5+3=8$. In general, for any two natural numbers n and m, their sum $n+m$ is also a natural number (or in shorthand, $\forall\, n,\, m \in \mathbb{N} \;\Rightarrow\; n+m \in \mathbb{N}$).[1] However, subtraction is not always possible, e.g., $3-5$ is not a natural number (i.e., there is no natural number that would give 3 when added to 5).

Integer numbers

include all natural numbers and a special number called *zero* (0), such that $n+0=n$ for any number n. They also contain *negative* numbers, such that for any $n \in \mathbb{N}$ there is a number denoted $-n$ such that $n+(-n)=0$. The latter equality is commonly written as $n-n=0$, i.e., subtracting a number is the same as adding its negative.

The set of all integers is denoted $\mathbb{Z}$. Subtraction is always possible in $\mathbb{Z}$, i.e., $\forall\; n,\, m \in \mathbb{Z} \;\Rightarrow\; n-m \in \mathbb{Z}$; for example, $3-5=-2$.

To visualise the integers we can show them on the number line.



The integers can also be multiplied, e.g., $5 \times (-3) = -15$, or $2 \times 0 = 0$. However, division is not always possible, e.g., $4 \div 2 = 2$, but $5 \div 2$ is not an integer number (i.e., there is no integer number that would give 5 after being multiplied by 2). To allow division we need to extend our number system.

Rational numbers

are numbers of the form p/q, where $p,\, q \in \mathbb{Z}$ and $q \neq 0$. The set of rational numbers is denoted $\mathbb{Q}$. If $q=1$ then $p/1 \equiv p$ is an integer, which means that integer numbers are contained in rationals, i.e, $\mathbb{Z}$ is a *subset* of $\mathbb{Q}$, or in shorthand, $\mathbb{Z} \subset \mathbb{Q}$.

[1] Here $\forall$ means "for all" or "given any", and $\Rightarrow$ means "then" or "hence".

In addition to the integers, rational numbers contain all fractions (i.e., proper and improper fractions, e.g., $3/5$ and $5/3 = 1\frac{2}{3}$, including any finite and repeating decimals, e.g., -1.2 and $0.333\ldots$).

Rational numbers can be added and subtracted,

$$\frac{p}{q} + \frac{r}{s} = \frac{ps + qr}{qs}, \qquad \frac{p}{q} - \frac{r}{s} = \frac{ps - qr}{qs}, \tag{2.1}$$

where p, q, r, $r \in \mathbb{Z}$ and q, $s \neq 0$, as well as multiplied and divided,

$$\frac{p}{q} \times \frac{r}{s} = \frac{pr}{qs}, \qquad \frac{p}{q} \div \frac{r}{s} = \frac{p}{q} \times \frac{s}{r} = \frac{ps}{qr}, \tag{2.2}$$

where $r \neq 0$ in the last equation.[2] The right-hand sides of the four equations in (2.1) and (2.2) are of the form m/n where m and n are integers and $n \neq 0$, hence these are rational numbers.

Note that although it seems that are about "twice as many" integers as natural numbers, and "many more" rational numbers than integers (and they are subsets of each other, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$), the number of elements in each of these three sets is in fact the same. Indeed, one can count all integers by using natural numbers, e.g., like this: 0 is 1, 1 is 2, -1 is 3, 2 is 4, -2 is 5, 3 is 6, etc. In this way every positive integer n will be counted at number $2n$ and its negative counterpart $-n$ will be $(2n+1)$th. Rational numbers can also be counted in a similar way. (BBC Sounds App 'A brief History of Mathematics' listen to the episode on Georg Cantor. There are 10 episodes in all)

All three sets, $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ (and any other infinite set whose elements can be counted by numbers 1, 2, 3, etc.) are said to be *countably infinite* (or denumerable).

Any linear algebraic equation with rational coefficients has a solution in $\mathbb{Q}$. Consider, however, quadratic equations, e.g., $x^2 = 4$ and $x^2 = 2$. The first one has two rational (in fact, integer) roots: $x = \pm 2$. The second equation has no solutions in rational numbers.

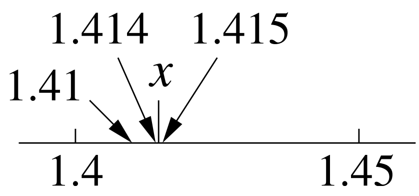
To prove this we are directed to Examples 2.1.5 and 2.1.6 (Additional Examples 1)) on Canvas where we assume to the contrary that there is a rational number $x = p/q$ such that $(p/q)^2 = 2$. It follows then that

$$p^2 = 2q^2. \tag{2.3}$$

By contradiction we show that equation (2.3) cannot hold, and $x^2 = 2$ has no solutions in $\mathbb{Q}$.

However, we can *approximate* the roots of $x^2 = 2$ as closely as we want with rational numbers, in particular, with decimals. Thus, for the positive root x, working to one,

[2] As almost always in algebra, the multiplication sign $\times$ (or sometimes $\cdot$) is omitted, unless dealing with numbers. Addition and multiplication are *commutative*, i.e., $a + b = b + a$ and $ab = ba$, and multiplication is *distributive* over addition: $(a + b)c = ac + bc$.



two and three decimal places, $1.4^2 = 1.96$, $1.41^2 = 1.9881$, and $1.414^2 = 1.999396$, while $1.415^2 = 2.002225$, so the root is bracketed by $1.414 < x < 1.415$. Continuing this process we are able to get infinitely closely to x with rational numbers. However, we can never "hit" this point exactly in $\mathbb{Q}$. (The exact positive root is denoted $\sqrt{2}$, which simply means 'a number whose square is equal to 2'.) In spite of the rational numbers being infinitely dense on the number line, they leave infinitely many "holes", such as $\sqrt{2}$. All such points are known as *irrational* numbers.

Real numbers

represent *all* points on the number line, i.e., both rational and irrational numbers. Real numbers populate the number line continuously, leaving no "holes". Hence, the set of real numbers $\mathbb{R}$ forms a *continuum.* Unlike $\mathbb{Z}$ and $\mathbb{Q}$, the set of real numbers is *uncountably infinite* (or nondenumerable), i.e., its members cannot be counted by the natural numbers.

Real numbers are important in applications, e.g., in mechanics, since time and position are both taken to be *continuous* variables. Famous numbers, such as $\pi = 3.14\ldots$ and $e = 2.71\ldots$, are irrational numbers. (They are also *transcendental*, i.e., they are not solutions of any algebraic equation.)

Intervals.

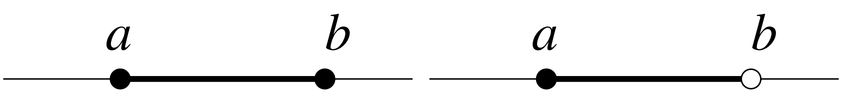
It is often necessary to consider segments or *intervals* of the real line, both finite and infinite. The following notation is used. A *closed* interval contains all real numbers x between a and b, including the end points a and b is denoted $[a, b]$. It can also be represented by a double inequality,

$$a \le x \le b \iff x \in [a, b],$$

where the double-arrow $\iff$ means 'is equivalent'. If one of the end points, say, b, is not included, the interval is *half-open* and is denoted $[a, b)$,

$$x \in [a, b) \iff a \le x < b.$$

Finally, the *open* interval (a, b) does not contain both ends, $a < x < b$.



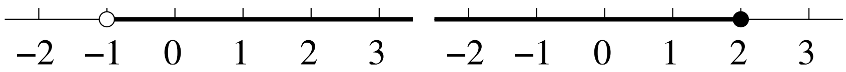
Intervals can also stretch from $-\infty$ or to $+\infty$, e.g.,

$$x > -1 \iff x \in (-1, +\infty),$$

where the interval does not contain both ends, or

$$x \leq 2 \iff x \in (-\infty, 2].$$

Note that brackets adjacent to infinity are always round, as $+\infty$ and $-\infty$ are not numbers, and hence are not included in the intervals. We can even write the set of real numbers as an interval: $\mathbb{R} = (-\infty, +\infty)$.



2.2 Functions

The notion of function is quite general, but in applied mathematics we usual deal with real functions of real variables, e.g., position, time, etc.

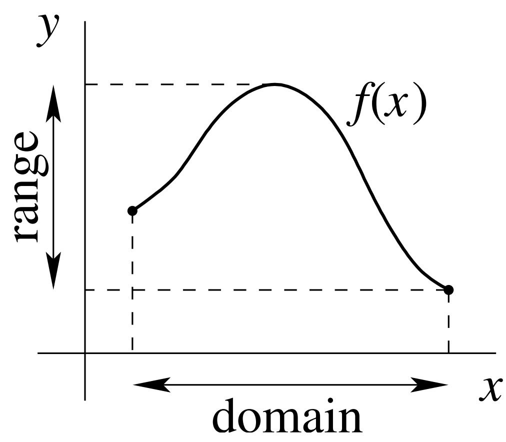
Definition: A *function* $y = f(x)$ is a rule or relation, which for each allowed value of x, gives the corresponding value of y.

Here x is the *independent* variable, y is the *dependent* variable, and the function itself is the "recipe" of what needs to be done with x to obtain y.

For example, the square function $y = x^2$ tells you to multiply x by itself to obtain the value of y.

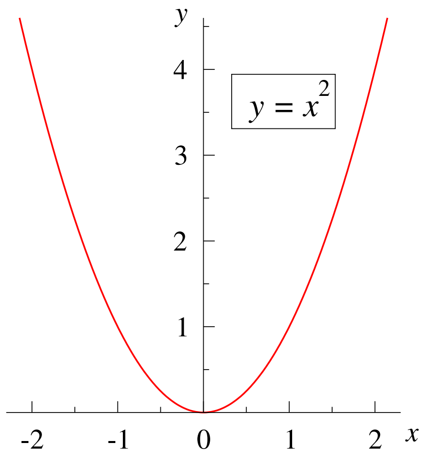
The set of all allowed values of x is called the *domain* of the function, and the set of all values of y is the *range* of the function.

A function can often be given by a formula (e.g., $y = x^2$). Real functions can also be represented graphically using the x-y coordinate plane.



Examples:

1. $y = x^2$ is the simplest quadratic function shown by the graph below.

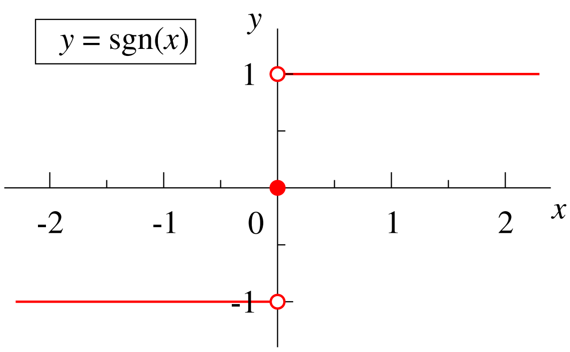


The domain of this function is all $\mathbb{R}$, i.e., $-\infty < x < \infty$, or $(-\infty, +\infty)$, and the range is $y \geq 0$, or $[0, +\infty)$.

2. The sign function $y = \operatorname{sgn}(x)$ is defined as follows:

$$\operatorname{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}.$$

For example, $\operatorname{sgn}(2) = 1$, $\operatorname{sgn}(-\pi) = -1$, $\operatorname{sgn}(0) = 0$, and it is easy to sketch this function.



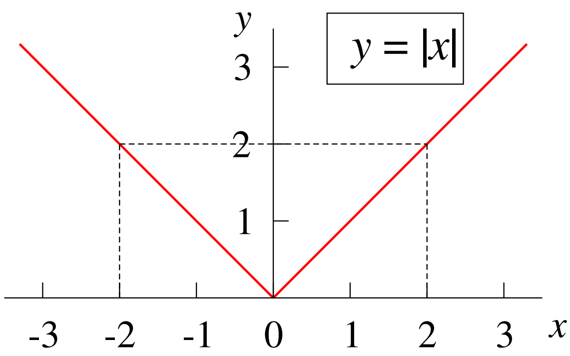
The points $(0, 1)$ and $(0, -1)$ are shown as empty circles, as they do not belong to the graph, while $(0, 0)$ belongs to the graph and is shown by the solid circle.

The function $\mathrm{sgn}(x)$ is defined for all real x, hence its domain is $(-\infty, +\infty)$, while its range is $\{-1,\, 0,\, 1\}$.

3. The modulus function $y = |x|$ (or the absolute value),

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases},$$

gives the distance from the origin (0) to point x on the number line. Hence, it is non-negative. In order to sketch it, note that for $x \geq 0$ is is represented by the straight line which bisects the first quadrant, and for $x < 0$ it is given by the straight line which bisects the second quadrant.



The domain of the modulus function is $(-\infty, +\infty)$, and the range $[0, +\infty)$.

Problem: solve the equation $|x| = 2$. From the graph, we immediately see that y equals 2 for two values of x, namely, $x = 2$ and $x = -2$.

In general, the equation $|x| = a$, for $a > 0$, has two roots: $x = \pm a$.

4. $y = \sqrt{1 - x^2}$.

Let us first find the domain and range of this function. The expression under the square root must be non-negative, so this function is defined only for $1 - x^2 \geq 0$. We thus have

$$x^2 \leq 1,$$

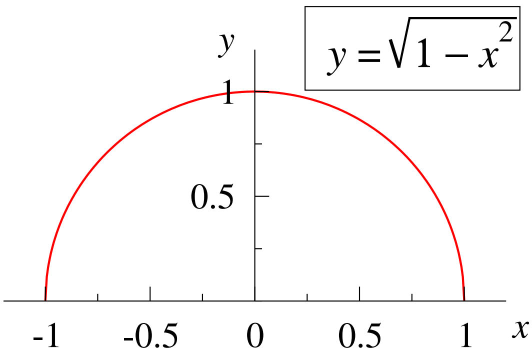
and taking square roots of both sides,[3]

$$|x| \leq 1.$$

Hence, the domain of the function is $-1 \leq x \leq 1$, or the closed interval $[-1, 1]$.

By inspection, the function takes its largest value $y = 1$ when $x = 0$. As x moves away from zero, the function decreases and for $x = \pm 1$ we have $y = 0$. Hence the range of this function is $0 \leq y \leq 1$, or $[0, 1]$.

Using this information and noticing that the function takes equal values for values of x symmetric with respect to zero (because of x^2), we obtain the following graph.



The graph of $y = \sqrt{1 - x^2}$ is in fact a semicircle. Indeed, squaring this expression gives $y^2 = 1 - x^2$, and rearranging, $x^2 + y^2 = 1$, which is the equation of a unit circle with the centre at the origin. Since the square root is non-negative, restricting the graph to the upper half-plane, $y \geq 0$.

In examples 1, 3 and 4 the graphs are symmetric about the y axis. This makes sketching the functions easier and motivates the following definition.

Definition: A function whose domain is symmetric with respect to the origin is called *even* if $f(-x) = f(x)$, and is called *odd* if $f(-x) = -f(x)$ for all x from the domain.

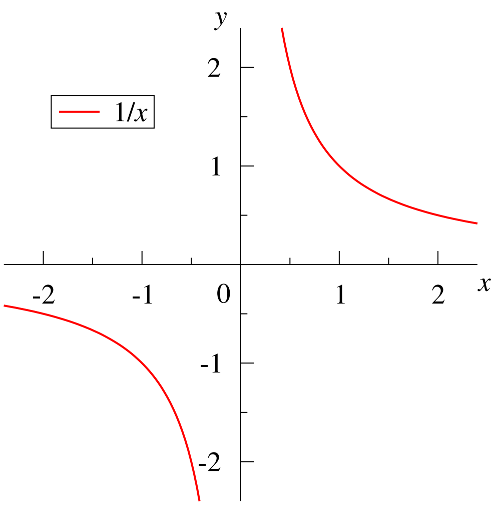
The graph of an even function is always symmetric with respect to reflection in the y axis. The graph of an odd function is symmetric with respect to rotation through 180° about the origin.

[3] Note that the function 'square root' is defined so that it is non-negative, e.g., $\sqrt{4} = 2$. Hence, in general, $\sqrt{a^2} = |a|$. To convince oneself in this, try $a = 2$ and then $a = -2$.

It is usually easy to check if a function is even or odd (or neither). Thus, for $y = x^2$ we have $(-x)^2 = x^2$, so this function is even. The functions $y = |x|$ and $y = \sqrt{1 - x^2}$ are also even. On the other hand, the function $y = \mathrm{sgn}(x)$ is odd since it changes from $+1$ to -1 or vice versa when x changes sign.

5. $y = 1/x$.

This function is defined for all $x \neq 0$, so its domain is a *union* of two intervals, $(-\infty, 0) \cup (0, +\infty)$, and is symmetric. We also have $f(-x) = 1/(-x) = -1/x = -f(x)$, so this function is odd. For $x > 0$, $y = 1/x$ decreases monotonically as x increases, which gives the following graph.



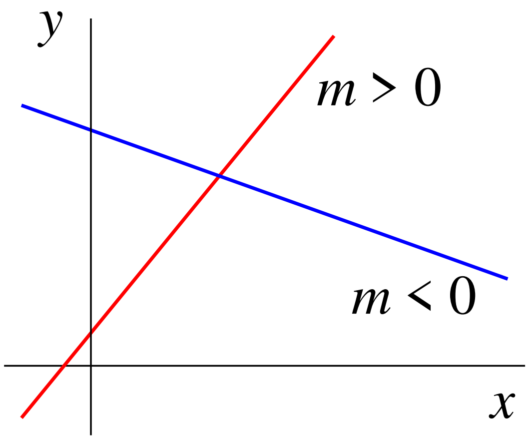
The graph of the function for $x < 0$ is obtained by rotating the $x > 0$ branch around the origin through 180°. Note that as x tends to $+\infty$ ($x \to +\infty$), values of y become ever closer to zero, so y tends to zero ($y \to 0$) in this limit.

2.3 Linear and quadratic functions

1. Linear function $y = mx + c$.

The graph of the linear function is a *straight line*. For $m > 0$ the function is increasing, for $m < 0$ it is decreasing, and for $m = 0$ the line is horizontal.

The coefficient m is known as the *gradient*. To clarify its meaning, let us solve the following problem: find the coefficient m for a straight line through the points P_1 (x_1, y_1)



and P_2 (x_2, y_2).

Since both points lie on the line, their coordinates satisfy the equations:

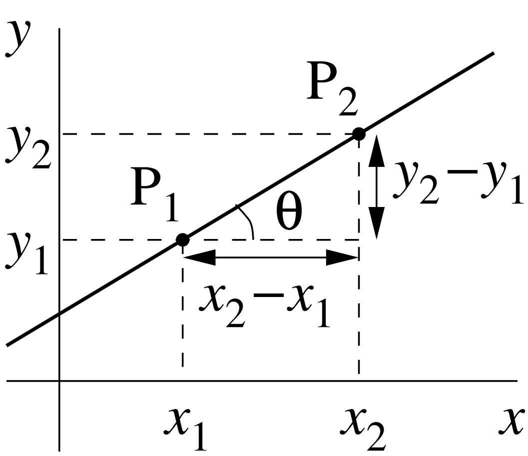
$$y_1 = mx_1 + c,$$
$$y_2 = mx_2 + c.$$

Subtracting the first equation from the second one, we have

$$y_2 - y_1 = m(x_2 - x_1),$$

which gives the gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \tag{2.4}$$



It is clear from the diagram that the ratio on the right-hand side of equation (2.4) is the 'opposite over adjacent' in the right-angle triangle with the hypotenuse formed by the line $y = mx + c$, and hence,

$$m = \tan\theta,$$

where θ is the angle that the line makes with the horizontal x axis.

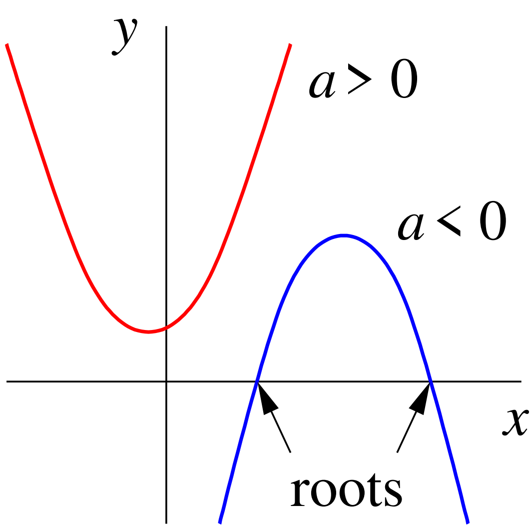
From the same triangle, the distance l between points P_1 and P_2 is

$$l = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}, \tag{2.5}$$

by Pythagoras's theorem.

2. Quadratic function $y = ax^2 + bx + c$.

The graph of the quadratic function ($a \neq 0$) is a *parabola*. For large positive or large negative x, the ax^2 term is dominant. It determines whether the "branches" of the parabola are upwards ($a > 0$) or downwards ($a < 0$).



The intercepts of the parabola with the x axis are the *roots* of the quadratic equation

$$ax^2 + bx + c = 0. \tag{2.6}$$

They are given by the familiar expression

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{2.7}$$

In this formula the quantity $b^2 - 4ac$ under the square root is the *discriminant* of the quadratic equation. Equation (2.6) has real roots only if $b^2 - 4ac \geq 0$. For $b^2 - 4ac = 0$ the two roots coincide, becoming a root of *multiplicity* two. For $b^2 - 4ac < 0$, the right-hand side of (2.7) contains the square root of a negative number, which cannot be found in $\mathbb{R}$. There is, however, a further extension of the set of numbers where this becomes possible (see Sec. 3.4).

It is useful to remember how (2.7) is found. First, divide equation (2.6) by a,

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

This is followed by *completing the square*. This is an equivalent transformation aimed at collecting all terms that contain x inside squared brackets,[4]

$$\begin{aligned} x^2 + 2\frac{b}{2a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a} &= 0, \\ \left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} &= 0, \\ \left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2}. \end{aligned}$$

Assuming that $b^2 - 4ac \geq 0$, we have

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}},$$

where we use the $\pm$ sign to account for both possible values, since the square root function itself is non-negative. Using $\sqrt{4a^2} = 2a$ in the denominator on the right-hand side,[5] transferring $b/(2a)$ to the right-hand side and writing the result as a single fraction, gives (2.7).

[4] The pattern we want to match is given by $(x + d)^2 = x^2 + 2xd + d^2$.

[5] Strictly speaking, we must have $\sqrt{4a^2} = 2|a|$; however, the $\pm$ sign allows for both signs of this expression, so we do not need to worry whether a is positive or negative.

2.4 Exponential and logarithmic functions

1. **Exponential function $y = e^x$.**

In the exponential function $y = e^x$, the number e introduced by Leonard Euler,[6] is the base of natural logarithms. This is a transcendental number, whose first 16 digits are quite easy to remember:

$$e = 2.7\underbrace{1828}_{\text{LT}}\underbrace{1828}_{\text{LT}}\underbrace{459045}_{\text{angles}}\ldots$$

Here 2.7 is followed by 1828, the year of birth of Leo Tolstoy[7] (twice!), and then by the angles of the right-angle isosceles triangle.

The graph of $y = e^x$ is shown in figure 1. One point on this graph that can be determined without a calculator is $x = 0$, $y = e^0 = 1$. The domain and range of this function are $\mathbb{R}$ and $(0, +\infty)$, respectively.

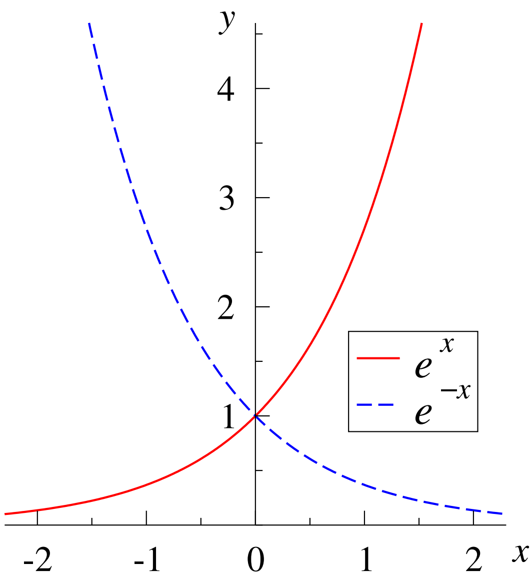


Figure 1: The exponential function $y = e^x$ and $y = e^{-x}$.

It is useful to plot another function, $y = e^{-x}$, in the same axes. Its graph is obtained by "switching" the positive and negative directions of the x axis, i.e., by flipping the graph of $y = e^x$ about the y axis (see figure 1). While $y = e^x$ is an increasing function, $y = e^{-x}$

[6] Leonard Euler was born in 1707 in Basel, Switzerland. For much of his adult life he worked in St Petersburg, Russia, where he died in 1783.

[7] Leo Tolstoy (1828–1910), Russian writer, author of *Anna Karenina*, *War and Peace*, etc.

is a decreasing one.

Exponential identities:

$$e^{a+b} = e^a e^b$$

$$(e^a)^n = e^{an}$$

$$e^{-a} = \frac{1}{e^a}$$

2. **Logarithmic function $y = \ln x$.**

Considering $y = e^x$ as an equation with a given y and unknown x, we write its solution as $x = \ln y$. The exponential and natural logarithm functions are inverse functions of each other.[8] The graph of $y = \ln x$ is obtained from the graph of $y = e^x$ (figure 1) by "switching" the axes x and y, see figure 2. One point on this graph that is found without a calculator is $x = 1$, $y = \ln 1 = 0$. For $y = \ln x$, the domain and range are $x > 0$ and $-\infty < y < \infty$, respectively.

Logarithmic identities:

$$\ln(ab) = \ln a + \ln b$$

$$\ln \frac{a}{b} = \ln a - \ln b$$

$$\ln (a^n) = n \ln a$$

It is interesting to use the graph of $y = \ln x$ to produce the graph of a related function $y = \ln(2x)$. This function achieves the same values "twice earlier", i.e., when its argument is a factor of 2 smaller than that of $\ln x$. Hence, its graph is obtained by "squeezing" the graph of $y = \ln x$ towards the y axis by a factor of 2. Alternatively, since $\ln(2x) = \ln x + \ln 2$, the graph of $y = \ln(2x)$ is obtained by shifting the graph of $y = \ln x$ upwards by $\ln 2 \approx 0.693$.

[8] We have $e^{\ln y} = y$ and $\ln(e^x) = x$; in particular, $\ln e = 1$.

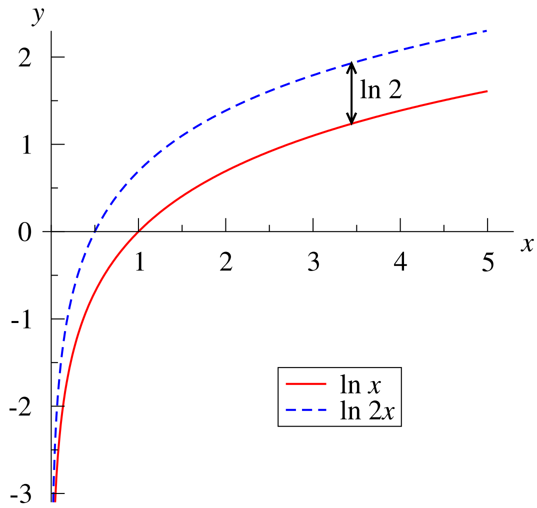


Figure 2: The logarithmic function $y = \ln x$ and $y = \ln(2x)$.

Inverse functions.

In general, if the equation $y = f(x)$ may be solved uniquely for the variable x, then this solution defines the *inverse function* denoted $x = f^{-1}(y)$.[9]

A function and its inverse satisfy the identity $y = f(f^{-1}(y))$. For example, we have $e^{\ln y} = y$ and also $\ln(e^x) = x$.

For a function to have an inverse, it must be 'one-to-one'. If a function is not one-to-one, as, e.g., $y = x^2$, then a particular one-to-one branch of the function can be chosen to define the inverse. Thus, if we restrict the domain of $y = x^2$ to $x \geq 0$ (the right-hand-side branch of the parabola) then the equation $y = x^2$ has a unique solution denoted by $x = \sqrt{y}$. This square-root function serves as the inverse of the square function. Using it, one can find *all* solutions of the equation $y = x^2$ as $x = \pm\sqrt{y}$ (for $y \geq 0$).

2.5 Trigonometric functions

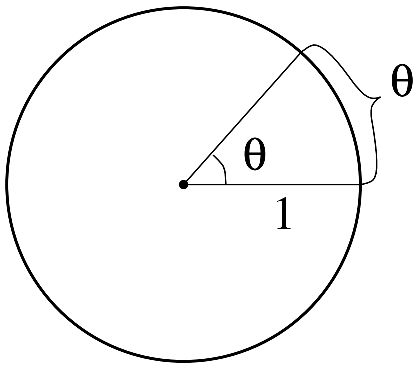
2.5.1 Radians

Angles are commonly measured in degrees. Degrees are obtained by dividing the circle into 360 equal parts. However, this measure is essentially *arbitrary*. This number is a convenient approximation for the number of days in which the Earth travels around the Sun (365.26 days). It makes the full angle divisible into many useful parts: 2, 3, 4, 5, 6,

[9] The notation f^{-1} should not be confused with the reciprocal, i.e., the multiplicative inverse, e.g., as in $x^{-1} = 1/x$. This is why inverse functions usually have distinct names and notation, as with $y = e^x$ and $x = \ln y$.

8, 12, etc.[10] One degree is approximately the angle travelled by the Earth in its motion around the Sun in one day.

A *natural* measure for the angles is *radians*. Radians can be defined by considering a circle of unit radius (*unit circle*) and stating that the magnitude of the angle, say, θ, is equal to the length of the arc that it subtends.



Since all circles are similar, the length of the arc subtended by the angle of θ on a circle of radius r is then $l = r\theta$.

The length of the circumference of the unit circle is $2\pi \times 1 = 2\pi$. Hence the full angle is 2π radians, and the straight angle (180 degrees) is π radians.

Conversion of angles measured in degrees and radians is done by

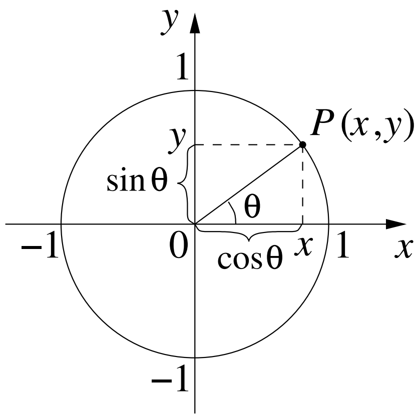
$$\theta[\text{degrees}] = \frac{\theta[\text{radians}] \times 180}{\pi}, \qquad \theta[\text{radians}] = \frac{\theta[\text{degrees}] \times \pi}{180}.$$

2.5.2 Cosine and sine

Consider a unit circle with the centre at the origin of the x-y plane, and a point P that lies on the unit circle and corresponds to angle θ.

By convention, this angle is measured from the *positive direction* of the x axis, and is positive if the rotation is in the *anticlockwise* direction. Rotation in the clockwise direction corresponds to a negative angle.

[10] Mars orbits the Sun in 687 days, but the day on Mars is about 3% longer than the day on Earth, so the Marsian year is 667 Martian days. So, if we lived on Mars, what number of degrees we would have divided the circle into? 640? 720?... And the year on Venus is about 225 Earth days, but only about two “Venerean days”...



Definition: $\cos\theta$ and $\sin\theta$ are the x and y coordinates of point P,

$$\cos\theta = x, \quad \sin\theta = y. \tag{2.8}$$

Properties of cosine and sine.

1. Note that since 2π is the full angle, all angles of the form

$$\theta + 2\pi n, \quad n \in \mathbb{Z}$$

correspond to the *same point* on the unit circle. Therefore,

$$\cos(\theta + 2\pi n) = \cos\theta, \quad \sin(\theta + 2\pi n) = \sin\theta, \tag{2.9}$$

which means that the cosine and sine functions are *periodic* with period 2π.

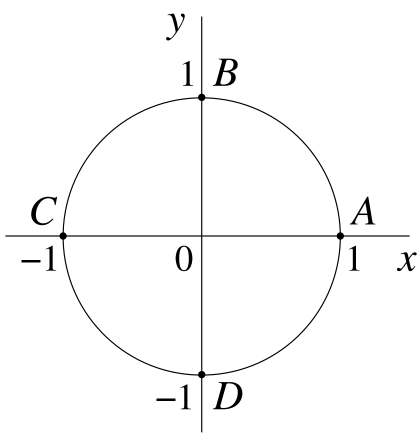
2. For all points on the unit circle, $x^2 + y^2 = 1$ (Pythagoras's theorem). Hence,

$$\cos^2\theta + \sin^2\theta = 1. \tag{2.10}$$

3. For $0 < \theta < \pi/2$, the point P is in the first quadrant, where both $\cos\theta$ and $\sin\theta$ are positive (see diagram above). In this case, x and y are the *adjacent* and *opposite* sides, respectively, to the angle θ in the right-angled triangle with a unit hypotenuse. Hence, our definition (2.8) of $\cos\theta$ and $\sin\theta$ is in agreement with the usual "triangle definition",

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}.$$

4. From the definition (2.8), we see that $\cos\theta$ is positive for all angles which correspond to points that lie on the right-hand-side semicircle of the unit circle (since $x > 0$ here), and negative for the angles that give points on the left-hand-side semicircle ($x < 0$). Similarly, $\sin\theta$ is positive for all angles which give points on the top semicircle ($y > 0$), and negative for the points on the bottom semicircle ($y < 0$).



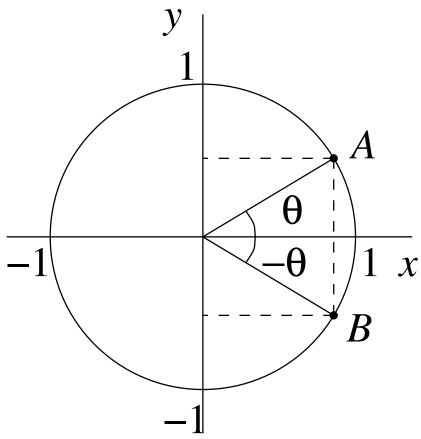
Using (2.8), it is also easy to determine the values of $\cos\theta$ and $\sin\theta$ for particular angles, such as 0, $\pi/2$, π and $3\pi/2$, shown by points A, B, C and D, respectively. Looking at the x and y coordinates of these points, we obtain:

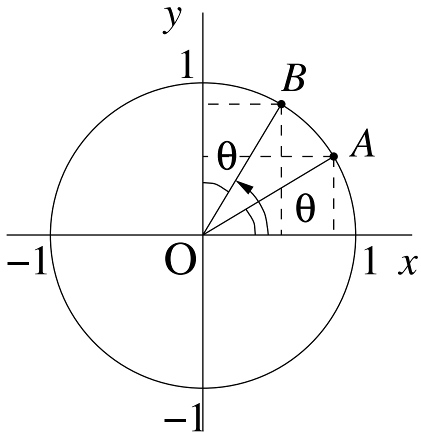
$$\begin{array}{llll} \cos 0 = 1, & \cos\dfrac{\pi}{2} = 0, & \cos\pi = -1, & \cos\dfrac{3\pi}{2} = 0, \\ \sin 0 = 0, & \sin\dfrac{\pi}{2} = 1, & \sin\pi = 0, & \sin\dfrac{3\pi}{2} = -1. \end{array}$$

5. Let us examine two angles, θ and $-\theta$, with the corresponding points A and B on the unit circle. The diagram shows that their x coordinates are the same, while their y coordinates have equal magnitudes but opposite signs. Hence,

$$\cos(-\theta) = \cos\theta, \quad \sin(-\theta) = -\sin\theta.$$

This means that $\cos\theta$ is an *even* function and $\sin\theta$ is an *odd* function.





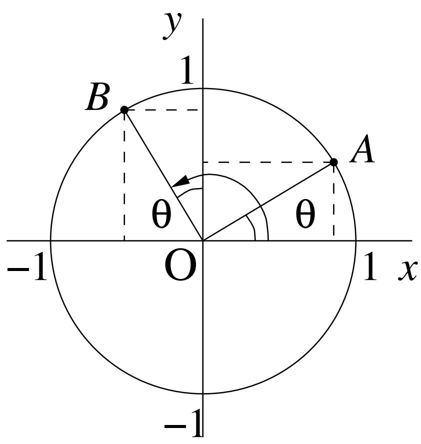
6. There are other pairs of angles for which useful relations between their sines and cosines can be established. Consider, for example, the angles θ and $\frac{\pi}{2} - \theta$, represented by points A and B on the unit circle. [The position of point B can be found by first rotating through $\pi/2$ (i.e., 90 degrees) anticlockwise and then by θ clockwise.]

   In this case we see that the x coordinate of point B is equal to the y coordinate of point A, and that the y coordinate of point B is equal to the x coordinate of point A. Hence,]

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta, \quad \sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta.$$

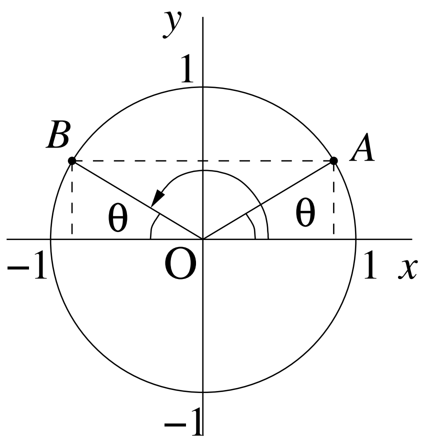
7. Consider now the angles θ (point A) and $\frac{\pi}{2} + \theta$ (point B). The position of point B can be found by first rotating through $\pi/2$ anticlockwise and then by a further rotation through θ in the same direction. In this case we see that the x coordinate of point B is equal to the magnitude of the y coordinate of point A, but has the opposite sign (i.e., is negative), and that the y coordinate of point B is equal to the x coordinate of point A. Hence,

$$\cos\left(\frac{\pi}{2}+\theta\right) = -\sin\theta, \quad \sin\left(\frac{\pi}{2}+\theta\right) = \cos\theta.$$



8. Another useful pair of angles are θ and $\pi-\theta$, shown by points A and B, respectively. We see that the two points have equal y coordinates (hence, the sines of the corresponding angles are the same), while their x coordinates have equal magnitudes but opposite signs. Hence,

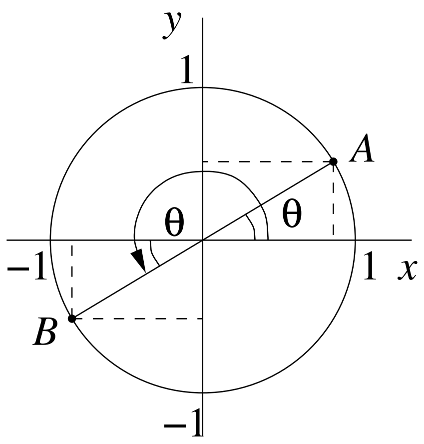
$$\cos(\pi-\theta) = -\cos\theta, \quad \sin(\pi-\theta) = \sin\theta.$$



9. The last pair of angles that we consider are θ and $\pi+\theta$, again shown by points A and B, respectively. This is probably one of the easiest case, as we can see that both pair of coordinates of the points A and B are equal in magnitude but opposite in sign. Therefore,

$$\cos(\pi+\theta) = -\cos\theta, \quad \sin(\pi+\theta) = -\sin\theta.$$

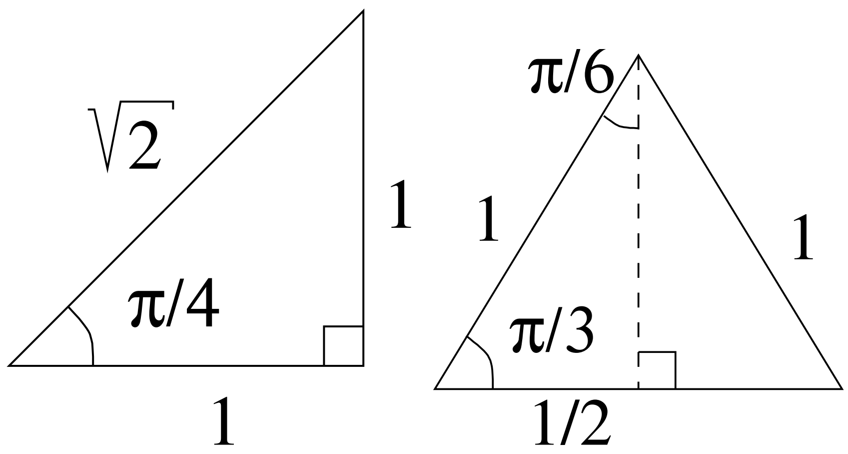
Both the cosine and the sine functions change their signs if the angle is increased by π (or by 180 degrees).



10. Finally, it is useful to know the exact values of the trigonometric functions for some common angles. They are shown in table 1.

Table 1: Values of $\cos\theta$ and $\sin\theta$ for some common angles.

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
θ (degrees)	0	30	45	60	90	180	270	360
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0



Note that the values of $\cos\dfrac{\pi}{4}$ and $\sin\dfrac{\pi}{4}$ are obtained by considering am isosceles right-angled triangle with sides of lengths 1, 1 and $\sqrt{2}$ (see diagram). The values of $\cos\dfrac{\pi}{3}$ and $\sin\dfrac{\pi}{6}$ are obtained by considering an equilateral triangle with unit sides, while $\sin\dfrac{\pi}{3}$ and $\cos\dfrac{\pi}{6}$ are obtained using equation (2.10) as $\sqrt{1-(1/2)^2}=\sqrt{3}/2$.

11. We conclude by sketching $\cos\theta$ and $\sin\theta$. This is done by observing how each of this functions changes and what values it takes as the angle increases from zero through $\pi/2$ to π. We also use the fact that $\cos\theta$ is an even function, while $\sin\theta$ is odd. Finally, once we have plotted both functions for θ between $-\pi$ and π we apply this "pattern" to the whole real axis, as the functions are periodic (see figure 3).

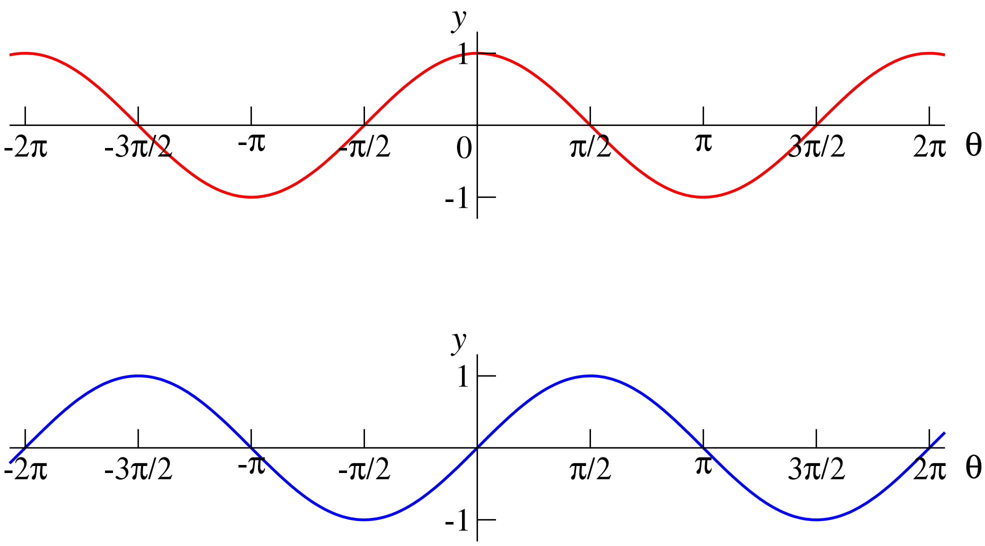


Figure 3: Graphs of $y=\cos\theta$ (top) and $y=\sin\theta$ (bottom).

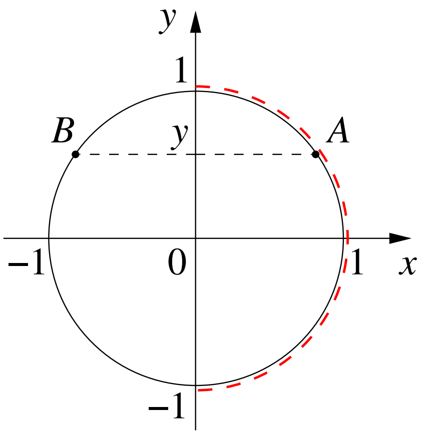
There are a number of useful formulae for the sine and cosine of sums and differences of angles, for $\cos 2\theta$ and $\sin 2\theta$ (double angle formulae), as well as for the products of sines or cosines (see the Trig Cheat Sheet available on Canvas).

2.5.3 Inverse trigonometric functions

(1) Consider the equation for α,

$$\sin\alpha = y, \tag{2.11}$$

where y is given. Its solution is given by *all* angles α that correspond to the two points (A and B) on the unit circle whose horizontal coordinate equals x.



Since equation(2.11) for a given y is satisfied by (infinitely) many values of α, they cannot be given by a function. (A function gives only one value for a given value of the argument.)

By convention, the *inverse sine* function $\arcsin y$ is defined for all $-1 \leq y \leq 1$ (domain) and gives angles in the interval $[-\pi/2, \pi/2]$ (range). This range is shown by the dashed line on the diagram above.

Using this function,[11] all angles corresponding to point A can be written as

$$\alpha = \arcsin y + 2\pi n \quad (n \in \mathbb{Z}),$$

and the angles corresponding to point B are given by

$$\alpha = \pi - \arcsin y + 2\pi n \quad (n \in \mathbb{Z}).$$

All angles given by these two expressions can be written concisely as

$$\boxed{\alpha = (-1)^n \arcsin y + \pi n\,,} \quad n \in \mathbb{Z}. \tag{2.12}$$

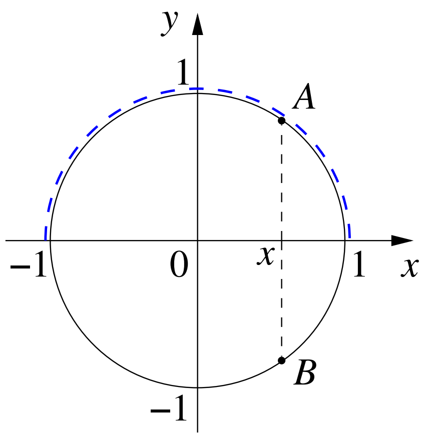
In this form even n give angles shown by point A, and odd n by point B.

(2) Similarly, consider the equation for α,

$$\cos \alpha = x, \tag{2.13}$$

[11] It is represented by the $\boxed{\sin^{-1}}$ button on the calculator.

where x is given. Its solution is given by *all* angles α that correspond to the two points (A and B) on the unit circle whose vertical coordinate equals y.



By convention, the *inverse cosine* function $\arccos x$ is defined for all $-1 \le x \le 1$ (domain) and gives angles in the interval $[0, \pi]$ (range). This range is shown by the dashed line on the diagram.

Using this function,[12] all angles corresponding to point A can be written as

$$\alpha = \arccos x + 2\pi n \quad (n \in \mathbb{Z}),$$

and the angles corresponding to point B are given by

$$\alpha = -\arccos x + 2\pi n \quad (n \in \mathbb{Z}).$$

All angles given by these two expressions can be written concisely as

$$\boxed{\alpha = \pm \arccos x + 2\pi n\,,} \quad n \in \mathbb{Z}. \tag{2.14}$$

The graphs of $y = \arcsin x$ and $y = \arccos x$ are shown in figure 4. They are obtained from the graphs of $\sin\theta$ and $\cos\theta$ (figure 3) by plotting parts of these curves (for the angles $-\pi/2 \le \theta \le \pi/2$ and $0 \le \theta \le \pi$, respectively), with the angle axis vertically up.

2.5.4 Tangent and cotangent

The tangent is not a "completely new" function, but a useful combination of the sine and cosine functions that appears naturally in various problems.

[12] It is represented by the $\boxed{\cos^{-1}}$ button on the calculator.

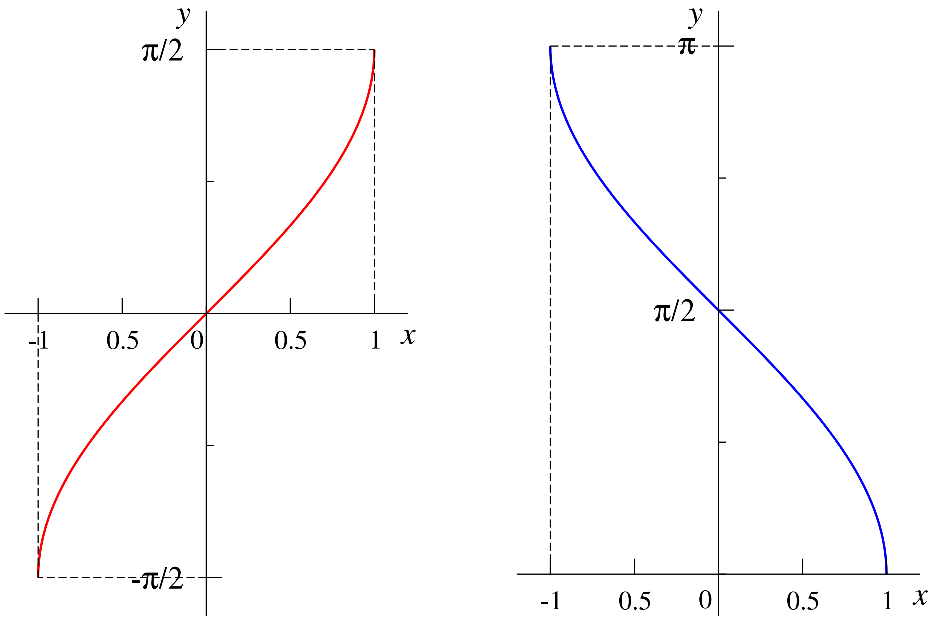
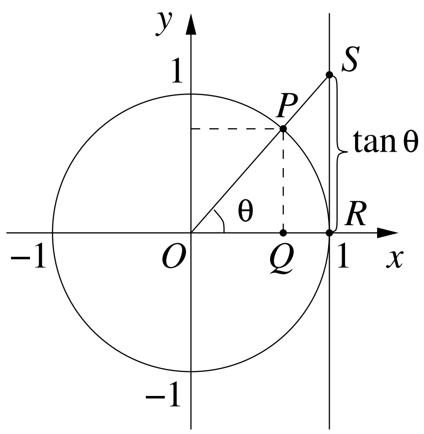


Figure 4: Graphs of $y = \arcsin x$ (left) and $y = \arccos x$ (right).

Definition:

$$\tan\theta = \frac{\sin\theta}{\cos\theta}. \tag{2.15}$$

This ratio is defined for any angle for which $\cos\theta \neq 0$. Hence, its domain includes all θ such that $\theta \neq \pi/2 + \pi n$ ($n \in \mathbb{Z}$).



We can see what $\tan\theta$ is geometrically using the unit circle diagram. We start by drawing a tangent to the circle at point R $(1, 0)$; it is parallel to the y axis. We then continue the line from the origin O through point P (corresponding to angle θ) until it meets the tangent at point S.

Since PQ is parallel to the y axis, the two triangles, OPQ and OSR, are similar. Hence, the ratios of their corresponding sides are equal:

$$\frac{|PQ|}{|OQ|} = \frac{|SR|}{|OR|}.$$

Since $|PQ| = \sin\theta$ and $|OQ| = \cos\theta$ (by definition), and $|OR| = 1$, we have

$$|SR| = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

The diagram shows that as the angle θ increases from 0 to $\pi/2$, $\tan\theta$ increases from zero to infinity (point S heading upwards along the tangent[13]).

Properties of $\tan\theta$.

1. Changing θ to $\theta + \pi$ in equation (2.15), we obtain

$$\tan(\theta + \pi) = \frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta,$$

   where we have used property 9 of sine and cosine.

   This shows that $\tan\theta$ is *periodic* with period π.

2. Changing θ to $-\theta$ in equation (2.15), and using property 5 of sine and cosine, we have:

$$\tan(-\theta) = \frac{\sin(-\theta)}{\cos(-\theta)} = \frac{-\sin\theta}{\cos\theta} = -\tan\theta,$$

   which means that $\tan\theta$ is an *odd* function.

3. Using data from table 1, we find the following useful exact values:

$$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}, \quad \tan\frac{\pi}{4} = 1, \quad \tan\frac{\pi}{3} = \sqrt{3}.$$

[13] This explains the name of the function.

To sketch the tangent function, we first show how it increases from 0 towards infinity as θ increases from 0 to $\pi/2$. We then use the fact that $\tan\theta$ is an odd function, so its graph for $-\pi/2 < \theta < 0$ is obtaining by rotating the graph for $0 \le \theta < \pi/2$ through 180 degrees about the origin. We then use the fact that $\tan\theta$ is periodic with period π and copy the graph from $(-\pi/2, \pi/2)$ to all other intervals $(-\pi/2 + \pi n, \pi/2 + \pi n)$, $n \in \mathbb{Z}$ (see figure 5).

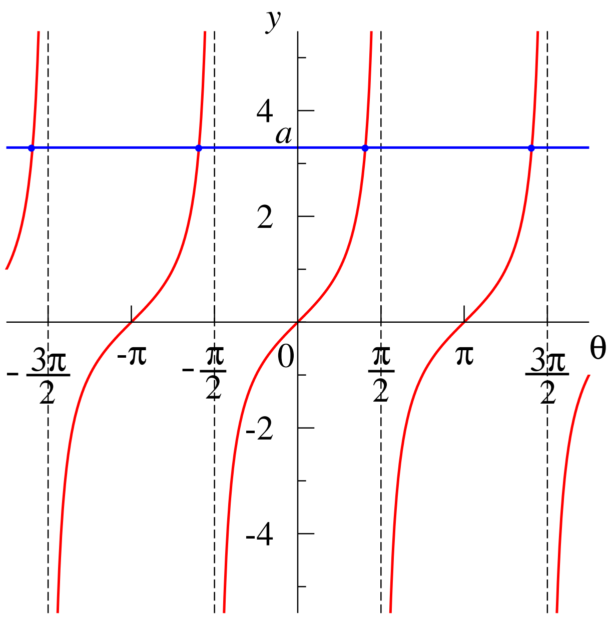


Figure 5: Graph of the function $y = \tan\theta$ and graphical solution of the equation $\tan\theta = a$.

Suppose we need to solve the equation

$$\tan\theta = a. \tag{2.16}$$

Let us first do this graphically. The solutions are given by the points where the graph of $y = \tan\theta$ meets the horizontal line $y = a$ (see figure 5). We see that equation (2.16) has solutions for any $-\infty < a < \infty$, and there is exactly one solution for every interval $(-\pi/2 + \pi n, \pi/2 + \pi n)$, $n \in \mathbb{Z}$.

By convention, the principal solution is taken from the interval $-\pi/2 < \theta < \pi/2$, and is denoted by $\arctan a$. The arctangent function[14] is thus defined for all $\mathbb{R}$ (domain) and its range is $(-\pi/2, \pi/2)$. The complete solution of equation (2.16) is then given by

[14] It is represented by the $\boxed{\tan^{-1}}$ button on the calculator.

$$\boxed{\theta = \arctan a + \pi n\,,} \qquad n \in \mathbb{Z}. \tag{2.17}$$

The graph of the function $y = \arctan x$ is shown in figure 6. It is obtained from central branch of the graph of the tangent (figure 5) by switching the vertical and horizontal axes.

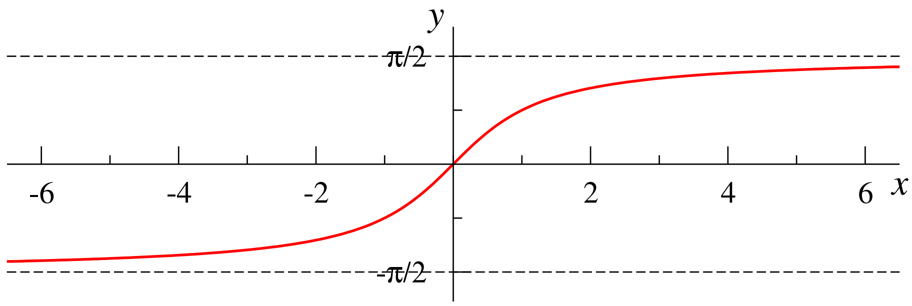


Figure 6: Graph of the function $y = \arctan x$ has two horizontal asymptotes, $y = -\pi/2$ and $y = \pi/2$, approached for $x \to -\infty$ and $x \to \infty$, respectively.

The cotangent is the reciprocal of tangent.

Definition:

$$\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}. \tag{2.18}$$

This function is defined for all angles such that $\sin\theta \neq 0$. Hence, its domain includes all $\theta \neq \pi n$ ($n \in \mathbb{Z}$). It follows from the properties of $\tan\theta$ that $\cot\theta$ is also an *odd* function with *period* π.

Another useful relation between the cotangent and tangent can be obtained by using the relations between sine and cosine of the angles θ and $\pi/2 - \theta$ (property 6 from Sec. 2.5.2):

$$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{\sin\left(\frac{\pi}{2} - \theta\right)}{\cos\left(\frac{\pi}{2} - \theta\right)} = \tan\left(\frac{\pi}{2} - \theta\right).$$

Hence, the graph of $\cot\theta$ can be obtained from the graph of $\tan\theta$ (figure 5) by first, reflecting the graph in the y axis (i.e., swapping θ and $-\theta$), and then shifting the graph by $\pi/2$. The result is shown in figure 7.

Solving the equation

$$\cot\theta = a \tag{2.19}$$

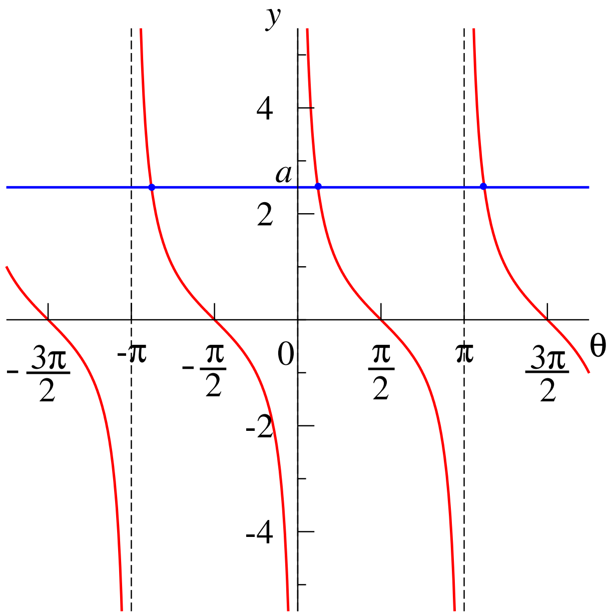


Figure 7: Graph of the function $y = \cot\theta$ and graphical solution of the equation $\cot\theta = a$.

graphically, we find the points where the graph of $y = \cot\theta$ meets the horizontal line $y = a$ (see figure 7). Equation (2.19) has solutions for any $-\infty < a < \infty$, and there is exactly one solution in every interval $(\pi n, \pi + \pi n)$, $n \in \mathbb{Z}$.

By convention, the principal solution is taken from the interval $0 < \theta < \pi$, and is denoted by $\operatorname{arccot} a$. The arccot function[15] is thus defined for all $\mathbb{R}$ (domain) and its range is $(0, \pi)$. The complete solution of equation (2.19) is then given by

$$\boxed{\theta = \operatorname{arccot} a + \pi n\,,} \qquad n \in \mathbb{Z}. \tag{2.20}$$

The graph of the function $y = \operatorname{arccot} x$ is shown in figure 8. It is obtained from $(0, \pi)$ branch of the cotangent graph (figure 7) by switching the vertical and horizontal axes.

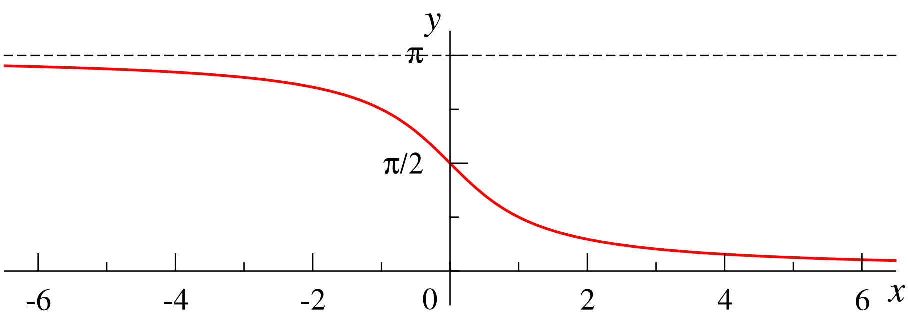


Figure 8: Graph of the function $y = \operatorname{arccot} x$ has two horizontal asymptotes, $y = \pi$ and $y = 0$, approached for $x \to -\infty$ and $x \to \infty$, respectively.

It is important to be able to sketch all these graphs!

[15] Sometimes denoted $\cot^{-1}$.

3 Basic calculus

3.1 Limits

This short section is intended to provide intuitive understanding of limits. Let us start with an example. Consider the following sequence:

$$1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\ldots,$$

whose nth term is $a_n = 1/n$. With increasing n the terms in the sequence become smaller and smaller, getting ever closer to zero (though never actually "hitting" it). In fact, for any positive number ε, however small, one can always find a sufficiently large number N so that $a_n < \varepsilon$ for all $n > N$ (take, e.g., $N = 1/\varepsilon$). So, the terms in the above sequence are getting arbitrarily close to zero. In this case we say that zero is the *limit* of the sequence (a_n). We write $\lim\limits_{n\to\infty} a_n = 0$, or using the explicit form of a_n,

$$\lim_{n\to\infty} \frac{1}{n} = 0,$$

where $n \to \infty$ means that n tends to infinity.[16]

Not all limits are so easy to find. For example, consider a sequence[17]

$$a_n = \left(1 + \frac{1}{n}\right)^n, \quad n = 1,\ 2,\ 3,\ldots. \tag{3.1}$$

As n increases, the expression in brackets becomes close to unity, and since we know that $1^n = 1$, we may guess (incorrectly!) that $\lim\limits_{n\to\infty} a_n = 1$. On the other hand, we know that for any number $p > 1$, even if p is close to unity, e.g., $p = 1.01$, the sequence p^n displays unbounded growth, i.e., arbitrarily large value are obtained for sufficiently high n (e.g., $1.01^{1\,000\,000} \approx 2.36 \times 10^{4321}$, which is a pretty large number.[18]). In fact, the sequence (3.1) is finely poised, as it converges to a well-known number between unity and infinity:

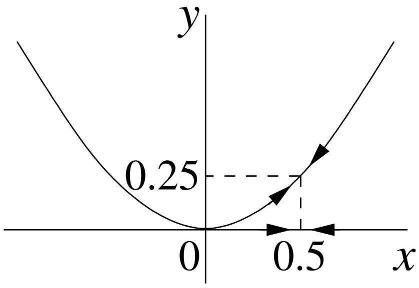
$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e. \tag{3.2}$$

Proving this remarkable result requires a careful investigation.

[16] In general, the number a is the limit of the sequence $\{a_n\}$ ($\lim\limits_{n\to\infty} a_n = a$) if for any $\varepsilon > 0$ there exists a number N (which depends on ε), such that $|a_n - a| < \varepsilon$ for all $n > N$.

[17] This sequence was first considered in 1683 by Jacob Bernoulli who looked at the problem of compound interest, but the limit (3.2) was obtained by Euler in 1748.

[18] Can you verify this result without "breaking" your calculator?



One can also consider limits of functions. For example, take $y = x^2$ and let x tend to 0.5, i.e., $x \to 0.5$. As x slides towards 0.5 from either side, the value of the function approaches $0.5^2 = 0.25$, so we write $\lim_{x\to 0.5} x^2 = 0.25$. In general, if a function $f(x)$ is continuous at point $x = a$, then it approaches its value $f(a)$ as x tends to a, i.e., $\lim_{x\to a} f(x) = f(a)$.[19]

Not all functions are continuous at all points, though. For example, the function $\mathrm{sgn}(x)$ (Sec. 2.2) is continuous for $x > 0$ and for $x < 0$, but has a discontinuity ("jump") at $x = 0$, so that $\lim_{x\to 0} \mathrm{sgn}(x)$ does not exist.[20]

In addition to limits at finite values of x, one can consider limits for $x \to +\infty$ and $x \to -\infty$. For example, $\lim_{x\to+\infty} e^{-x} = 0$ (see graph of e^{-x} in Sec. 2.4), or

$$\lim_{x\to\infty} \frac{x+1}{x^2+5} = 0,$$

since the denominator increases faster than the numerator for large x, or

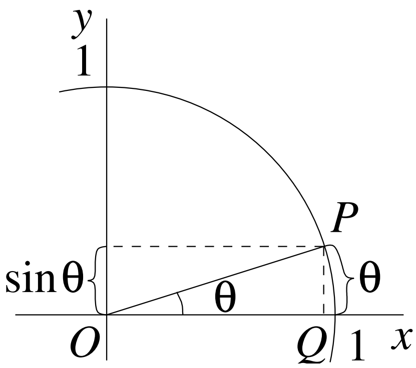
$$\lim_{x\to-\infty} \arctan x = -\frac{\pi}{2},$$

as can be seen from the graph of $\arctan x$ in Sec. 2.5.4.

Just as with sequences, not all limits are so easy to find. For example, in $\lim_{x\to 0} \frac{\sin x}{x}$, both the numerator and the denominator vanish for $x = 0$, so we have an indeterminate expression "0/0". To find this limit, one needs to consider values of $\sin\theta$ for small angles θ. (Small here means $\theta \ll 1$, i.e., angles much smaller than one radian.) The diagram shows that for such a small angle, the side PQ of the triangle OPQ (which gives $\sin\theta$), is very close to the length of the arc subtended by the angle θ, so $\sin\theta \simeq \theta$. (The sign $\simeq$ indicates that this equality holds asymptotically for $\theta \to 0$.) Hence,

[19] The number A is the limit of the function $f(x)$ for $x \to a$ (i.e., $\lim_{x\to a} f(x) = A$) if for every $\varepsilon > 0$ there exists $\delta > 0$ (which depends on ε) such that $|f(x) - A| < \varepsilon$ for all $|x - a| < \delta$.

[20] In such cases one can define the left- and right-hand side limits, so that $\lim_{x\to 0-} \mathrm{sgn}(x) = -1$ and $\lim_{x\to 0+} \mathrm{sgn}(x) = 1$, where, in general, $x \to a-$ and $x \to a+$ indicate that x approaches a from below and from above, respectively.

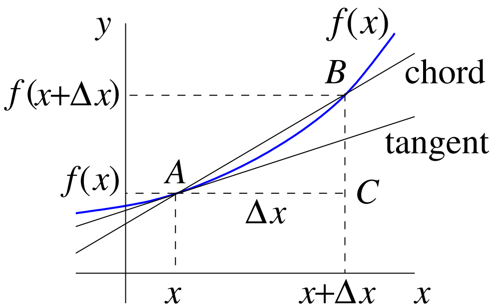


$$\lim_{x\to 0}\frac{\sin x}{x}=1.$$

3.2 Derivatives and differentials

3.2.1 Derivative

The derivative solves the problem of finding the gradient of a tangent to the graph of a function.



Consider the graph a function $y = f(x)$ and the tangent to it at point A with coordinates $(x, f(x))$. To find its gradient, consider a *chord* through points A and B, where B also lies on the graph and have the coordinates $(x + \Delta x, f(x + \Delta x))$. Here Δx ('delta x') is a single symbol that denotes the *change* in x.

In the triangle ABC, $|AC| = \Delta x$ and $|BC| = f(x + \Delta x) - f(x)$. Hence, the gradient of the chord is

$$\frac{|BC|}{|AC|} = \frac{f(x + \Delta x) - f(x)}{\Delta x}. \tag{3.3}$$

The chord will turn into the tangent at point A if we let point B slide along the graph towards A, i.e., if we make Δx smaller and smaller.

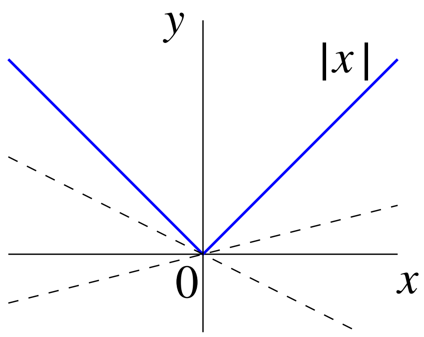
This is achieved by taking the *limit* $\Delta x \to 0$. In this limit the gradient of the chord (3.3) will turn into the gradient of the tangent at point x. This quantity is denoted $f'(x)$ and is called the *derivative* of the function. According to the above,

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}, \tag{3.4}$$

where $\lim\limits_{\Delta x \to 0}$ means taking the limit indicated below.

Besides $f'(x)$ and y', other notation for the derivative of the function $y = f(x)$ is[21]

$$\frac{dy}{dx} \quad \text{or} \quad \frac{df}{dx}. \tag{3.5}$$



Note that in order for the tangent and the derivative to exist, the function must be "smooth".[22] Consider, for example, the function $y = |x|$. For $x > 0$ its graph is represented by a straight line with gradient 1, and hence the derivative of $|x|$ for $x > 0$ is 1. For $x < 0$ its gradient and derivative are equal to -1. However, we cannot draw a unique tangent to the graph at $x = 0$, since many lines pass through this point. The derivative of $|x|$ at this point does not exist.

3.2.2 Table of standard derivatives

Derivatives of functions can be found directly from expression (3.4). Consider, for example, the quadratic function $f(x) = ax^2 + bx + c$. Its derivative is:

[21] This notation is due to Gottfried Wilhelm Leibniz (1746–1716).
[22] By definition, the function is *smooth* at point x if the limit (3.4) exists.

$$
\begin{aligned}
(ax^2+bx+c)' &= \lim_{\Delta x\to 0}\frac{a(x+\Delta x)^2+b(x+\Delta x)+c-(ax^2+bx+c)}{\Delta x}\\
&= \lim_{\Delta x\to 0}\frac{2ax\Delta x+a\Delta x^2+b\Delta x}{\Delta x}\\
&= \lim_{\Delta x\to 0}(2ax+b+a\Delta x)\\
&= 2ax+b,
\end{aligned}
$$

where the last term in the second last line vanishes in the limit $\Delta x \to 0$.

By using similar methods, one obtain the derivatives of common elementary functions shown in table 2.

Table 2: Table of standard derivatives.

Function	Derivative	Comment
C	0	C is a constant
x^n	nx^{n-1}	
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	using $n=\frac{1}{2}$ above
e^x	e^x	
$\ln x$	$\frac{1}{x}$	
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\frac{1}{\cos^2 x}$	
$\cot x$	$-\frac{1}{\sin^2 x}$	
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	
$\arctan x$	$\frac{1}{1+x^2}$	
$\operatorname{arccot} x$	$-\frac{1}{1+x^2}$	

3.2.3 Rules of differentiation

Finding the derivatives of combinations of elementary functions is done with the help of the following rules. (Below f and g are arbitrary differentiable functions of some independent variable.)

1. $(Cf)' = Cf'$, where C is a constant, i.e., a constant factor is taken outside when differentiating a function.

2. Sum rule:

$$(f+g)' = f' + g'.$$

   Note that $(f-g)' = f' - g'$ follows from the sum rule and rule 1.

3. Product rule:

$$(fg)' = f'g + fg'.$$

4. Quotient rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

5. Chain rule is used to differentiate a function of a function:

$$[f(g(x))]' = f'(g(x))g'(x),$$

   where f' is the derivative of f with respect to its argument (here, g). This rule can also be used when we have more than two "nested" functions. This rule looks especially natural in Leibniz's notation:

$$\frac{df}{dx} = \frac{df}{dg}\frac{dg}{dx}.$$

These rules can be proved starting with the definition of the derivative (3.4).

Examples.

Here we use the rules of differentiation and standard derivatives from table 2.

1. $(4x^7)' = 4(x^7)' = 4(7x^6) = 28x^6$. (Using rule 1 here.)

2. $(\sqrt{x} + \cos x)' = (x^{1/2})' + (\sin x)' = \dfrac{1}{2}x^{-1/2} - \sin x = \dfrac{1}{2\sqrt{x}} - \sin x$. (Using the sum rule.)

3. $(x \ln x)' = x' \ln x + x(\ln x)' = 1 \times \ln x + x\,\dfrac{1}{x} = \ln x + 1$. (Using the product rule.)

4. $(\tan x)'$ is a standard derivative, but let us verify it here using the quotient rule:

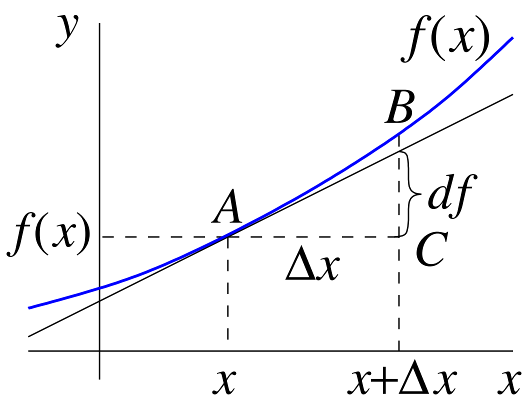
$$\begin{aligned}(\tan x)' = \left(\frac{\sin x}{\cos x}\right)' &= \frac{(\sin x)' \cos x - \sin x(\cos x)'}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}\,,\end{aligned}$$

   where we used the identity $\cos^2 x + \sin^2 x = 1$.

5. $\left(e^{x^2}\right)' = e^{x^2}\,(x^2)' = e^{x^2} \times 2x = 2xe^{x^2}$. (Using the chain rule.)

6. And here we have to use the chain rule twice:

$$\begin{aligned}(\ln[\sin(2x^3+5)])' &= \frac{1}{\sin(2x^3+5)}\,(\sin(2x^3+5))' = \\ &= \frac{1}{\sin(2x^3+5)}\,\cos(2x^3+5)(2x^3+5)' \\ &= \frac{\cos(2x^3+5)}{\sin(2x^3+5)}\,(2 \times 3x^2) \\ &= 6x^2 \cot(2x^3+5).\end{aligned}$$



3.2.4 Differential

Consider the graph of the function $y = f(x)$ and the tangent to it at point A with coordinates $(x, f(x))$. If we increase x by Δx the value of the function will change from $f(x)$ to $f(x+\Delta x)$, so the *change* in the function, denoted by Δf, is

$$\Delta f = f(x+\Delta x) - f(x).$$

This corresponds to the segment BC in the diagram.

For small Δx, the tangent follows the function very closely. It is much easier to evaluate the change in the tangent. This is given by $f'(x)\Delta x$, since $f'(x)$ is the gradient of the tangent at point x [see equation (2.4)]. This quantity is denoted df. It represents the *principal linear* part of the change in the function, and is called the *differential*:

$$df = f'(x)\Delta x. \tag{3.6}$$

If we consider $f(x) = x$, then $f'(x) = 1$, and equation (3.6) gives

$$dx = \Delta x.$$

Hence, the differential of the independent variable is simply equal to its change.

We can then re-write the **definition** of the differential as

$$\boxed{df = f'(x)dx.} \tag{3.7}$$

Equivalently, we can now write

$$\frac{df}{dx} = f'(x).$$

The definition (3.7) gives *meaning* to the common notation (3.5) for the derivative as the *ratio* of the differentials of the function and independent variable. (If the dependent variable is $y = f(x)$, then obviously $dy = df$.)

3.2.5 Applications of derivatives

One application of the derivative is in investigating the behaviour of functions.

If $f'(x) > 0$ then the function near point x is increasing. If $f'(x) < 0$, the function is decreasing.

If $f'(x) = 0$ then the tangent to the graph of the function at point $(x, f(x))$ is horizontal. Such point is known as the *stationary point*, since the function is neither increasing not decreasing here.

This point be can a *maximum*, if the function increases to the left of x and decreases to the right of x. Alternatively, this point can be a *minimum*, if the function decreases to the left of x and increase to the right of it. A third option is a *point of inflection*, when the function either increases or decreases both to the left and to the right of x, but "pauses" here instantaneously (see figure 9).

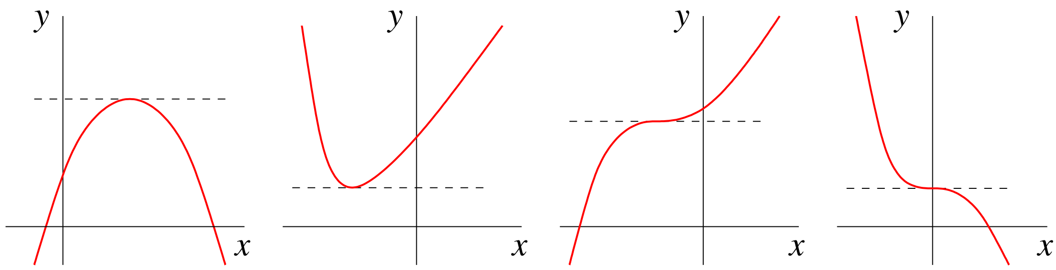


Figure 9: Stationary points (from left to right): maximum, minimum, points of inflection.

It is sometimes useful to use the *second derivative*, i.e., the derivative of the first derivative,

$$f''(x) = (f'(x))',$$

also denoted

$$\frac{d^2 f}{dx^2}.$$

In a similar way, one can find the third, forth and higher derivatives. The notation for the nth derivative is

$$f^{(n)}(x) \quad \text{or} \quad \frac{d^n f}{dx^n},$$

where in the first form n is in brackets, to distinguish it from the nth power.

In particular, the second derivative can be used to investigate the nature of stationary points (where $f'(x) = 0$). Namely, if $f''(x) > 0$ at this point (which means that the first derivative is increasing), then this point is a *minimum*. If $f''(x) < 0$ (the first derivative is decreasing) then this point is a *maximum*.

Note that these conditions are *sufficient*, but not necessary. For example, the function $f(x) = x^4$ is positive for all $x \neq 0$, and $f(0) = 0$, so $x = 0$ is a minimum. Accordingly, the first derivative $f'(x) = 4x^3$ equals zero at $x = 0$. However, the second derivative $f''(x) = 12x^2$ also vanishes at $x = 0$, thereby not allowing one to determine the type of the stationary point.

3.3 Integration

3.3.1 Primitives and indefinite integrals

Definition: A function $F(x)$ is called a *primitive* for the function $f(x)$ if

$$\frac{dF(x)}{dx} = f(x).$$

Example: a primitive for $f(x) = 2x$ is $F(x) = x^2$.

Note, however, that x^2+1 is also a primitive for $2x$, since $(x^2+1)' = 2x$. So the primitive $F(x)$ is not unique. Consider another function

$$F_1(x) = F(x) + C,$$

where C is a constant. Taking the derivative, we have

$$\frac{dF_1(x)}{dx} = \frac{dF(x)}{dx} + \frac{dC}{dx} = f(x) + 0 = f(x),$$

so, according to the above definition, $F_1(x)$ is also a primitive.

Definition: The set of all primitives of a function $f(x)$ is called the *indefinite integral* of $f(x)$:

$$\underbrace{\int f(x)dx}_{\text{notation}} = F(x) + C,$$

where $F(x)$ is a primitive of $f(x)$ and C is an arbitrary constant.

Properties:

1. $$\frac{d}{dx}\left(\int f(x)dx\right) = \frac{d}{dx}(F(x) + C) = f(x).$$

2. $$\int dF(x) \underset{\text{by (3.7)}}{=} \int F'(x)dx = \int f(x)dx = F(x) + C.$$

   We see that integration and differentiation are the opposite of each other, and the differential sign d and the integral sign $\int$ cancel each other out.

3. $$\int Af(x)dx = A\int f(x)dx, \text{ where } A \text{ is a constant.}$$

4. $$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx.$$

   The last two properties follow from the differentiation rules 1 and 2, see Sec. 3.2.3. They are consequences of the fact that both integration and differentiation are *linear* operations.

3.3.2 Table of standard integrals

Most of the entries below are obtained by looking at the table of standard derivatives (Sec. 3.2.2), and they can all be verified by taking the derivative of the function on the right-hand side.

(i) $$\int dx = x + C \quad \text{(this is an example of property 2 above).}$$

(ii) $$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1).$$

(iii) $\displaystyle\int \frac{dx}{x} = \ln|x| + C.$

(iv) $\displaystyle\int e^x dx = e^x + C.$

(v) $\displaystyle\int \cos x\, dx = \sin x + C.$

(vi) $\displaystyle\int \sin x\, dx = -\cos x + C.$

(vii) $\displaystyle\int \frac{dx}{\cos^2 x} = \tan x + C.$

(viii) $\displaystyle\int \frac{dx}{\sin^2 x} = -\cot x + C.$

(ix) $\displaystyle\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C = -\arccos x + \tilde{C}.$

(x) $\displaystyle\int \frac{dx}{1+x^2} = \arctan x + C = -\operatorname{arccot} x + \tilde{C}.$

(xi) $\displaystyle\int \frac{dx}{\sqrt{x^2+1}} = \ln\left(x + \sqrt{x^2+1}\right) + C.$

(xii) $\displaystyle\int \frac{dx}{\sqrt{x^2-1}} = \ln\left|x + \sqrt{x^2-1}\right| + C.$

The last two integrals are in fact the inverse hyperbolic functions $\operatorname{arcsinh} x$ and $\operatorname{arccosh} x$; these integrals are also known as "long logarithms".

3.3.3 Methods of integration

1. Integration under the differential sign.

If we know the indefinite integral

$$\int f(x)dx = F(x) + C,$$

then

$$\int f(u(x))d(u(x)) = F(u(x)) + C. \tag{3.8}$$

Since $d(u(x)) = u'(x)dx$, the above relation can be written as

$$\int f(u(x))u'(x)dx = F(u(x)) + C. \tag{3.9}$$

Its validity can be checked by differentiation and is due to the chain rule.

<u>Example 1.</u>

We use $dx = \dfrac{1}{5}d(5x-2)$

$$\begin{aligned}\int \frac{dx}{\sqrt{5x-2}} &= \frac{1}{5}\int (5x-2)^{-1/2}d(5x-2)\\ &= \frac{1}{5}\int u^{-1/2}du = \frac{1}{5}\frac{u^{1/2}}{\frac{1}{2}} + C\\ &= \frac{2}{5}(5x-2)^{1/2} + C = \frac{2}{5}\sqrt{5x-2} + C,\end{aligned}$$

where we let $u = 5x-2$ and used the standard integral (ii).

<u>Example 2.</u>

$$\int \frac{xdx}{\sqrt{1+x^4}} = \frac{1}{2}\int \frac{d(x^2)}{\sqrt{1+(x^2)^2}} = \ln\left(x^2 + \sqrt{1+x^4}\right) + C,$$

where we used $xdx = \frac{1}{2}d(x^2)$ and standard integral (xi).

Example 3.

$$\int \sin(ax+b)dx = \frac{1}{a}\int \sin(ax+b)d(ax+b) = -\frac{1}{a}\cos(ax+b) + C,$$

where we used $dx = \frac{1}{a}d(ax+b)$ $(a \neq 0)$ and the standard integral (vi).

Example 4.

Using $\frac{dx}{x} = d(\ln x)$

$$\int \frac{\ln x}{x}dx = \int \ln x \, d(\ln x) = \frac{1}{2}(\ln x)^2 + C,$$

where we used $\int udu = \frac{1}{2}u^2 + C$ with $u = \ln x$ in the last step.

2. Integration by substitution.

An integral $\int f(x)dx$ can be transformed using $x = \varphi(t)$, i.e., by making x a function φ of the new variable t, which gives

$$\int f(x)dx = \int f(\varphi(t))\varphi'(t)dt, \tag{3.10}$$

where we used $dx = \varphi'(t)dt$. The choice of the function φ is made in such a way that the right-hand side of (3.10) is more convenient for integration.

Example 5.

To find

$$\int x\sqrt{x-1}dx,$$

we let $t = \sqrt{x-1}$, so that $x = t^2 + 1$ and $dx = 2tdt$, so that

$$\int x\sqrt{x-1}dx = \int (t^2+1)t \cdot 2tdt = 2\int (t^4+t^2)dt = 2\left(\frac{t^5}{5}+\frac{t^3}{3}\right)+C$$
$$= \frac{2}{5}(x-1)^{5/2} + \frac{2}{3}(x-1)^{3/2} + C.$$

Example 6: trigonometric substitutions.

When an integral contains $\sqrt{a^2-x^2}$, one often uses $x = a\sin t$; for integrals with $\sqrt{x^2-a^2}$, use $x = a/\cos t$, and for those with $\sqrt{a^2+x^2}$ use $x = a\tan t$.

Thus, to find

$$\int \frac{dx}{(a^2-x^2)^{3/2}},$$

we choose $x = a\sin t$, which gives $a^2 - x^2 = a^2(1-\sin^2 t) = a^2\cos^2 t$ and $dx = a\cos t\, dt$, so that

$$\int \frac{dx}{(a^2-x^2)^{3/2}} = \int \frac{a\cos t\, dt}{(a^2\cos^2)^{3/2}} = \frac{1}{a^2}\int \frac{dt}{\cos^2 t}$$
$$= \frac{1}{a^2}\tan t + C = \frac{1}{a^2}\frac{a\sin t}{a\cos t} + C$$
$$= \frac{1}{a^2}\frac{x}{\sqrt{a^2-x^2}} + C.$$

3. Integration by parts.

The method of integration by parts originates in the product rule,

$$(uv)' = u'v + uv',$$

where u and v are some functions. Multiplying this by the differential of the independent variable dx, and taking indefinite integrals on both sides, we obtain

$$\int (uv)'dx = \int u'v\, dx + \int uv'dx,$$
$$uv = \int v\, du + \int u\, dv,$$

where we introduced the differentials $du = u'dx$ and $dv = v'dx$ (see Sec. 3.2.4). Rearranging, we have

$$\int u\,dv = uv - \int v\,du, \tag{3.11}$$

which is the formula for *integration by parts.*

Example 7.

Here $u = \ln x$, $v = x$

$$\begin{aligned}\int \ln x\,dx &= x\ln x - \int x\,d(\ln x) = x\ln x - \int x\,\frac{1}{x}\,dx\\ &= x\ln x - \int dx = x\ln x - x + C.\end{aligned}$$

Example 8.

To find the integral

$$\int e^x \cos x\,dx$$

we integrate by parts twice and obtain an equation for the integral sought:

Here we make use of $\cos x\,dx = d(\sin x)$, $\sin x\,dx = -d(\cos x)$, and $d(e^x) = e^x dx$

$$\begin{aligned}\int e^x \cos x\,dx = \int e^x\,d(\sin x) &= e^x \sin x - \int \sin x\,d(e^x)\\ &= e^x \sin x - \int e^x \sin x\,dx = e^x \sin x + \int e^x\,d(\cos x)\\ &= e^x \sin x + e^x \cos x - \int \cos x e^x dx.\end{aligned}$$

Hence

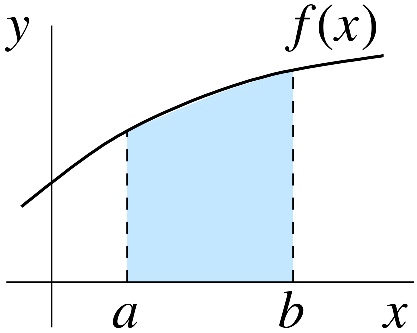
$$\int e^x \cos x\,dx = e^x(\sin x + \cos x) - \int e^x \cos x\,dx,$$

which gives

$$\int e^x \cos x \, dx = \frac{e^x}{2}(\sin x + \cos x) + C,$$

where we have added the usual constant on the right-hand side.

3.3.4 Definite integrals



The *definite integral* of a function $f(x)$ from $x = a$ to $x = b$ is denoted by

$$\int_a^b f(x)dx, \tag{3.12}$$

and gives the *area* between the graph of $y = f(x)$ and the x axis, and between the two vertical lines $x = a$ and $x = b$.[23]

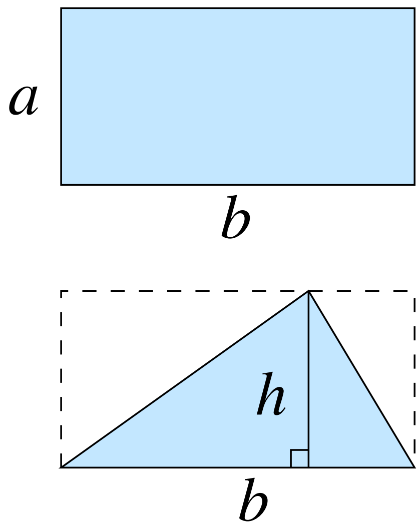
However, what is the meaning of "area" for a shape with a curved boundary?

The area of a rectangle with sides a and b *is defined* as $A = ab$.[24] Using this, one can work out the area of a triangle with base b and height h as $\frac{1}{2}bh$ (by building it up to a rectangle consisting of two pairs of equal triangles).[25] Hence, the area can be found for any shape that can be divided into a number of triangles. This is possible for shapes

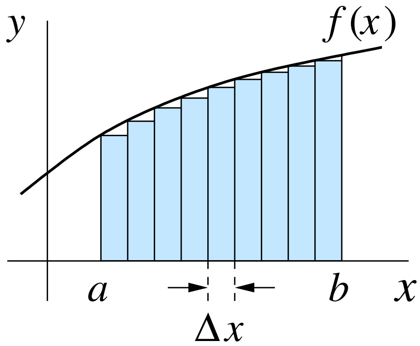
[23] Note that the definite integral does not depend on the letter used for the independent variable, i.e., $\int_a^b f(x)dx = \int_a^b f(\xi)d\xi = \int_a^b f(t)dt$. ($\xi$ is the lowercase Greek letter xi.)

[24] People learned from experience that this is a useful quantity when they realised that two rectangular plots of land yield the same amount of crop if the product 'length $\times$ width' is the same for them. On the other hand, if someone wanted to build a fence around their plot of land then this quantity is useless, and another one (the perimeter) should be used.

[25] This implies that if a shape is cut into several parts then its area is equal to the sum of the areas of the parts.



whose boundaries consist of straight segments, but not for those whose boundaries are curved, e.g., for a circle.[26]



To find the area under the graph of $y = f(x)$, let us divide the interval $[a, b]$ into n equal subintervals of width $\Delta x = (b - a)/n$. We then construct n rectangles, each of width Δx, with vertical sides at points $x_1 = a$, $x_2 = x_1 + \Delta x$, etc., up to $x_{n+1} = b$. The height of each rectangle is such that its top left corner lies on the graph, i.e., the height of ith rectangle is equal to $f(x_i)$ ($i = 1, 2, \ldots, n$). As we can see, the total area covered by the rectangles approximates the area under the graph, and the larger the value of n, the more accurate this approximation becomes. The definite integral (3.12) can then be defined as the limit of the sum of the areas of the rectangles:

[26] To find the area of a circle of radius r, consider a regular polygon with n vertices inscribed in the circle. This polygon can be split into n isosceles triangles with heights $r\cos(\pi/n)$ and bases $2r\sin(\pi/n)$, giving the area of the polygon $A_n = r^2 n \sin(\pi/n)\cos(\pi/n)$. In the limit $n \to \infty$, the polygon becomes closer and closer to the circle, and using $\sin(\pi/n) \simeq \pi/n$ (see end of Sec. 3.1) and $\lim\limits_{n\to\infty} \cos(\pi/n) = 1$ (why?), we have $\lim\limits_{n\to\infty} A_n = \pi r^2$.

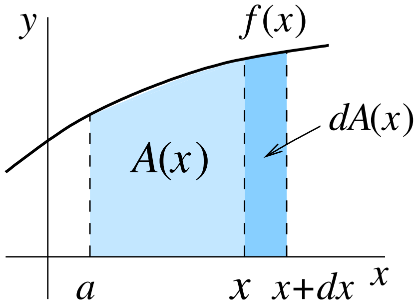
$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x. \tag{3.13}$$

The expression on the right-hand side explains the origins of the symbol used for the definite integral on the left-hand side. Here $f(x)dx$ is the area of a rectangle with width dx (assumed small) and height $f(x)$. The integration sign $\int$ is the modified letter S, which implies summation, and the limits a and b indicate the interval of x over which the sum is taken.

Using (3.13) directly to find the area under a curve is not easy, except for simple functions such as $f(x) = x$ or $f(x) = x^2$. There is, however, a "magic formula" that gives the required quantity immediately if the primitive $F(x)$ of the function $f(x)$ is known. This formula is

$$\int_a^b f(x)dx = F(b) - F(a), \tag{3.14}$$

and is sometimes called "the fundamental theorem of calculus".



To prove (3.14), consider the area under the graph between points a and x,

$$A(x) = \int_a^x f(t)dt, \tag{3.15}$$

as a function of the upper limit x.[27] If we increase x by a small amount dx, the change in the area can be approximated by the area of a rectangle,

$$A(x + dx) - A(x) \simeq f(x)dx.$$

Dividing both sides by dx and letting $dx \to 0$ we obtain (see Sec. 3.2.1)

$$\frac{dA(x)}{dx} = f(x),$$

[27] Here we use t as the independent variable to avoid confusion with the upper limit x.

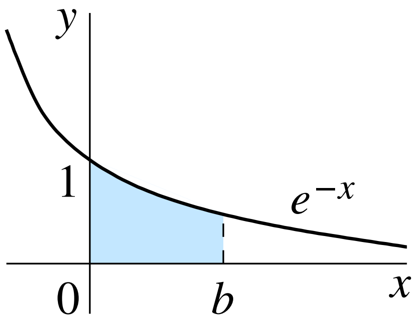
hence, $A(x)$ is a primitive for the function $f(x)$. We know that all primitives for a given function differ by a constant, which means that

$$A(x) = F(x) + C. \tag{3.16}$$

To find the constant C, note that $A(a) = 0$ (since for $x = a$ the area under the graph shrinks to zero), which gives $C = -F(a)$. Using this in (3.16) and combining with (3.15), we obtain

$$\int_a^x f(t)dt = F(x) - F(a),$$

which, upon changing x to b and t to x, gives (3.14).



Example.

Find the area under the curve $y = e^{-x}$ between $x = 0$ and $x = b > 0$.

The area is given by the definite integral

$$\int_0^b e^{-x}dx = \left[-e^{-x}\right]_0^b = -e^{-b} - (-e^0) = 1 - e^{-b}.$$

Note that for $b \to \infty$ the shape becomes infinitely long, but its area remains finite: $\lim_{b\to\infty}(1 - e^{-b}) = 1$. Hence, we see that $\int_0^\infty e^{-x}dx = 1$.

Finally, note that there are indefinite integrals, even innocent looking ones, e.g.,

$$\int \frac{dx}{\ln x}, \quad \int \frac{\sin x}{x}\,dx, \quad \int \frac{e^x}{x}\,dx, \quad \int e^{-x^2}dx,$$

that cannot be expressed in terms of elementary functions. For a few of such integrals that are important, new *special* functions are introduced to provide the answer, e.g., the integral sine $\mathrm{Si}(x) = \int_0^x \frac{\sin t}{t}\,dt$. Surprisingly, some of the integrals can be found as definite integrals with specific limits, e.g.,

$$\int_0^\infty \frac{\sin x}{x} = \frac{\pi}{2} \quad \text{and} \quad \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

3.4 Complex numbers

Consider the quadratic equation

$$z^2 = -1 \tag{3.17}$$

We know that it has no solutions in $\mathbb{R}$. However, let us *imagine*[28] that there is a number whose square equals -1. Denoting such "number" i, we see that $z = i$ solves equation (3.17), since

$$i^2 = -1. \tag{3.18}$$

There is also another solution $z = -i$, since $(-i)^2 = (-1)^2 i^2 = -1$. Thus, we see that the quadratic equation (3.17) has two solutions,

$$z = \pm i.$$

The new number i is known as the *imaginary unit*. Using it we can now solve any quadratic equation. For example, for

$$z^2 - 2z + 5 = 0 \tag{3.19}$$

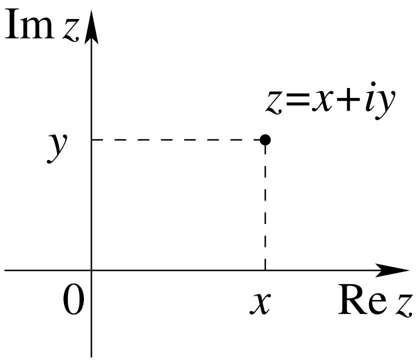
we have

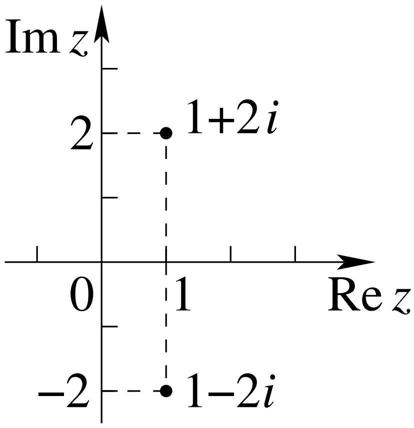
$$z = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2}.$$

Using i we can easily find $\sqrt{-16}$, which is a number whose square equals -16. Indeed, $(4i)^2 = 4^2 i^2 = -16$, so the roots are

$$z = \frac{2 \pm 4i}{2} = 1 \pm 2i. \tag{3.20}$$

[28] When the great German mathematician David Hilbert (1862-1943) was asked about one of his former students, he said: "Oh, that one? He became a poet. He did not have enough imagination to be a mathematician".





By introducing i we have just expanded the set of numbers beyond $\mathbb{R}$ to what is known as *complex numbers*.

Definition. A *complex number* is a number of the form

$$z = x + iy,$$

where x, $y \in \mathbb{R}$.

Here x is known as the *real part*, denoted $x = \operatorname{Re} z$, and y is the *imaginary part*, $y = \operatorname{Im} z$, of the complex number z. The set of all complex numbers is denoted $\mathbb{C}$.

While real numbers correspond to all points of the line, complex numbers "live" in the plane. The horizontal (x) axis of the *complex plane* is know as the real axis, and the vertical (y) axis is the imaginary axis. A complex number $z = x + iy$ is shown as a point with coordinates (x, y).

For example, we can use the complex plane to show the two roots of the quadratic equation (3.19), $z = 1 \pm 2i$. Note that these complex numbers are positioned symmetrically about the real axis, because their imaginary parts are the opposite of each other. This prompts the following definition.

Definition. For a complex number $z = x + iy$, the complex number

$$z^* = x - iy$$

is its *complex conjugate*.[29]

The complex conjugate is obtained by changing the sign of the imaginary part, or, putting simply, by replacing i by $-i$.

Example: $(1+2i)^* = 1-2i$ and $(1-2i)^* = 1+2i$.

In general,

$$(z^*)^* = (x-iy)^* = x+iy = z, \tag{3.21}$$

i.e., the complex conjugate of a complex conjugate is the number itself.

Note that the product of a complex number with its complex conjugate is real:

$$zz^* = (x+iy)(x-iy) = x^2-(iy)^2 = x^2+y^2. \tag{3.22}$$

Their sum is also real:

$$z+z^* = (x+iy)+(x-iy) = 2x = 2\mathrm{Re}\, z. \tag{3.23}$$

Complex arithmetics.

Addition, multiplication and division obey the usual rules of algebra and we use $i^2=-1$. For two complex numbers, $z_1 = x_1+iy_1$ and $z_2 = x_2+iy_2$,

1. Addition:

$$z_1+z_2 = (x_1+x_2)+i(y_1+y_2).$$

2. Multiplication:

$$z_1z_2 = (x_1+iy_1)(x_2+iy_2) = (x_1x_2-y_1y_2)+i(x_1y_2+y_1x_2).$$

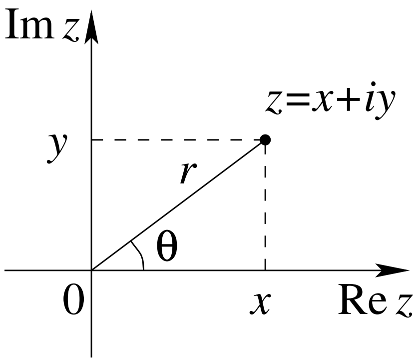
3. Division: this is done by multiplying the numerator and denominator by the complex conjugate of the denominator and using (3.22)

$$\frac{z_1}{z_2} = \frac{x_1+iy_1}{x_2+iy_2} = \frac{(x_1+iy_1)(x_2-iy_2)}{(x_2+iy_2)(x_2-iy_2)} = \frac{x_1x_2+y_1y_2}{x_2^2+y_2^2}+i\,\frac{y_1x_2-x_1y_2}{x_2^2+y_2^2}.$$

Polar form of complex numbers.

The position of a point in the plane can be specified by giving its Cartesian coordinates x and y. Alternatively, one can set the distance r from the origin to the point, and the

[29] We denote the complex conjugate by an asterisk. Another common symbol for complex conjugates is a bar over the number ($\bar{z}$).



angle θ between the direction to the point and the x axis (see diagram). The variables r and θ are the *polar coordinates* of the point.

The Cartesian and polar coordinates are linked by

$$x = r\cos\theta, \quad y = r\sin\theta.$$

Using these in $z = x + iy$, gives the *polar form* of the complex number z,

$$z = r(\cos\theta + i\sin\theta). \tag{3.24}$$

In this form

$$r = \sqrt{x^2 + y^2}, \tag{3.25}$$

is known as the *modulus* of the complex number, $r = |z|$, and the angle θ is the *argument* of the complex number, $\theta = \arg z$. It satisfies the equation

$$\tan\theta = \frac{y}{x}.$$

By convention, the argument is taken in the interval

$$-\pi < \theta \leq \pi.$$

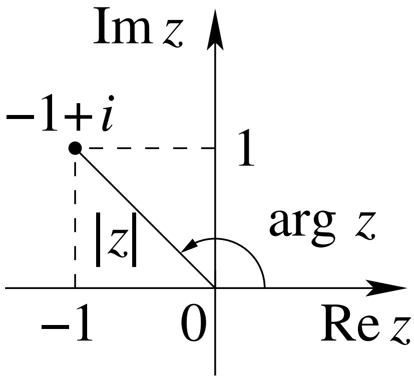
Example: For $z = -1 + i$, the modulus is $|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$ and the argument is $\arg z = 3\pi/4$ (which is easy to see from the diagram).[30]

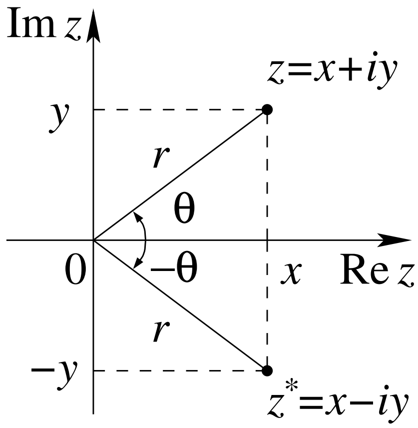
Taking the complex conjugate of the complex number in polar form (3.24),

$$z^* = r(\cos\theta - i\sin\theta) = r[\cos(-\theta) + i\sin(-\theta)],$$

shows that $|z^*| = |z|$ and $\arg z^* = -\arg z$, as also seen from the diagram.

[30] To obtain this analytically, we solve $\tan\theta = -1$, which gives $\theta = -\pi/4$ and $\theta = 3\pi/4$ and choose the angle for which $\cos\theta < 0$ (since $\mathrm{Re}\, z < 0$ for $z = -1 + i$).





y

Euler's formula.

Let us work out the quantity $e^{i\theta}$ by using the Maclaurin series for e^x (4.26):

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots$$

Using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, etc., and grouping the real terms and the imaginary terms together,

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right)}_{\cos\theta} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right)}_{\sin\theta}$$

where we recognise the real part as the Maclaurin series for $\cos\theta$ and the imaginary part as the Maclaurin series for $\sin\theta$. We thus have

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{3.26}$$

which is known as *Euler's formula.*

Using complex numbers thus reveals a deep relation between exponential and trigonometric functions. In particular, for $\theta = \pi$ we obtain from (3.26)

$$e^{i\pi} = -1,$$

which links π, e, i and 1 in a single formula!

Using (3.26) in (3.24) gives the *exponential form* of the complex number:

$$z = re^{i\theta}. \tag{3.27}$$

The exponential form is especially convenient for *multiplying* complex numbers. Taking $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ gives

$$z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)}.$$

We see that $|z_1 z_2| = r_1 r_2 = |z_1||z_2|$ and $\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg z_1 + \arg z_1$, which shows that upon multiplication the moduli of the complex numbers multiply, while their arguments add.

In particular, multiplying $z = re^{i\theta}$ by its complex conjugate $z^* = re^{-i\theta}$ gives

$$zz^* = re^{i\theta} re^{-i\theta} = r^2 = |z|^2,$$

which can also be seen from equations (3.22) and (3.25).

As a further application of Euler's formula, taking the nth power of the right-hand side of (3.26) immediately proves *de Moivre's formula*:

$$\begin{aligned}(\cos\theta + i\sin\theta)^n &= (e^{i\theta})^n = e^{in\theta} \\ &= \cos n\theta + i\sin n\theta.\end{aligned}$$

Finally, starting from the identity

$$e^{i(\alpha+\beta)} = e^{i\alpha} e^{i\beta},$$

then using Euler's formula on both sides,

$$\cos(\alpha+\beta) + i\sin(\alpha+\beta) = (\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)$$

and equating the real and imaginary parts, gives the *trigonometric addition formulae*

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

and

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta,$$

(again refer to the Trig Cheat Sheet on Canvas).

4 Infinite Series

4.0.1 Introduction

The sum of an infinite sequence of numbers $\{a_n\}$ is called an *infinite series* and is defined as

$$a_1 + a_2 + a_3 + \cdots + a_n + \dots \tag{4.1}$$

The number a_n is the **nth** term of the series. The goal of this section is to understand the meaning of such an infinite series and to develop methods to calculate it. Since there are infinitely many terms to add in an infinite series, we cannot just keep adding to see what comes out. Instead we look at the result of summing the first n terms of the sequence and stopping. The sum of the first n terms

$$a_1 + a_2 + a_3 + \cdots + a_n = S_n \tag{4.2}$$

is an ordinary finite sum, labelled S_n, and can be calculated by normal addition. It is called the **nth partial sum**. The sequence $\{S_n\}$ is the **sequence of the partial sums** of the series and is defined by

$$\begin{aligned} S_1 &= a_1 \\ S_2 &= a_1 + a_2 \\ S_3 &= a_1 + a_2 + a_3 \\ S_n &= a_1 + a_2 + a_3 + \cdots + a_n \\ &\dots \end{aligned}$$

As n gets larger, we expect the partial sums to get closer and closer to a limiting value in the same sense that the terms of a sequence approach a value. If the sequence of partial sums converges to a limit, L, we say that the series **converges** and that its **sum** is L. In other words if this sequence is convergent, there exists a number L such that

$$\lim_{n\to\infty} S_n = L \tag{4.3}$$

then the series (4.1) is called **convergent** and L is called its **sum**. If on the other hand

$$\lim_{n\to\infty} S_n \tag{4.4}$$

does not exist, the series is called **divergent**. We continue our discussion of infinite series with a special form called **geometric series**. These series arise more frequently than any other infinite series, they are used in many practical problems and they illustrate all the essential features of infinite series in general. Consider the sequence

$$1,\ x,\ x^2,\ x^3,\ \dots, \tag{4.5}$$

where x is any number. This is a **geometric progression** with the nth term x^n ($n = 0,\ 1,\ 2, \dots$). Let us consider the sum of the first $n+1$ terms of the progression,

$$S_n = 1 + x + x^2 + \cdots + x^n.$$

It is easy to obtain a closed-form expression for S_n (i.e., without '$\cdots$'), by multiplying and dividing the sum on the right-hand side by $1-x$:

$$\begin{aligned} S_n &= \frac{(1 + x + x^2 + \cdots + x^n)(1-x)}{1-x} \\ &= \frac{1 + x + x^2 + \cdots + x^n - (x + x^2 + x^3 + \cdots + x^{n+1})}{1-x}. \end{aligned}$$

In the numerator, all terms except 1 and $-x^{n+1}$ cancel, so we obtain

$$S_n = \frac{1-x^{n+1}}{1-x}. \tag{4.6}$$

This formula is value for any $x \neq 1$. (For $x = 1$, $S_n = n + 1$, isn't it?) If we now consider $|x| < 1$, then the successive terms in the sequence (4.5) are getting closer to zero with increasing n, so that $\lim\limits_{n\to\infty} x^n = 0$. In this case we can find the sum of the infinite geometric series

$$S = 1 + x + x^2 + x^3 + x^4 + \dots,$$

by taking the limit of S_n,

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1-x^{n+1}}{1-x} = \frac{1}{1-x},$$

since the limit of x^{n+1} for $n \to \infty$ is zero. Hence we have proved the following identity for the geometric series:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots, \tag{4.7}$$

which holds for $|x| < 1$. For example, for $x = 1/2$ we have

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots \ = \frac{1}{1-\frac{1}{2}} = 2.$$

Replacing x by $-x$ in equation (4.7) we obtain

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - \ldots, \tag{4.8}$$

and replacing x by x^2 in (4.8) gives another useful relation:

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + x^8 - \ldots. \tag{4.9}$$

A more usual form of the geometric series is given by

$$a + ar + ar^2 + ar^3 + \cdots + ar^n + \ldots \tag{4.10}$$

It differs from (4.5) by an extra factor a and the use of r instead of x for the common ratio. As before our goal is to find a formula for the value of the geometric sum

$$S_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} \tag{4.11}$$

for any value of a, r and the positive integer n. In a procedure similar to before we multiply both sides of (4.11) by the ratio r to give

$$rS_n = r(a + ar + ar^2 + \cdots + ar^{n-1}) = ar + ar^2 + ar^3 + \cdots + ar^n \tag{4.12}$$

We now subtract (4.12) from (4.11) and note that most of the terms on the right hand side cancel to leave

$$S_n - rS_n = a - ar^n \tag{4.13}$$

Solving for S_n gives the formula

$$S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1) \tag{4.14}$$

If $|r| < 1$ then $r^n \to 0$ as $n \to \infty$ and hence taking the limit as $n \to \infty$ gives

$$\lim_{n\to\infty} S_n = \frac{a}{1-r} \tag{4.15}$$

comparable to equation (4.7) with $a = 1$ and $r = x$. If on the other hand $|r| \geq 1$ then $|r^n| \to \infty$ and the series diverges.

Example 4.1 Investigate whether the following infinite series are convergent or divergent

$$(a) \quad \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ \dots$$

This represents a geometric series with the first term $a = 1/2$ and the common ratio $|r| = 1/2 < 1$. Hence S_n = the sum of the first n terms is given by

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1/2(1-\frac{1}{2^n})}{1-1/2} = 1 - \frac{1}{2^n}$$

and since

$$\lim_{n\to\infty}\left(1 - \frac{1}{2^n}\right) = 1$$

the series is convergent and has sum $S = 1$.

$$(b) \quad \sum_{n=1}^{\infty}(-1)^{n-1} = 1 - 1 + 1 - 1 + 1 \ldots$$

Here $S_n = 0$ or $S_n = 1$ according as n is even or odd. Hence

$$\lim_{n\to\infty} S_n$$

does not exist and the series is divergent.

4.0.2 Maclaurin series

In the preceding section we investigated infinite series and in particular geometric (or power) series of the form

$$a + ar + ar^2 + ar^3 + \cdots = \sum_{n=1}^{\infty} ar^{n-1} \tag{4.16}$$

This section shows how functions that are infinitely differentiable can generate power series, called **Maclaurin Series** or **Taylor Series**. In many cases, these series can provide useful polynomial approximations of the generating functions and are used routinely by mathematicians and scientists. We start with the Maclaurin Series named after the Scottish mathematician Colin Maclaurin (1698-1746). Consider the exponential function $y = e^x$. The only value of this function that we know *exactly* is for $x = 0$: $e^0 = 1$. How to find values of e^x for other x? Let us draw a tangent to the graph of $y = e^x$ at the point $x = 0$, $y = 1$. Its gradient is given by the derivative, $(e^x)' = e^x$, evaluated at $x = 0$, i.e., it is equal to 1. The required tangent $y = x + 1$ is shown in Figure 1.

For small x the graph of $y = e^x$ stays close to the tangent,

$$e^x \approx x + 1 \tag{4.17}$$

For example, for $x = 0.01$ we have

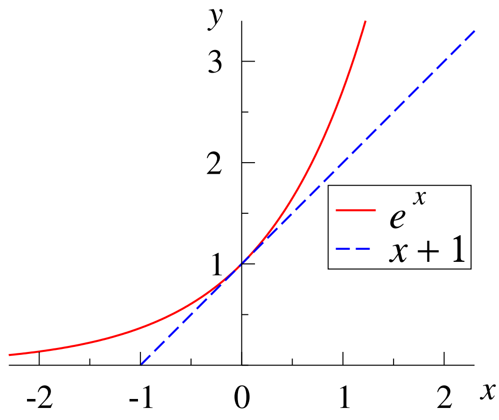


Figure 10: Graph of $y = e^x$ and the tangent to it at $x = 0$: $y = x + 1$.

$$x + 1 = 1.01, \text{ while } e^{0.01} \approx 1.01005,$$

where the latter value is found using the calculator.[31] Further away from $x = 0$, the agreement gets worse, e.g., for $x = 0.3$,

$$x + 1 = 1.3, \text{ while } e^{0.3} \approx 1.34986.$$

To find a better approximation of e^x than the linear function (4.17), let us represent it by an infinite polynomial,

$$e^x = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots, \tag{4.18}$$

where a_0, a_1, $a_2, \dots$ are coefficients. We will find the coefficients by requiring that the two sides of equation (4.18) and their 1st, 2nd, etc., derivatives at $x = 0$ are equal. Setting $x = 0$ in equation (4.18) gives

$$e^0 = a_0 + 0 + 0 + 0 + \dots,$$

so that

$$a_0 = 1.$$

Taking the derivative of equation (4.18), we have

[31] How does the calculator "know" the values of e^x for a wide range of x?

$$e^x = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots, \tag{4.19}$$

and setting $x = 0$, we find

$$a_1 = 1.$$

Differentiating both sides of equation (4.19) gives

$$e^x = 2a_2 + 3 \cdot 2a_3x + 4 \cdot 3a_4x^2 + \dots, \tag{4.20}$$

which, upon setting $x = 0$, yields

$$1 = 2a_2 \quad \Rightarrow \quad a_2 = \frac{1}{2}.$$

Taking the derivative of equation (4.20), we find

$$e^x = 3 \cdot 2a_3 + 4 \cdot 3 \cdot 2a_4x + \dots, \tag{4.21}$$

which, upon setting $x = 0$, yields[32]

$$1 = 3 \cdot 2a_3 \quad \Rightarrow \quad a_3 = \frac{1}{2 \cdot 3}.$$

Repeating the procedure once more will give us

$$a_4 = \frac{1}{2 \cdot 3 \cdot 4}.$$

We can now see the emergence of a pattern:

$$a_0 = 1,\ a_1 = 1 = \frac{1}{1},\ a_2 = \frac{1}{1 \cdot 2},\ a_3 = \frac{1}{1 \cdot 2 \cdot 3},\ a_4 = \frac{1}{1 \cdot 2 \cdot 3 \cdot 4}. \tag{4.22}$$

This suggests that the nth coefficient is given by[33]

[32] Of course, $3 \cdot 2 = 6$, but keeping this as a product will help us see the pattern.
[33] This is proved by differentiating equation (14.18) n times and setting $x = 0$.

$$a_n = \frac{1}{n!}, \tag{4.23}$$

where

$$n! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n \tag{4.24}$$

is n-factorial. Note that by convention, $0! = 1$, which means that equation (4.23) is valid for all n from zero. Putting these coefficients into equation (4.18) gives

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots + \frac{x^n}{n!} + \ldots \tag{4.25}$$

This equation is the *Maclaurin series* for e^x. It can be written in compact form using the summation sign:[34]

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \tag{4.26}$$

The Maclaurin series for e^x converges for all x.[35] In particular, for $x = 1$,

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \frac{1}{720} + \ldots,$$

and the sum converges quite rapidly. Including the first 11 terms (up to $1/10!$) gives $e \approx 2.718281801\ldots$, thus providing correctly the first 8 digits of e. A similar expansion can be developed for any "good" function $f(x)$:

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_n x^n + \ldots \tag{4.27}$$

For $x = 0$ this gives

$$f(0) = a_0.$$

[34] It uses the Greek letter Σ (sigma), which is the first letter of the Greek word "sum".

[35] This can be see from the comparison of successive terms in the series. The ratio of the $(n+1)$th term $x^{n+1}/(n+1)!$ term to the nth term is $x/(n+1)$, so for $n > |x|$ this ratio is smaller than unity, and for larger n the terms in the series decrease very quickly.

Differentiating equation (4.27), we have

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \ldots + na_nx^{n-1} + \ldots,$$

which yields

$$f'(0) = a_1.$$

The next derivative,

$$f''(x) = 2a_2 + 6a_3x^2 + \ldots + n(n-1)a_nx^{n-2} + \ldots,$$

gives

$$f''(0) = 2a_2,$$

and after taking the derivative n times and setting $x = 0$ we find

$$f^{(n)}(0) = n(n-1)(n-2)\ldots \cdot 3 \cdot 2a_n,$$

which gives the expression for the nth coefficient:

$$a_n = \frac{f^{(n)}(0)}{n!}. \tag{4.28}$$

Hence we have the Maclaurin expansion:

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots + \frac{f^{(n)}(0)}{n!}x^n + \ldots, \tag{4.29}$$

Let us use equation (4.28) to find the coefficients of the Maclaurin expansion for $f(x) = \cos x$. We obtain them by filling the table (Table 1), where the second column shows the function and its successive derivatives, the third column shows their values at $x = 0$, and the last column contains the coefficients.

Table 3: Finding the coefficients of the Maclaurin series for $\cos x$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	a_n
0	$\cos x$	1	1
1	$-\sin x$	0	0
2	$-\cos x$	-1	$-1/2!$
3	$\sin x$	0	0
4	$\cos x$	1	$1/4!$
5	$-\sin x$	0	0
6	$-\cos x$	-1	$-1/6!$

The coefficients in the table follow a simple pattern and we have

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots . \tag{4.30}$$

Notice that this Maclaurin series contains only even powers of x, because $\cos x$ is an even function. In a similar way, we find the Maclaurin expansion for $\sin x$. Table 2 shows how the first six coefficients (three of which are nonzero) are obtained.

Table 4: Finding the coefficients of the Maclaurin series for $\sin x$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	a_n
0	$\sin x$	0	0
1	$\cos x$	1	1
2	$-\sin x$	0	0
3	$-\cos x$	-1	$-1/3!$
4	$\sin x$	0	0
5	$\cos x$	1	$1/5!$

The coefficients in Table 2 follow a clear pattern, which gives

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots . \tag{4.31}$$

This series contains only odd powers of x, since $\sin x$ is an odd function. These Maclaurin series for $\cos x$ (4.30) and $\sin x$ (4.31) converge for any value of x. They can also be written in the compact form:

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \tag{4.32}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \tag{4.33}$$

To write the $\cos x$ and $\sin x$ series, we use $2n$ as the generic *even* number and $2n + 1$ for the generic *odd* number.

Two more useful Maclaurin series are

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \tag{4.34}$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}. \tag{4.35}$$

These series converge for $-1 < x \leq 1$ and $-1 \leq x \leq 1$, respectively. The easiest way to derive them is to take the derivative of the function on the left, expand it in a power series using either (4.8) or (4.9), and then integrates both sides of the equation obtained,[36] choosing the integration constant by checking the value at $x = 0$. Another useful Maclaurin series is the *binomial* series,

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + \ldots. \tag{4.36}$$

It converges for $|x| < 1$ and can be derived using equation (4.28), similarly to the series for $\cos x$ and $\sin x$. Note that for $\alpha = n$, $n \in \mathbb{N}$, equation (4.36) is the usual binomial expansion, while $\alpha = -1$ gives the geometric series (4.8).

4.0.3 Taylor series

A Taylor series is an infinite series expansion of a function about a specific point which is not the origin. A one-dimensional Taylor series is thus an expansion of a real function $f(x)$ about a point $x = a$ and is given by

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ \ldots \ + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ \ldots \tag{4.37}$$

[36] This requires *uniform convergence* of the series, which exists for $|x| < 1 - \varepsilon$, $\forall \varepsilon > 0$.

Remember if the point $x = 0$ then the expansion is known as the Maclaurin series. As you increase the degree of the Taylor polynomial, ie the more terms you include on the r.h.s of equation (4.37), the more accurate the approximation of the function $f(x)$.

Example 14.2 Find the Taylor series for the function $f(x) = \ln x$ about the point $x = 1$. Note $x = 0$ would be an unwise choice!

$$\begin{aligned} f(x) = \ln x \to f(a) = f(1) = \ln 1 = 0 \\ f'(x) = \frac{1}{x} \to f'(1) = +1 \\ f''(x) = \frac{-1}{x^2} \to f''(1) = -1 \\ f'''(x) = \frac{2}{x^3} \to f'''(1) = 2 \\ f^{(4)} = \frac{-2.3}{x^4} \to f^{(4)}(1) = -2.3 \\ \ldots \end{aligned}$$

Do you see a pattern evolve? Can you see that the next term would be

$$f^{(5)} = 2.3.4 = +4!$$

We thus obtain

$$f^{(n)} = \pm(n-1)!$$

The alternating sign can be accommodated by inserting a term of the form

$$(-1)^n \quad or \quad (-1)^{n+1}$$

depending on whether the first term for $n = 1$ is negative or positive. Hence we see that for this Example 4.2

$$f^{(n)} = (-1)^{n+1}(n-1)!$$

Check this works for $n = 1, 2, 3, \ \ldots$. This yields the Taylor series

$$\begin{aligned}
\ln x &= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} + \ \ldots \\
&= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(n-1)!}{n!}(x-1)^n \\
&= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^n}{n}
\end{aligned}$$

5 Differential equations

5.1 Introduction

The study of differential equations (DE's) lies at the very heart of mathematical modelling and is used in engineering, the natural and biological sciences, economics, management and finance. Basically a DE involves an unknown function y and its derivatives. The unknown in a DE is **not** a number (as in an algebraic equation), but rather a function or a relationship. Here are some examples of DE's:

$$(a)\ \frac{dy}{dx} + 4y = \cos x \qquad (b)\ \frac{d^2y}{dx^2} + 16y = 0 \qquad (c)\ y'(t) = 0.1y(100-y)$$

The goal is to find the function y that satisfies the equation. The **order** of the DE is the order of the highest-order derivative that appears in the equation, so (a) and (c) are first-order DE's while (b) is second-order. A DE is **linear** if the unknown function and its derivatives appear only to first power and are not composed with other functions, hence (a) and (b) are linear but (c) is non-linear as the r.h.s contains y^2. Solving a first-order DE requires integration - you must undo the derivative y' to find y. Integration introduces an arbitrary constant, so the **general solution** of the first-order DE involves one arbitrary constant. Similarly the general solution for a second-order DE contains two arbitrary constants. It is possible to determine specific values for the arbitrary constants if the DE is accompanied by additional information which we call **initial conditions**. Lets look at some phenomena in nature that are described by means DE's.

Example 5.1: Population growth.

The growth rate of a population is proportional to its size N. Predict how N depends on time t.

Mathematically the rate of change is the derivative dN/dt, so we have

$$\frac{dN}{dt} = kN, \tag{5.1}$$

where $k > 0$ is growth rate constant. This is a *differential equation.*

Let us solve this equation. Dividing both sides by N and multiplying by dt,

$$\frac{dN}{N} = kdt,$$

and taking indefinite integrals on both sides,

$$\int \frac{dN}{N} = \int k dt,$$

we have

$$\ln N = kt + C, \tag{5.2}$$

where C is an arbitrary constant,[37] and we use $\ln N$ instead of $\ln |N|$ since $N > 0$. Taking exponents on both sides of (5.2),

$$\begin{aligned} e^{\ln N} &= e^{kt+C}, \\ N &= e^C e^{kt}. \end{aligned}$$

Since C is an arbitrary constant, we can replace e^C with just C, which gives

$$N(t) = Ce^{kt}, \tag{5.3}$$

where we now show explicitly on the left that N is a function of time.

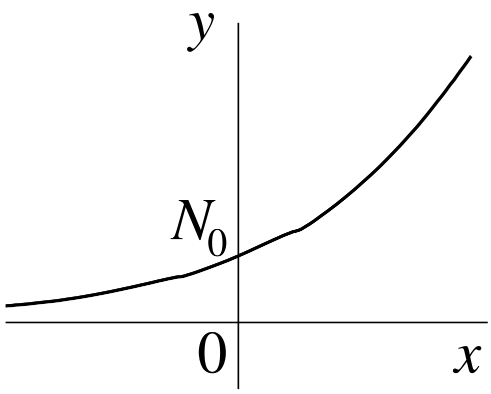
Note that the solution (5.3) of the differential equation (5.1) is a *function*, not a number or set of numbers, as for algebraic equations. Note also that this solution contains an arbitrary constant, so it describes a whole "family of functions", each of which is a solution of (5.1).

Suppose we know that at time $t = 0$ the size of the population is N_0. This can be written as $N(0) = N_0$. Using this *initial condition*, we can find the constant C in (5.3) from

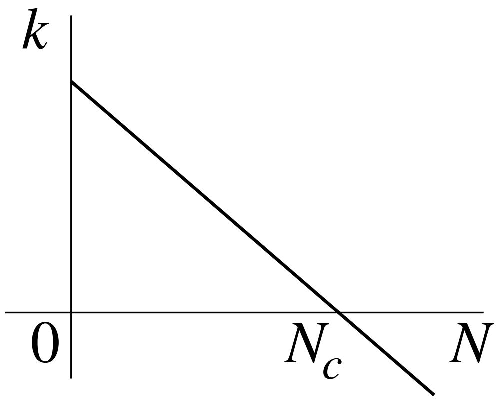
$$N_0 = Ce^0 \quad \Rightarrow \quad C = N_0,$$

so that (5.3) becomes

$$N(t) = N_0 e^{kt}. \tag{5.4}$$



This dependence is illustrated on the graph for some value of k. (To sketch $N(t)$ it is sufficient the recognise that the dependence of N on t is exponential. Larger k will lead to a more rapid dependence of N on t, while smaller k will make $N(t)$ "flatter".) As we see, the simplest model described by equation (5.1) predicts an unbounded growth of the population. This is only possible if there are unlimited resources (both food and territory).



A more realistic model should account for the fact that resources are usually limited. Mathematically, this will lead to a *decrease* of the growth rate constant k with increasing N. The simplest dependence of this type is linear, with a negative gradient, as shown on the graph. This dependence of k on N can be described by the equation

$$k = b(N_c - N), \tag{5.5}$$

where N_c is the *critical* size of the population (for which the growth rate vanishes), and b

[37] There is no point to add two different constants on both sides of (5.2), since one of them can always be moved to the other side, so that the two constants combine into one.

is a positive constant which determines the gradient of k. Using k from (5.5) in equation (5.1), we obtain

$$\frac{dN}{dt} = b(N_c - N)N. \tag{5.6}$$

This equation can be simplified by introducing a new variable $x = N/N_c$ which gives the size of the population in terms of the critical size. Using $N = N_c x$ and $dN/dt = N_c\, dx/dt$ in (5.6), we obtain, after some simpler algebra,

$$\frac{dx}{dt} = ax(1 - x), \tag{5.7}$$

where $a = bN_c$. This differential equation is know as the *logistic equation*. It was proposed as an improved model of the population growth by a Belgian mathematician Pierre François Verhulst (1804–1849).

Example 5.2: Newton's second law.

Consider a particle moving in one dimension along the x axis. Its motion is descibed by

position x,

velocity $v = \dfrac{dx}{dt}$,

acceleration $a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}$.

According to Newton's second law,

$$ma = F$$

where m is the mass of the particle and F is the net force acting on it. This force may depend on the position and velocity of the particle and on time t, i.e., $F = F(x, dx/dt, t)$. Hence, Newton's second law can be written as

$$m\frac{d^2x}{dt^2} = F\left(x, \frac{dx}{dt}, t\right). \tag{5.8}$$

By solving this differential equation we can determine how the particle moves, i.e., find $x(t)$.

Definition. An equation which relates the dependent variable (say, y) and its n derivatives, y', y'', ..., $y^{(n)}$, and the independent variable (say, x),

$$F(y, y', y'', \ldots, y^{(n)}, x) = 0,$$

is the nth-order *differential equation*.

In the examples above, the logisitic equation (5.7) is a first-order differential equation (DE), while Newton's second law (5.8) is a second-order DE.

The general solution of a nth-order differential equation contains n arbitrary constants (often denoted C_1, C_2,..., C_n).[38]

These constants can be determined, e.g., from the *initial conditions* which fix the values of the function and its $n-1$ derivatives, $y(0)$, $y'(0)$, ..., $y^{(n-1)}(0)$ at $x = 0$ (or at some other value of x).

5.2 First-order differential equations: variable separable type

Consider a first-order DE in the form

$$\frac{dy}{dx} = F(x, y).$$

If the right-hand side is a product of two functions, $F(x, y) = f(x)g(y)$, then

$$\frac{dy}{dx} = f(x)g(y) \tag{5.9}$$

is a *variable separable* differential equation. Indeed, dividing both side of (5.9) by $g(y)$ and multiplying by dx, we have

[38] This can be seen easily from an example. Thus, finding the solution of the simplest nth-order DE, $y^{(n)} = 0$, will involve integrating this expression consecuitively n times. Each time a new arbitrary integration constant will emerge, i.e., after integrating once we will have $y^{(n-1)} = C_1$; after integrating again, $y^{(n-2)} = C_1x + C_2$, etc., untill y is found as $y = C_1x^{n-1}/(n-1)! + C_2x^{n-2}/(n-2)! + \cdots + C_{n-1}x + C_n$.

$$\frac{dy}{g(y)} = f(x)dx.$$

The variables x and y have now been been separated, and indefinite integrals can be taken on both sides,

$$\int \frac{dy}{g(y)} = \int f(x)dx,$$

leading to a relation between y and x, which solves the equation.

Example 5.3: The equation

$$\frac{dy}{dx} + 2y^2 \cos x = 0$$

does not have the form of (5.9), but can easily be rearranged into it:

$$\frac{dy}{dx} = -2y^2 \cos x.$$

Hence,

$$\begin{aligned} \frac{dy}{y^2} &= -2\cos x dx, \\ \int \frac{dy}{y^2} &= -2\int \cos x dx, \\ -\frac{1}{y} &= -2\sin x + C, \end{aligned}$$

or

$$y = \frac{1}{2\sin x - C}.$$

Finally, since C is arbitrary, we replace C by $-C$, which gives a nicer-looking final answer:

$$y = \frac{1}{2\sin x + C}\,.$$

Example 5.4: Radioactive decay.

Consider a certain large number N of atoms whose nuclei are unstable and can undergo *radioactive* decay (producing some energetic particles – electrons or positrons, α-particles, or γ-rays). The law of radioactive decay states that in unit time a certain fraction of the nuclei, say λ, will decay.[39] This means that in unit time λN of nuclei decay, and in time dt the number of atoms whose nuclei decay is $\lambda N dt$. Hence, the change in the number of atoms is given by

$$dN = -\lambda N dt,$$

where the minus sign is included because N decreases, so that dN must be negative. We thus have

$$\frac{dN}{dt} = -\lambda N, \tag{5.10}$$

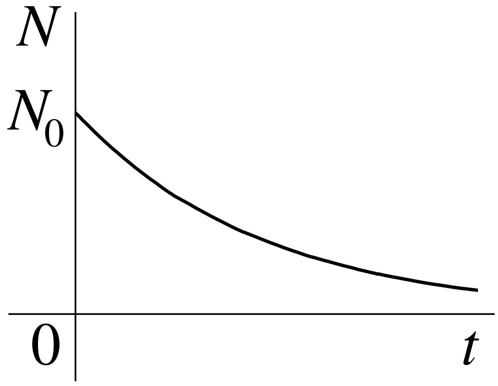
which is the *equation of radioactive decay*. Equation (5.10) is similar to the population growth equation (5.1), except for the minus sign on the right-hand side, and is variable separable. Solving it, we obtain:

$$\begin{aligned} \frac{dN}{N} &= -\lambda dt, \\ \int \frac{dN}{N} &= -\lambda \int dt, \\ \ln N &= -\lambda t + C, \\ N &= e^C e^{-\lambda t}, \end{aligned}$$

and replacing e^C by C, we obtain the number of un-decayed atoms as a function of time as

$$N(t) = Ce^{-\lambda t}. \tag{5.11}$$

[39] The value of λ depends on the type of nucleus, and varies a lot, e.g., for a long-lived radioactive nucleus, such as uranium ^{235}U, $\lambda = 3.12 \times 10^{-17}$ s^{-1}, while for a short-lived isotope of fluorine ^{18}F used in medicine for positron emission tomography (PET scans), $\lambda = 1.05 \times 10^{-4}$ s^{-1}.



If we know that at $t = 0$ the number of atoms was N_0, i.e., $N(0) = N_0$, then, using this in (5.11) we find that $C = N_0$, so that (5.11) can be written as

$$N(t) = N_0 e^{-\lambda t}. \tag{5.12}$$

The time dependence of N is exponentially decreasing and is sketched for $t \geq 0$ on the graph.

Example 3: Body falling under gravity with resistance.

A body is falling from rest near the Earth's surface. Apart from gravity, it is acted upon by the air resistance force proportional to the square of the velocity. Find how the velocity of the body changes with time.

The gravitation force acting on the body is vertically down and equals mg, where m is the mass of the body and g is the acceleration due to gravity. The resistance force can be written as mkv^2, where v is the velocity and $k > 0$ is a constant. Choosing the downwards direction as positive, we can write Newton's second law for the body as

$$m\frac{dv}{dt} = mg - mkv^2.$$

Dividing through by m we obtain a variable separable DE

$$\frac{dv}{dt} = g - kv^2,$$

and proceed as follows:

$$\frac{dv}{g-kv^2}=dt,$$

$$\int\frac{dv}{g-kv^2}=\int dt. \tag{5.13}$$

The integral on the right-hand side is trivial, and the integral on the left-hand side is similar to a known integral which is solved by using partial fractions[40]

$$\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\int\left[\frac{1}{a+x}+\frac{1}{a-x}\right]dx=\frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|+C.$$

Using this to do the integral on the left-hand side of (5.13),

$$\int\frac{dv}{g-kv^2}=\frac{1}{k}\int\frac{dv}{g/k-v^2}=\frac{1}{2k\sqrt{g/k}}\ln\left|\frac{\sqrt{g/k}+v}{\sqrt{g/k}-v}\right|+C,$$

we obtain from (5.13)

$$\frac{1}{2\sqrt{gk}}\ln\left|\frac{\sqrt{g/k}+v}{\sqrt{g/k}-v}\right|=t+C, \tag{5.14}$$

where we kept the integration constant on the right-hand side.

Since the body falls from rest, $v(0)=0$, using $v=0$ and $t=0$ in (5.14) gives

$$\frac{1}{2\sqrt{gk}}\ln\left|\frac{\sqrt{g/k}}{\sqrt{g/k}}\right|=0+C,$$

$$\frac{1}{2\sqrt{gk}}\ln 1=C,$$

so that $C=0$. Equation (5.14) then gives

$$\ln\left|\frac{\sqrt{g/k}+v}{\sqrt{g/k}-v}\right|=2\sqrt{gk}t.$$

[40] See problem sheet 3 for details.

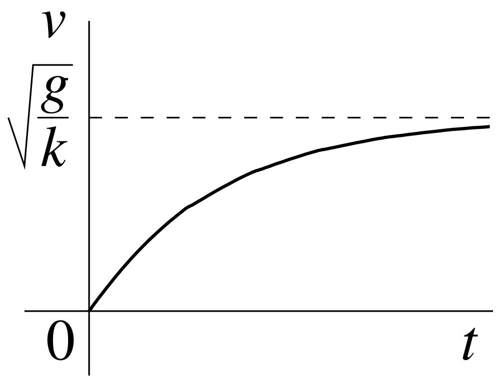
The modulus sign above can be dropped since the expression inside remains positive at all times, and we find the velocity as follows:

$$\frac{\sqrt{g/k}+v}{\sqrt{g/k}-v} = e^{2\sqrt{gk}t},$$

$$\sqrt{g/k}+v = e^{2\sqrt{gk}t}(\sqrt{g/k}-v),$$

$$v(e^{2\sqrt{gk}t}+1) = \sqrt{g/k}\,(e^{2\sqrt{gk}t}-1),$$

$$v = \sqrt{\frac{g}{k}}\,\frac{e^{2\sqrt{gk}t}-1}{e^{2\sqrt{gk}t}+1}.$$



Finally, multiplying the numerator and denominator by $e^{-\sqrt{gk}t}$, we have

$$v(t) = \sqrt{\frac{g}{k}}\,\frac{e^{\sqrt{gk}t}-e^{-\sqrt{gk}t}}{e^{\sqrt{gk}t}+e^{-\sqrt{gk}t}} = \sqrt{\frac{g}{k}}\,\tanh\left(\sqrt{gk}\,t\right),$$

where $\tanh x = \sinh x/\cosh x$ is the hyperbolic tangent.[41]

The function $\tanh x$ is odd, monotonically increasing and $\lim_{x\to\infty}\tanh x = 1$, which allows for an easy sketch of $v(t)$ for $t \geq 0$. We see that at large times the velocity approaches a constant value, $\lim_{t\to\infty} v(t) = \sqrt{g/k}$, known as the terminal velocity.

[41] Recalling the definitions of the hyperbolic cosine and sine

$$\cosh x = \frac{1}{2}(e^x+e^{-x}), \quad \sinh x = \frac{1}{2}(e^x-e^{-x}) \quad \Rightarrow \quad \tanh x = \frac{e^x-e^{-x}}{e^x+e^{-x}}.$$

5.3 First-order linear differential equations

Consider a first-order differential equation

$$\frac{dy}{dx} = F(x, y), \tag{5.15}$$

in which the right-hand side is a *linear* function of y, i.e.

$$F(x, y) = q(x) - p(x)y$$

where the "coefficients" $p(x)$ and $q(x)$ can depend on x.[42] Substituting this into equation (5.15),

$$\frac{dy}{dx} = q(x) - p(x)y,$$

and moving the term containing y to the left, we have

$$\frac{dy}{dx} + p(x)y = q(x), \tag{5.16}$$

which is the standard form of the *first-order linear DE.*

The equation for $q(x) = 0$,

$$\frac{dy}{dx} + p(x)y = 0. \tag{5.17}$$

is known as the *homogeneous* first-order linear DE. For $q(x) \neq 0$ equation (5.16) is called *inhomogeneous.* The homogeneous equation has an important property. Suppose we have found a function $y_1(x)$ which satisfies equation (5.17), i.e.,

$$y_1'(x) + p(x)y_1(x) = 0.$$

In this case *any* function of the form

$$y(x) = Cy_1(x), \tag{5.18}$$

[42] The term containig y is traditionally written with a minus sign.

where C is a constant, is also a solution of (5.17). Indeed,

$$\frac{d}{dx}\big[Cy_1(x)\big] + p(x)Cy_1(x) = C\big[y_1'(x) + p(x)y_1(x)\big] = 0.$$

We know that the general solution of a first-order differential equation must always contain one arbitrary constant. This means that (5.18) represents the general form of the solution of the homogeneous equation (5.17). Suppose we have found a particular solution of the inhomogeneous equation (5.16), say, y_p, such that

$$y_p' + p(x)y_p = q(x).$$

If we now consider

$$y = Cy_1 + y_p, \tag{5.19}$$

and substitute it into (5.16), we obtain

$$\begin{aligned} \frac{d}{dx}(Cy_1 + y_p) + p(x)(Cy_1 + y_p) &= Cy_1' + y_p' + p(x)(Cy_1 + y_p) \\ &= C(\underbrace{y_1' + p(x)y_1}_{=0}) + y_p' + p(x)y_p \\ &= q(x). \end{aligned}$$

Given that (5.19) contains an arbitrary constant, we conclude that the general solution of the inhomogeneous equation (5.16) will always have the form (5.19). Two methods can be used for solving first-order linear DE. We will apply them to solve equation (5.16) in its general form.

1. Method of variation of the constant.

In this method, the problem is solved in two steps.

Step 1: solve the homogeneous version of the equation.

Equation (5.17) is variable separable. We have

$$\begin{aligned}
\frac{dy}{dx} &= -p(x)y, \\
\int \frac{dy}{y} &= -\int p(x)dx, \\
\ln y &= -\int p(x)dx + C.
\end{aligned}$$

Taking exponents of both sides,[43]

$$y = e^C e^{-\int p(x)dx},$$

and renaming e^C by C,

$$y = Ce^{-\int p(x)dx}.$$

Step 2: replace the constant C in the above by a function $C(x)$, and seek solution of the inhomogeneous equation in the form

$$y = C(x)e^{-\int p(x)dx}. \tag{5.20}$$

Substituting (5.20) into (5.16), we have

$$\frac{d}{dx}\left[C(x)e^{-\int p(x)dx}\right] + p(x)C(x)e^{-\int p(x)dx} = q(x).$$

Using the product rule and chain rule (to differentiate the exponent),

$$\begin{aligned}
C'(x)e^{-\int p(x)dx} + C(x)\frac{d}{dx}e^{-\int p(x)dx} + p(x)C(x)e^{-\int p(x)dx} &= q(x), \\
C'(x)e^{-\int p(x)dx} + C(x)e^{-\int p(x)dx}\frac{d}{dx}\left[-\int p(x)dx\right] + p(x)C(x)e^{-\int p(x)dx} &= q(x),
\end{aligned}$$

[43] Strictly speaking, we must use $\int dy/y = \ln|y|$, which will lead to $|y| = e^C e^{-\int p(x)dx}$ and hence, $y = \pm e^C e^{-\int p(x)dx}$. Given that C is an arbitrary constant, the $\pm$ sign gets absorbed when $\pm e^C$ is renamed C. This shows that it is not necessary to use modulus under ln when solving first-order linear DE.

$$C'(x)e^{-\int p(x)dx} + \cancel{C(x)e^{-\int p(x)dx}[-p(x)] + p(x)C(x)e^{-\int p(x)dx}} = q(x).$$

Here the second and third terms on the left-hand side cancel and we have

$$\frac{dC}{dx} = e^{\int p(x)dx}q(x),$$

so that

$$C(x) = \int e^{\int p(x)dx}q(x)dx + C_1,$$

where C_1 is a new arbitrary constant. Substituting this into (5.20) gives the solution sought:

$$y = e^{-\int p(x)dx}\int e^{\int p(x)dx}q(x)dx + C_1 e^{-\int p(x)dx}. \tag{5.21}$$

Here the first term is a particular solution of the inhomogeneous equation and the second term is the general solution of the homogeneous equation (sometimes called the *complementary function*), cf. equation (5.19). Equation (5.21) shows that the first-order liner DE (5.16) can be solved for any $p(x)$ and $q(x)$ as long as one can find the indefinite integrals involved. However, there is no need to memorise equation (5.21), as one only needs to remember the *two steps* involved in finding the solution.

<u>Example:</u> $xy' + 3y = x^3$.

First, change this DE into the standard form by diving through by x: (Comparing with (5.16), we see that in this eqn. $p(x) = \frac{3}{x}$ and $q(x) = x^2$).

$$y' + \frac{3}{x}y = x^2. \tag{5.22}$$

<u>Step 1:</u> solve the homogeneous equation

$$y' + \frac{3}{x}y = 0.$$

Using variable separation,

$$\frac{dy}{dx} = -\frac{3}{x}\,y,$$
$$\int \frac{dy}{y} = -\int \frac{3}{x}\,dx,$$
$$\ln y = -3\ln x + C.$$

Taking exponents of both sides, (Note $e^{-3\ln x} = \left(e^{\ln x}\right)^{-3}$)

$$y = e^{C}e^{-3\ln x},$$

and renaming e^{C} by C, we obtain

$$y = \frac{C}{x^3}.$$

Step 2: replace $C \longrightarrow C(x)$ and seek solution of (5.22) in the form

$$y = \frac{C(x)}{x^3}. \tag{5.23}$$

Substituting this into (5.22),

$$\left[\frac{C(x)}{x^3}\right]' + \frac{3}{x}\,\frac{C(x)}{x^3} = x^2,$$

and using the product rule to differentiate $[\dots]$, we have

$$C'(x)\frac{1}{x^3} + \cancel{C(x)\left(-\frac{3}{x^4}\right)} + \cancel{\frac{3}{x}\,\frac{C(x)}{x^3}} = x^2,$$

where the two terms containing $C(x)$ cancel,[44] so that

[44] Such cancellation should always happen if the proceeding steps are done correctly.

$$C'(x)\frac{1}{x^3} = x^2.$$

Hence,

$$\frac{dC}{dx} = x^5,$$
$$dC = x^5 dx,$$

and integrating,

$$C(x) = \int x^5 dx = \frac{x^6}{6} + C_1,$$

where C_1 is an arbitrary constant. Substituting into (5.23) we obtain

$$y = \frac{x^3}{6} + \frac{C_1}{x^3}, \tag{5.24}$$

which is the general solution of equation (5.22).

2. The integrating factor method.

In this method, we multiply equation (5.16) by a function $\mu(x)$, (Recall $(fg)' = f'g + fg'$)

$$\mu(x)\frac{dy}{dx} + \underbrace{\mu(x)p(x)}_{=\frac{d\mu}{dx}} y = \mu(x)q(x). \tag{5.25}$$

Noticing that the first term on the left-hand side matches that which results from the product rule for the derivative of of $\mu(x)y$, we choose $\mu(x)$ so that

$$\frac{d\mu}{dx} = \mu(x)p(x), \tag{5.26}$$

to make the product rule expression complete. In this case (5.25) becomes

$$\frac{d}{dx}\left[\mu(x)y\right] = \mu(x)q(x). \tag{5.27}$$

Solving equation (5.26),

$$\frac{d\mu}{\mu} = p(x)dx,$$
$$\int \frac{d\mu}{\mu} = \int p(x)dx,$$
$$\ln\mu = \int p(x)dx,$$

so that

$$\mu(x) = e^{\int p(x)dx}, \tag{5.28}$$

which is the *integrating factor*.[45] Going back to equation (5.27) and taking indefinite integrals on both sides gives

$$\mu(x)y = \int \mu(x)q(x)dx + C,$$

so that

$$y = \frac{1}{\mu(x)}\int \mu(x)q(x)dx + \frac{C}{\mu(x)}.$$

Finally, using the expression (5.28) for the integrating factor,

$$y = e^{-\int p(x)dx}\int e^{\int p(x)dx}q(x)dx + Ce^{-\int p(x)dx}. \tag{5.29}$$

[45] Note that we do not need to include an arbitrary constant when finding the integrating factor, or we can choose this constant as we like, e.g., set it to zero. This is because we need just one function $\mu(x)$ that satisfies (5.25), ideally the simplest.

Note that this answer is exactly the same as that we obtained by variation of the constant (5.21) (except for the constant C_1 being called C here). As with the first method, there is no need to memorise this result, but only to remember the idea that leads to the answer.

Example: let us solve the same equation (5.22),

$$y' + \frac{3}{x}y = x^2.$$

Multiplying this equation by $\mu(x)$,

$$\mu(x)y' + \underbrace{\mu(x)\frac{3}{x}}_{=\mu'(x)} y = \mu(x)x^2,$$

we choose $\mu(x)$ so that the left-hand side matches the product rule form, and the equation becomes

$$\frac{d}{dx}[\mu(x)y] = \mu(x)x^2. \tag{5.30}$$

To find the integrating factor we solve

$$\begin{aligned}
\mu'(x) &= \mu(x)\frac{3}{x},\\
\frac{d\mu}{dx} &= \mu(x)\frac{3}{x},\\
\int \frac{d\mu}{\mu} &= 3\int \frac{dx}{x},\\
\ln\mu &= 3\ln x,
\end{aligned}$$

so that the integrating factor is (Choosing $C = 0$ is usually best for the integrating factor)

$$\mu(x) = e^{3\ln x} = \left(e^{\ln x}\right)^3 = x^3.$$

Substituting this into (5.30),

$$\frac{d}{dx}[x^3 y] = x^3 x^2,$$

and integrating both sides,

$$x^3 y = \int x^5 dx,$$

$$x^3 y = \frac{x^6}{6} + C,$$

we obtain

$$y = \frac{x^3}{6} + \frac{C}{x^3},$$

which is the same answer as (5.24).

5.4 Second-order linear DE with constant coefficients: homogeneous case

In this section we consider differential equations of the form

$$ay'' + by' + cy = 0, \tag{5.31}$$

where $a \neq 0$, b and c are some constants. (We assume a, b, $c \in \mathbb{R}$ here, although the theory for complex a, b and c is quite similar.) This is a *second-order linear homogeneous differential equation with constant coefficients.*

Suppose we know that a certain function, say y_1, is a solution of (5.31), i.e.,

$$ay_1'' + by_1' + cy_1 = 0.$$

We can then show that any function of the form

$$y = C_1 y_1, \tag{5.32}$$

where C_1 is a constant, is also a solution of (5.31). Indeed,

$$a(C_1y_1)'' + b(C_1y_1)' + cC_1y_1 = C_1(ay_1'' + by_1' + cy_1) = 0.$$

We know that the general solution of a second-order DE must contain *two* arbitrary constants. This means that (5.32) does not qualify for the general solution of equation (5.31). However, if there exist some other function, say y_2, which also satisfies (5.31), then we can construct the *general solution* as

$$y = C_1y_1 + C_2y_2, \tag{5.33}$$

where both C_1 and C_2 are arbitrary constants.(The two functions y_1 and y_2 must be *linearly independent*, i.e., one should not be equal to the other times a constant. Otherwise, the two terms in (5.33) combine into one.)

<u>Exercise:</u> verify that (5.33) satisfies equation (5.31).

How to find y_1 and y_2? Let us seek solution of equation (5.31) in the form

$$y = e^{\lambda x}.$$

We then have $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substituting these into (5.31),

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0,$$

and dividing by $e^{\lambda x}$, we obtain the *auxiliary equation* (AE)

$$a\lambda^2 + b\lambda + c = 0. \tag{5.34}$$

This quadratic equation has, in general, two roots given by

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{5.35}$$

Denoting these roots λ_1 and λ_2, we can set $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$, and use these to form the general solution (5.33). Depending on the value of the *discriminant* $b^2 - 4ac$, the roots of the AE can be real, complex, or coincide. Let us consider these *three cases* separately.

(1) <u>For $b^2 - 4ac > 0$</u> equation (5.34) has two distinct real roots,

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a},$$

and the general solution of the differential equation is

$$\boxed{y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}.} \tag{5.36}$$

(2) <u>For $b^2 - 4ac < 0$</u> the roots of the auxiliary equation are complex,

$$\lambda = \frac{-b \pm \sqrt{-(4ac - b^2)}}{2a} = \frac{-b \pm i\sqrt{4ac - b^2}}{2a} \equiv \alpha \pm i\beta,$$

where we denoted by α and β the real and imaginary parts of the roots. Hence, $\lambda_1 = \alpha + i\beta$ and $\lambda_2 = \alpha - i\beta$, and the general solution is

$$y = C_1 e^{(\alpha + i\beta)x} + C_2 e^{(\alpha - i\beta)x}. \tag{5.37}$$

The drawback of this form is that this function is in general complex, while the differential equation that it solves is real. For example, the differential equation may desrcibe the oscillations of a pendulum, so we will be interested in the solutions that are real rather than complex! However, we can transform (5.37) using Euler's formula

$$\begin{aligned} y &= C_1 e^{\alpha x} e^{i\beta x} + C_2 e^{\alpha x} e^{-i\beta x} \\ &= C_1 e^{\alpha x}(\cos\beta x + i\sin\beta x) + C_2 e^{\alpha x}(\cos\beta x - i\sin\beta x) \\ &= \underbrace{(C_1 + C_2)}_{\text{constant}} e^{\alpha x}\cos\beta x + \underbrace{i(C_1 - C_2)}_{\text{constant}} e^{\alpha x}\sin\beta x\,. \end{aligned}$$

Renaming the two constants C_1 and C_2, we can write the solution as

$$\boxed{y = C_1 e^{\alpha x}\cos\beta x + C_2 e^{\alpha x}\sin\beta x.} \tag{5.38}$$

(3) <u>For $b^2 - 4ac = 0$</u> the two roots of the AE coincide, $\lambda_1 = \lambda_2$, i.e.,

$$\lambda = -b/2a$$

is a root of multiplicity 2. In this case the two exponents in (5.36) are identical. So, while $y_1 = e^{\lambda x}$ is a valid solution, what is the second solution that would allow us to form the general solution (5.33)? It turns out (see below) that $y_2 = xe^{\lambda x}$, so that the general solution is

$$\boxed{y = C_1 e^{\lambda x} + C_2 x e^{\lambda x}.} \tag{5.39}$$

<u>Derivation of the solution for $b^2 - 4ac = 0$.</u>

Recall that any quadratic function can be factorised if we know its roots, (Expanding the right-hand side of (5.40) and comparing with the left-hand side, we see that $\lambda_1 + \lambda_2 = -b/a$ and $\lambda_1\lambda_2 = c/a$, which is a particular case of Vieta's theorem. François Viète (Vieta in Latin) (1540–1603) was a French lawyer turned mathematician.) e.g.,

$$a\lambda^2 + b\lambda + c = a(\lambda - \lambda_1)(\lambda - \lambda_2). \tag{5.40}$$

Following this pattern, we can factorise the left-hand side of (5.31) as

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + c = a\left(\frac{d}{dx} - \lambda_2\right)\left(\frac{d}{dx} - \lambda_1\right)y,$$

where $\dfrac{d}{dx}$ is the *operator* that differentiates the function to its right, e.g.,

$$\frac{d}{dx}y = \frac{dy}{dx} = y', \qquad \frac{d}{dx}\frac{d}{dx}y = \frac{d^2y}{dx^2} = y''.$$

Hence, for the case of identical roots the differential equation (5.31) can be written as

$$\left(\frac{d}{dx} - \lambda\right)\left(\frac{d}{dx} - \lambda\right) y = 0, \tag{5.41}$$

where we dropped the factor a on the left.

Let us now introduce a new function

$$u = \left(\frac{d}{dx} - \lambda\right) y. \tag{5.42}$$

Equation (5.41) then is a first-order linear homogeneous DE for u,

$$\left(\frac{d}{dx} - \lambda\right) u = 0,$$

which we solve by variable separation:

$$\begin{aligned}
\frac{du}{dx} - \lambda u &= 0, \\
\frac{du}{dx} &= \lambda u, \\
\int \frac{du}{u} &= \lambda \int dx, \\
\ln u &= \lambda x + C, \\
u &= e^C e^{\lambda x},
\end{aligned}$$

and, replacing e^C by C,

$$u = Ce^{\lambda x}.$$

We can now find y from equation (5.42), which gives

$$\frac{dy}{dx} - \lambda y = Ce^{\lambda x}.$$

Using the integrating factor method, we multiply this equation by $\mu(x)$,

$$\mu(x)\frac{dy}{dx} - \lambda\mu(x)y = C\mu(x)e^{\lambda x}, \tag{5.43}$$

and choose $\mu(x)$ to make the left-hand side fit the product rule form, i.e.,

$$\begin{aligned}
\frac{d\mu}{dx} &= -\lambda\mu(x), \\
\int \frac{d\mu}{\mu} &= -\int \lambda dx, \\
\ln\mu &= -\lambda x, \\
\mu &= e^{-\lambda x}.
\end{aligned}$$

Equation (5.43) then becomes

$$\begin{aligned}
\frac{d}{dx}[e^{-\lambda x}y] &= Ce^{-\lambda x}e^{\lambda x}, \\
e^{-\lambda x}y &= C\int dx \\
e^{-\lambda x}y &= Cx + C_1, \\
y &= Cxe^{\lambda x} + C_1e^{\lambda x},
\end{aligned}$$

which is the general solution (5.39) (in which C_2 is used instead of C).

Example 1: $y'' + 5y' + 6y = 0$.

The auxiliary equation

$$\lambda^2 + 5\lambda + 6 = 0$$

has roots

$$\lambda = \frac{-5 \pm \sqrt{25-24}}{2} = \frac{-5 \pm 1}{2},$$

so that $\lambda_1 = -2$ and $\lambda_2 = -3$ [case (1)], and the general solution is given by (5.36):

$$y = C_1 e^{-2x} + C_2 e^{-3x}. \tag{5.44}$$

The arbitrary constants can be determined if some initial conditions are given, e.g., $y(0) = 0$:

$$C_1 e^0 + C_2 e^0 = 0 \iff C_1 + C_2 = 0 \iff C_2 = -C_1.$$

We see that one condition is not enough for finding two constants. Suppose we also know the derivative at $x = 0$, e.g., $y'(0) = 1$. Taking the derivative of (5.44),

$$y' = -2C_1 e^{-2x} - 3C_2 e^{-3x},$$

and then setting $x = 0$,

$$-2C_1 e^0 - 3C_2 e^0 = 1 \iff -2C_1 - 3C_2 = 1,$$

and using $C_2 = -C_1$, we have

$$-2C_1 + 3C_1 = 1 \iff C_1 = 1 \quad \text{and} \quad C_2 = -1.$$

The solution of the DE which satisfies $y(0) = 0$, $y'(0) = 1$ then is

$$y = e^{-2x} - e^{-3x}.$$

Example 2: $y'' - 4y' + 4y = 0$.

The auxiliary equation

$$\lambda^2 - 4\lambda + 4 = 0$$

has roots

$$\lambda = \frac{4 \pm \sqrt{16-16}}{2} = 2,$$

so that $\lambda = 2$ is a root of multiplicity two [case (3)]. The general solution is then given by (5.39)

$$y = C_1 e^{2x} + C_2 x e^{2x}.$$

Example 3: $y'' + 4y' + 13y = 0.$

The auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

has roots

$$\lambda = \frac{-4 \pm \sqrt{16-52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i.$$

This is case (2) [$\alpha = -2$, $\beta = 3$] and the general solution is of the form (5.38):

$$y = C_1 e^{-2x} \cos 3x + C_2 e^{-2x} \sin 3x.$$

5.5 Inhomogeneous linear second-order DE with constant coefficients

Let us now consider the *inhomogeneous* linear second-order DE with constant coefficients,

$$ay'' + by' + cy = f(x), \tag{5.45}$$

in which the right-hand side is nonzero. Suppose we have been given a function, say y_p, which satisfies this equation,

$$ay_p'' + by_p' + cy_p = f(x).$$

This function is known as a *particular solution*. It can be used to construct the *general* solution of the inhomogeneous equation (5.45) as

$$y = C_1 y_1 + C_2 y_2 + y_p. \tag{5.46}$$

Here $C_1 y_1 + C_2 y_2$ is the general solution of the homogeneous equation. In the context of (5.46) it is often called the *complementary function*.

Exercise: verify that (5.46) satisfies equation (5.45).

For an arbitrary $f(x)$ the solution of the inhomogeneous equation (5.45) can be found by using the **method of variation of the constants**.[46]

In this method we seek solution of equation (5.45) in the form

$$y = C_1(x) y_1 + C_2(x) y_2, \tag{5.47}$$

where y_1 and y_2 are two independent solutions of the homogeneous equation. The first derivative then is

$$\begin{aligned} y' &= C_1'(x) y_1 + C_1(x) y_1' + C_2'(x) y_2 + C_2(x) y_2' \\ &= \underbrace{C_1'(x) y_1 + C_2'(x) y_2}_{\text{set equal to } 0} + C_1(x) y_1' + C_2(x) y_2', \end{aligned}$$

where we can set the sum of the first two terms equal to zero,[47]

$$C_1'(x) y_1 + C_2'(x) y_2 = 0. \tag{5.48}$$

[46] This method was pioneered by Leonard Euler and Joseph-Louis Lagrange (1736-1813), an Italian of French descent, who lived and worked in Turin, Berlin and Paris, in his *Solution de différens problémes du calcul integral* (1766).

[47] We can do this because substitution of y, y' and y'' into equation (5.45) will provide us with *one* equation, which is not enough for finding *two* functions $C_1(x)$ and $C_2(x)$. So we can impose an extra condition (5.48), which turns out to be convenient.

The second derivative then is

$$\begin{aligned} y'' &= \big[C_1(x)y_1' + C_2(x)y_2'\big]' \\ &= C_1'(x)y_1' + C_1(x)y_1'' + C_2'(x)y_2' + C_2(x)y_2''. \end{aligned}$$

Substituting y'', y' and y into equation (5.45) gives:

$$\begin{aligned} a\big[C_1'(x)y_1' + C_1(x)y_1'' + C_2'(x)y_2' + C_2(x)y_2''\big] \\ + b\big[C_1(x)y_1' + C_2(x)y_2'\big] \\ + c\big[C_1(x)y_1 + C_2(x)y_2\big] = f(x) \end{aligned}$$

which can be re-arranged as

$$a\big[C_1'(x)y_1' + C_2'(x)y_2'\big] + C_1(x)(\underbrace{ay_1'' + by_1' + cy_1}_{=0}) + C_2(x)(\underbrace{ay_2'' + by_2' + cy_2}_{=0}) = f(x)$$

and gives

$$a\big[C_1'(x)y_1' + C_2'(x)y_2'\big] = f(x),$$

or

$$C_1'(x)y_1' + C_2'(x)y_2' = \frac{1}{a}f(x). \tag{5.49}$$

Equations (5.48) and (5.49) are simultaneous equations for $C_1'(x)$ and $C_2'(x)$. Multiplying (5.48) by y_2' and (5.49) by $-y_2$ and adding them up, we have

$$C_1'(x)(y_1y_2' - y_1'y_2) = -\frac{1}{a}f(x)y_2,$$

which gives

$$C_1'(x) = -\frac{1}{a}\frac{f(x)y_2}{y_1y_2' - y_2y_1'},$$

so that

$$C_1(x) = -\frac{1}{a}\int \frac{f(x)y_2}{y_1y_2' - y_2y_1'}dx. \tag{5.50}$$

Multiplying (5.48) by $-y_1'$ and (5.49) by y_1 and adding them up, gives

$$C_2'(x)(-y_2y_1' + y_2'y_1) = \frac{1}{a}f(x)y_1,$$

which results in

$$C_2(x) = \frac{1}{a}\int \frac{f(x)y_1}{y_1y_2' - y_2y_1'}dx. \tag{5.51}$$

Equations (5.50) and (5.51) allow one to find the "varying constants" $C_1(x)$ and $C_2(x)$ and thus obtain the solution of the inhomogeneous equation from (5.47).

Note that the combination of the solutions y_1 and y_2 and their derivatives in the denominators or (5.50) and (5.51) is a 2×2 determinant,

$$y_1y_2' - y_2y_1' = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \equiv W, \tag{5.52}$$

known as the *Wronskian* and usually denoted W. It is not equal to zero as long as y_1 and y_2 are linearly independent.

Exercise: verify that if $W = 0$ then $y_1 = Cy_2$.

The method of varying constants requires calculation of the integrals (5.50) and (5.51), which may be a tedious task. There is an alternative method which can be used when $f(x)$ has a simple form (see below). It is known as the **method of undetermined coefficients** (or alternatively, the '*method of judicious guessing*').

This method involves seeking a particular solution y_p of equation (5.45) in the form that matches the right-hand side $f(x)$. Initially this particular solution contains a number of coefficients which are found by substitution of y_p and its derivatives y_p' and y_p'' into the differential equation.

After finding this particular solution, it is added to the complementary function to obtain the general solution (5.46).

The method of undetermined coefficients is used when $f(x)$ has one of these two forms.

(1) $f(x) = e^{px}P_n(x)$, where P_n is a polynomial of degree n. In this case seek particular solution in the form

$$y_p = x^r e^{px} Q_n(x), \tag{5.53}$$

where $Q_n(x)$ is a polynomial of degree n with undetermined coefficients, and

- $r = 0$ if $p \neq \lambda_1$ and $p \neq \lambda_2$ (λ_1 and λ_2 being roots of the AE);
- $r = 1$ if $p = \lambda_1$ or $p = \lambda_2$, and $\lambda_1 \neq \lambda_2$;
- $r = 2$ if $p = \lambda_1 = \lambda_2$ (when the AE has a root of multiplicity two).

(2) $f(x) = e^{px}[P_n(x)\cos qx + Q_m(x)\sin qx]$, where P_n and $Q_m(x)$ are polynomials of degree n and m, respectively. In this case seek solution in the form

$$y_p = x^r e^{px}[S_N(x)\cos qx + T_N(x)\sin qx], \tag{5.54}$$

where $S_N(x)$ and $T_N(x)$ are polynomials of degree $N = \max\{n, m\}$ with undetermined coefficients, and

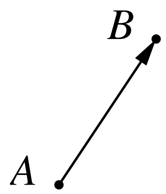
- $r = 0$ if $p \pm iq \neq \alpha \pm i\beta$ (complex roots of the AE);
- $r = 1$ if $p \pm iq = \alpha \pm i\beta$.

6 Vectors

There are quantities in nature that cannot be described by magnitude alone (e.g., wind, displacement, force, etc.), since direction is also important. To describe these we use *vectors*.

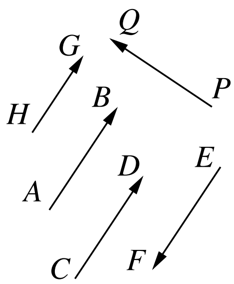
6.1 Definitions and notation

Definition. Consider two points A and B connected by a line segment, for which direction is also indicated by placing an arrow at one end. This is a now a *directed line segment*.



The notation for this directed line segment is $\overrightarrow{AB}$, and its length (or magnitude) is denoted $|\overrightarrow{AB}|$.

Definition. Two directed line segments are *equal* is they have equal lengths, lie on parallel lines and are in the same direction.[48]



For example, in this diagram, $\overrightarrow{AB} = \overrightarrow{CD}$. However, $\overrightarrow{AB} \neq \overrightarrow{EF}$ (have equal lengths and lie on parallel lines, but different directions), $\overrightarrow{AB} \neq \overrightarrow{HG}$ (lie on parallel lines and are in the same direction, but have different lengths), $\overrightarrow{AB} \neq \overrightarrow{PQ}$ (have equal lengths but do not lie on parallel lines).

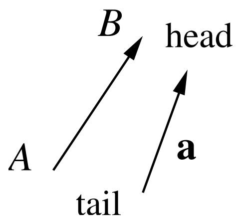
Definition. The class of *all* directed line segments equal to a given one, defines a *vector*.

A vector is often shown by a representative of the class, e.g., $\overrightarrow{AB}$. However, unlike directed line segments, vectors are not "nailed" to specific points in space.

[48] How to define "in the same direction" rigorously mathematically? Let us draw a line through the "tails" of the two directed line segments which lie on distinct parallel lines. This line will divide the plane in which the vectors lie into two half-planes. If the "heads" of the vectors are in the same half-plane, then they are in the same direction; if the "heads" are in different half-palnes than they have opposite directions.

A more compact notation for vectors is by using a single boldface (usually lower-case) letter, e.g., $\mathbf{a}$, $\mathbf{b}$, etc., and their magnitudes (lengths) are denoted $|\mathbf{a}|$, $|\mathbf{b}|$, etc. By convention, the magnitude of a vector is often given by the same letter in italics, e.g., $|\mathbf{a}| \equiv a$, $|\mathbf{v}| \equiv v$, etc.

In handwriting vectors are usually indicated by an arrow above the letter or by underlining, e.g., $\vec{a}$ or $\underline{a}$. In this case, the length is $|\vec{a}|$ or $|\underline{a}|$ or just a. Whatever your choice, it is important to *always* indicate vectors.

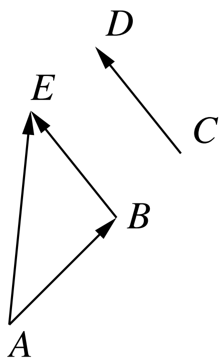


For a vector, the end with the arrow is called the 'head' and the other end is the 'tail'. Hence, for $\overrightarrow{AB}$ point B is the head and A is the tail.

6.2 Operations on vectors

A vector can be viewed as a *transformation* upon which every point is moved through a distance equal to the length of the vector in the direction given by the vector. Such transformation is known as *translation*.

Suppose we perfom two consecuitive translations given by the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$. By constructing a vector $\overrightarrow{BE} = \overrightarrow{CD}$, we see that the combined translation ($\overrightarrow{AB}$ followed by $\overrightarrow{BE}$) is equivalent to the translation given by the vector $\overrightarrow{AE}$.

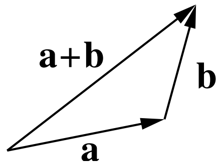


We say that vector $\overrightarrow{AE}$ is the *sum* of two vectors:

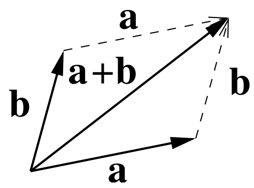
$$\overrightarrow{AE} = \overrightarrow{AB} + \overrightarrow{BE} = \overrightarrow{AB} + \overrightarrow{CD} .$$

This combination of translations defines the *addition of vectors*. To add two vectors, one needs to place the tail of the second vector at the head of the first vector, and connect the tail of the first vector with the head of the second one.

Plotting vectors **a** and **b** head-to-tail, we find $\mathbf{a}+\mathbf{b}$ as the third side of the triangle (with a proper direction). This method of adding vectors is known as the *triangle rule.*



Alternatively, one can construct a parallelogram with sides given by vectors **a** and **b**; the sum $\mathbf{a}+\mathbf{b}$ is then given by its diagonal. This is known as the *parallelogram rule.*



From the parallelogram rule we immediately see that

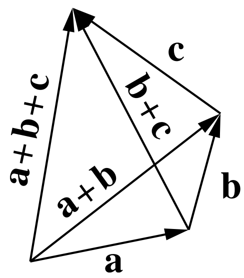
$$\mathbf{a}+\mathbf{b}=\mathbf{b}+\mathbf{a}, \tag{6.1}$$

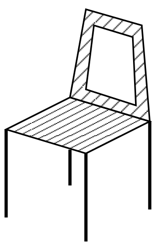
i.e., it proves that the addition of vectors is *commutative.*[49]

What if we need to add more than two vectors, e.g., **a**, **b** and **c**? This can be done in two ways, i.e., by first adding $(\mathbf{a}+\mathbf{b})$ and then adding **c**, or by adding $(\mathbf{b}+\mathbf{c})$ together and then adding it to **a**. As seen from the diagram, both ways give the same result, so that

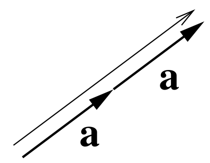
$$(\mathbf{a}+\mathbf{b})+\mathbf{c}=\mathbf{a}+(\mathbf{b}+\mathbf{c})=\mathbf{a}+\mathbf{b}+\mathbf{c}.$$

This means that addition of vectors is *associative*, i.e., brackets are not needed.





[49] Relation (6.1) looks so natural that one may wonder why it needs to be proved. However, not all operetions are commutative. For example, while translations are commutative, it is easy to check that rotations in general are not. Take a chair and rotate it forward by 90° and then left by 90°. Then, starting from the same initial position, do the same rotations in a different order, i.e., first left and then forward. Note the difference in the final positions of the chair!



Thus we see that if more than two vectors need to be added, they can all be placed head-to-tail, and the result is obtained by connecting the tail of the first with the head of the last.

Let us now add $\mathbf{a}$ and $\mathbf{a}$. The vector $\mathbf{a} + \mathbf{a}$ has the same direction as $\mathbf{a}$ and twice its length, so it seems natural to write it as $2\mathbf{a}$.

In general, we define multiplication of vectors by numbers (*scalars*) as follows.

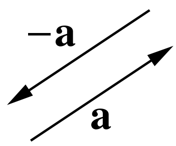
Definition. The product of a vector $\mathbf{a}$ and number λ is a vector denoted $\lambda\mathbf{a}$, whose length is given by

$$|\lambda\mathbf{a}| = |\lambda||\mathbf{a}|,$$

and whose direction is the same as $\mathbf{a}$ for $\lambda > 0$ and opposite for $\lambda < 0$.

If $\lambda = 0$, we obtain a zero-length vector $0\mathbf{a} = \mathbf{0}$ known as the *null vector*. It has no direction and is usually written simply as 0 without boldface or arrow.

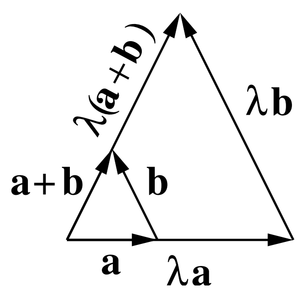
Example: the vector $(-1)\mathbf{a} \equiv -\mathbf{a}$ has the same length as $\mathbf{a}$ but its direction is opposite to that of $\mathbf{a}$.



Multiplication of vectors by numbers is *distributive*, i.e.,

$$\lambda(\mathbf{a} + \mathbf{b}) = \lambda\mathbf{a} + \lambda\mathbf{b}. \tag{6.2}$$

This familiar relation needs to proved, as multiplication of vectors by numbers is not the same as multiplication of numbers.



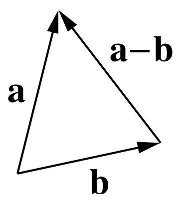
Proof. The proof of this relation is geometrical, based on considering similar triangles. One triangle is formed by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{a}+\mathbf{b}$. The other triangle has sides $\lambda\mathbf{a}$ and $\lambda\mathbf{b}$, which appear on right-hand side of identity (6.2). Its third side must then be λ times longer than $\mathbf{a}+\mathbf{b}$ and have the same direction, which is the vector on the left-hand side of (6.2).[50]

Subtraction of vectors.

Subtracting vector $\mathbf{b}$ from $\mathbf{a}$ is the same as adding $\mathbf{a}$ and $-\mathbf{b}$:

$$\mathbf{a}-\mathbf{b}=\mathbf{a}+(-\mathbf{b}).$$

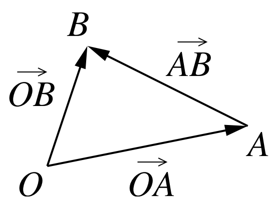
In practice this is achieved by plotting $\mathbf{a}$ and $\mathbf{b}$ with the same "tail" and connecting the head of $\mathbf{b}$ with the head of $\mathbf{a}$. Obtained in this way, the vector $\mathbf{a}-\mathbf{b}$ does show the *difference* between $\mathbf{a}$ and $\mathbf{b}$.



Using the directed line segment notation for the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$, we have

$$\overrightarrow{OB}-\overrightarrow{OA}=\overrightarrow{AB},$$

which is immediately verified by $\overrightarrow{OA}+\overrightarrow{AB}=\overrightarrow{OB}$.



Naturally, then

$$\overrightarrow{OA}-\overrightarrow{OB}=\overrightarrow{BA}=-\overrightarrow{AB}\,.$$

[50] The proof given is for $\lambda>0$. For $\lambda<0$ the ratio of the lengths of the sides of the two triangles will be $|\lambda|$ and the directions of $\lambda\mathbf{a}$, etc., must be reversed. As an exercise, construct the vector diagram for this case and convince yourself that (6.2) holds.

6.3 Applications to geometry

In geometrical applications vectors are often used to indicate the positions of various points with respect to a particular point known as the *origin*, commonly denoted O. Its own position is sometimes fixed (e.g., at the origin of a coordinate frame); in other cases it can be chosen arbitrarily.

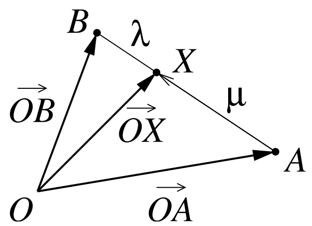
1. Ratio theorem.

The ratio theorem is useful relation which helps to solve problems involving lines and points where they meet.

Theorem. Point X lies on the segment AB and divides it in the ratio $\mu : \lambda$, counting from A, if and only if

$$(\lambda + \mu) \overrightarrow{OX} = \lambda \overrightarrow{OA} + \mu \overrightarrow{OB}, \tag{6.3}$$

where O is an arbitrary origin.[51]



Proof. We will prove the theorem in the forward direction, i.e., assume the position of the point X on AB and derive the vector identity required.

Since X divides AB in the ratio $\mu : \lambda$, we have

$$\frac{AX}{AB} = \frac{\mu}{\lambda + \mu},$$

which gives

$$\overrightarrow{AX} = \frac{\mu}{\lambda + \mu} \overrightarrow{AB} .$$

Then

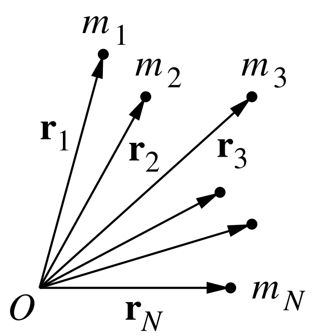
[51] Note that the order in which λ and μ appear on the right-hand side of (6.3) is the opposite to that in which they enter the ratio.

$$
\begin{aligned}
\overrightarrow{OX} &= \overrightarrow{OA} + \overrightarrow{AX} \\
&= \overrightarrow{OA} + \frac{\mu}{\lambda+\mu}\,\overrightarrow{AB} \\
&= \overrightarrow{OA} + \frac{\mu}{\lambda+\mu}(\overrightarrow{OB} - \overrightarrow{OA}) \\
&= \left(1 - \frac{\mu}{\lambda+\mu}\right)\overrightarrow{OA} + \frac{\mu}{\lambda+\mu}\,\overrightarrow{OB} \\
&= \frac{\lambda}{\lambda+\mu}\,\overrightarrow{OA} + \frac{\mu}{\lambda+\mu}\,\overrightarrow{OB},
\end{aligned}
$$

which, after multiplying by $\lambda + \mu$ gives (6.3).

To prove the theorem in the other direction, one repeats the steps that lead to (6.3) backwards, hence showing that this vector identity implies that X lies on AB and divides it as $\mu : \lambda$.

2. Centre of mass. Centroid of a triangle.



Consider N points with masses m_1, m_2, $\ldots$, m_N, whose positions are gives by the vectors $\mathbf{r}_1$, $\mathbf{r}_2$,$\ldots$, $\mathbf{r}_N$, with respect to some origin O.

Definition. The *centre of mass* of this system is the point with the position vector

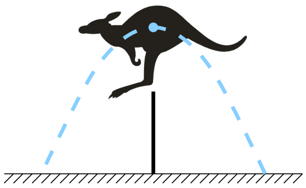
$$
\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2 + \cdots + m_N\mathbf{r}_N}{m_1 + m_2 + \cdots + m_N}. \tag{6.4}
$$

The importance of the centre of mass becomes clear when studying mechanics.

For example, when an extended object moves under gravity, e.g., a vase tumbles off a table or a kangaroo leaps a fence, it is its centre of mass that follows the familiar simple parabolic path (if the air resitance is neglected).

Example 1. For two masses m_1 and m_2 the centre of mass is at

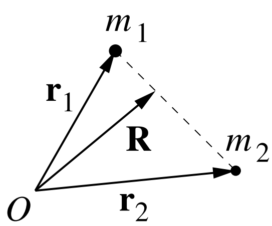
$$
\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}.
$$



We then have

$$(m_1 + m_2)\mathbf{R} = m_1\mathbf{r}_1 + m_2\mathbf{r}_2,$$

so by the ratio theorem (6.3), this means that the centre of mass lies on the line connecting m_1 and m_2 and divides this line in the ratio $m_2 : m_1$. In particular, if m_1 is greater than m_2, then $m_2/m_1 < 1$ and the centre of mass lies closer to that larger mass m_1.



Example 2. Let us put three equal masses m at the vertices of a triangle ABC. From (6.4), its centre of mass lies at point G such that

$$\overrightarrow{OG} = \frac{m\,\overrightarrow{OA} + m\,\overrightarrow{OB} + m\,\overrightarrow{OC}}{m + m + m},$$

where O is an arbitrary origin. Hence we have

$$\overrightarrow{OG} = \frac{1}{3}(\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}). \tag{6.5}$$

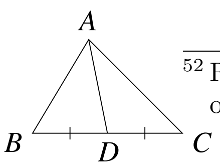
Definition. Point G defined by equation (6.5) is the *centroid* of $\triangle\, ABC$.

Where does it actually lie? The answer is provided by the following theorem.

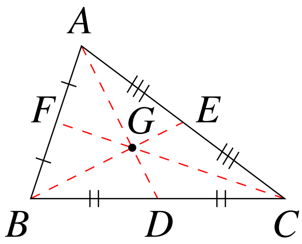
Theorem. The centroid of a triangle lies at the point where its three medians meet, and divides each median in the ratio $2 : 1$ counting from the vertex.[52]

Proof. Let us consider one of the medians, say AD. Let point X divide AD in the ratio $2 : 1$ counting from A. According to the ratio theorem,

$$(1 + 2)\,\overrightarrow{OX} = \overrightarrow{OA} + 2\,\overrightarrow{OD},$$



[52] Reminder: a *median* in a triangle is a line segment which joins a vertex with the midpoint of the opposite side. Thus, if $BD = DC$ then AD is a median of the triangle ABC.



where O is an arbitrary origin (not shown). This gives

$$\overrightarrow{OX} = \frac{1}{3}(\overrightarrow{OA} + 2\,\overrightarrow{OD}). \tag{6.6}$$

Since D is the midpoint of BC, it divides BC in the ratio $1 : 1$, so by the ratio theorem,

$$2\,\overrightarrow{OD} = \overrightarrow{OB} + \overrightarrow{OC}\,.$$

Substituting this into (6.6), we obtain

$$\overrightarrow{OX} = \frac{1}{3}(\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}).$$

Comparing this with the definition of the centroid (6.5) we see that point X coincides with the centroid G. This means that the centroid of the triangle ABC does lie on AD and divide it in the ratio $2 : 1$.

Similarly, if point Y divides the median BE in the ratio $2 : 1$, and since E is the midpoint of AC, we have

$$\overrightarrow{OY} = \frac{1}{3}(\overrightarrow{OB} + 2\,\overrightarrow{OE}) = \frac{1}{3}(\overrightarrow{OB} + \overrightarrow{OA} + \overrightarrow{OC}),$$

which shows that this point is the same as X and G.

Finally, if point Z divides the median CF in the ratio $2 : 1$, and since F is the midpoint of AB, we have

$$\overrightarrow{OY} = \frac{1}{3}(\overrightarrow{OC} + 2\,\overrightarrow{OF}) = \frac{1}{3}(\overrightarrow{OB} + \overrightarrow{OA} + \overrightarrow{OB}),$$

hence Z is the same point as X, Y and G, which completes the proof.

6.4 Linear independence and basis

Definition. Consider n vectors $\mathbf{a}_1$, $\mathbf{a}_2$, …, $\mathbf{a}_n$. These vectors are *linearly independent* if

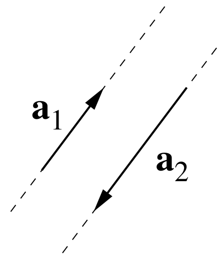
$$\lambda_1\mathbf{a}_1 + \lambda_2\mathbf{a}_2 + \cdots + \lambda_n\mathbf{a}_n = 0 \tag{6.7}$$

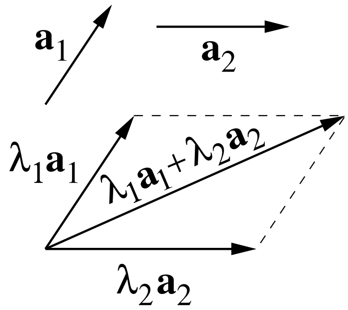
only if *all* $\lambda_i = 0$ ($i = 1, 2, \ldots, n$).

If, on the other hand, equation (6.7) can be satisfied with some λ_i being nonzero, then the vectors $\mathbf{a}_1, \ldots, \mathbf{a}_n$, are *linearly dependent*.

The expression on the left-hand side of (6.7) is a *linear combination* of the vectors $\mathbf{a}_1, \ldots, \mathbf{a}_n$, with $\lambda_1, \ldots, \lambda_n$ being the coefficients.

Note that if one of the vectors in (6.7) is a null vector, the vectors are linearly dependent. If, for example, $\mathbf{a}_k = \mathbf{0}$, then the corresponding coefficient can be chosen arbitrarily, e.g., $\lambda_k = 1$, and (6.7) is satisfied whithout all the coefficients being zero.





Example. Consider two vectors, $\mathbf{a}_1$ and $\mathbf{a}_2$, which lie on parallel lines. (Such vectors are known as *collinear*.) In this case $\mathbf{a}_1 = \lambda\mathbf{a}_2$, where $\lambda = |\mathbf{a}_1|/|\mathbf{a}_2|$ if the two vectors are in the same direction, or $\lambda = -|\mathbf{a}_1|/|\mathbf{a}_2|$ if they have opposite directions (as in the diagram). This means that

$$\mathbf{a}_1 - \lambda\mathbf{a}_2 = 0,$$

so the linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$ is zero with $\lambda_1 = 1$ and $\lambda_2 = -\lambda$ (i.e., nonzero coefficients). Hence, two collinear vectors are linearly dependent.

On the other hand, if $\mathbf{a}_1$ and $\mathbf{a}_2$ are *not collinear* then they are lineary independent. Indeed, any vector of the form $\lambda_1\mathbf{a}_1 + \lambda_2\mathbf{a}_2$ represents a diagonal in a parallelogram with sides parallel to $\mathbf{a}_1$ and $\mathbf{a}_2$. The only way to obtain $\lambda_1\mathbf{a}_1 + \lambda_2\mathbf{a}_2 = 0$, i.e., to make this diagonal vanish, is by making both sides of the parallelogram equal to zero, which requires $\lambda_1 = \lambda_2 = 0$.

However, if we consider three vectors, $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{a}_3$, that lie in the same plane (i.e., are *coplanar*) then they are linearly dependent. To verify this we can draw lines parallel to

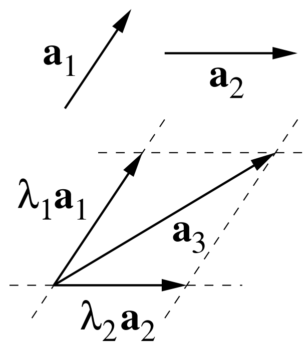
$\mathbf{a}_1$ and $\mathbf{a}_2$ through the end points of $\mathbf{a}_3$. These lines form a parallelogram in which $\mathbf{a}_3$ is a diagonal. This means that $\mathbf{a}_3$ can be represented as

$$\mathbf{a}_3 = \lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2 \tag{6.8}$$

with some coefficients λ_1 and λ_2, so that

$$\lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2 - \mathbf{a}_3 = 0.$$

Here the coefficient of $\mathbf{a}_3$ equals -1 and is *nonzero*, hence, the three vectors are linearly dependent.

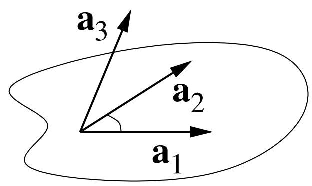


These considerations motivate the following definition.

Definition. The *dimension* of a vector space is equal to the maximum number of linearly independent vectors is allows.

Thus, if we consider a straight line, the maximum number of linearly independent vectors on this line is one, as all other vectors on the line are scalar multiples of a given vector. Hence, the line is *one-dimensional.*

In the plane, two non-collinear vectors are linearly independent, but any three vectors are lineary dependent. Hence, the plane is *two-dimensional.*



However, if the third vector ($\mathbf{a}_3$) "sticks out" of the plane in which $\mathbf{a}_1$ and $\mathbf{a}_2$ lie, then it cannot be expanded as (6.8), since any $\lambda_1 \mathbf{a}_1 + \lambda_2 \mathbf{a}_2$ remains in the plane. This makes such three vectors linearly independent, which means that the space in which we live is *three-dimensional.*[53]

[53] The theory of special relativity developed by Albert Einstein (1879–1955) considers *four-dimensional* space-time, in which time t is considered on an almost equal footing with the spatial coordinates x, y and z. However, physically time is special, since we cannot travel at our will in time in the same way we do in space! What we see around us is a three-dimensional "slice" of the four-dimensional space-time.

Suppose we have a set of n linearly independent vectors $\mathbf{a}_1$, $\mathbf{a}_2$, ..., $\mathbf{a}_n$ in n-dimensional space. If we take any other vector $\mathbf{r}$ from the same space, the $n+1$ vectors, $\mathbf{a}_1$, ..., $\mathbf{a}_n$, $\mathbf{r}$ will be linearly dependent, i.e., we will have

$$\lambda\mathbf{r} + \lambda_1\mathbf{a}_1 + \lambda_2\mathbf{a}_2 + \cdots + \lambda_n\mathbf{a}_n = 0,$$

with some coefficients being nonzero. In particular, $\lambda \neq 0$,[54] so that

$$\mathbf{r} = -\frac{\lambda_1}{\lambda}\,\mathbf{a}_1 - \frac{\lambda_2}{\lambda}\,\mathbf{a}_2 - \cdots - \frac{\lambda_n}{\lambda}\,\mathbf{a}_n. \tag{6.9}$$

This shows that *any* vector in this n-dimensional space can be presented as a linear combination of the vectors $\mathbf{a}_1$, ..., $\mathbf{a}_n$ with some coefficients.

The vectors $\mathbf{a}_1$, $\mathbf{a}_2$, ..., $\mathbf{a}_n$ are then said to form a *basis* in the n-dimensional space. The coefficients (here, $-\lambda_1/\lambda$, $-\lambda_2/\lambda$, ..., $-\lambda_2/\lambda$) are the *components* of vector $\mathbf{r}$ in this basis. The set of n components is unique for each vector. This allows one to work with components (i.e., numbers) rather then the vectors themselves when performing operation on vectors.

6.5 Basis and components

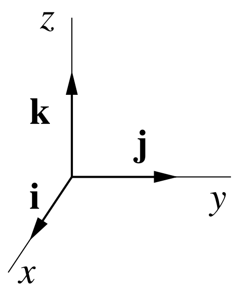
In three-dimensional space the maximum number of linearly independent vectors is three. Any three such vectors can be used as a basis.

The common and convenient choice links these vectors with the Cartesian axes x, y and z. The three vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, are *unit* vectors in the directions of the axes.[55] They are at right angles to each other and $|\mathbf{i}| = |\mathbf{j}| = |\mathbf{k}| = 1$.

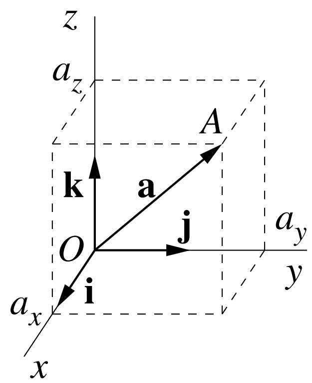
Note that by convention, we use the 'right-handed' coordinate system. In this system the rotation from the x axis towards the y axis is anticlockwise (i.e., in the positive direction) when viewed from the "top" of the z axis.

[54] If $\lambda = 0$ then we get back to (6.7), but with some $\lambda_i \neq 0$, which contradicts the linear independence of $\mathbf{a}_1$, ..., $\mathbf{a}_n$.

[55] The notation that uses letters i, j and k is due to Hamilton. William Rowan Hamilton (1805–1865) is probably the greatest Irish mathematician. For a number of years he was obsessed with the idea of extending complex numbers that "live" in the plane and have two components (real and imaginary), to some three-dimensional numbers he called 'triplets'. The problem proved difficult (in fact, impossible!). On the 16th of October 1843, Hamilton was walking along with his wife, from the Dunsink observatory, where he resided as Royal Astronomer of Ireland, to the meeting of the Royal Irish Academy in Dublin. As they were passing Brougham (or Broome) Bridge across the Royal Canal, he realised that the numbers had to be four-dimensional, of the form $q = a + bi + cj + dk$, with the three "imaginary units" obeying the basic rules $i^2 = j^2 = k^2 = ijk = -1$. Hamilton was so excited that he carved this equation in the stone of the bridge using his penknife! He called these numbers *quaternions* and presented a paper *On a new Species of Imaginary Quantities connected with a theory of Quaternions* at the next meeting of the Academy on 13 November 1943 [Proceedings of the Royal Irish Academy, vol. 2 (1844), pp. 424–434]. Although the quaternions did not quite live up to Hamilton's expectations, they are useful, e.g., for describing rotations in 3D computer animations. The bridge now carries a plaque which commemorates the event and shows the above equation.



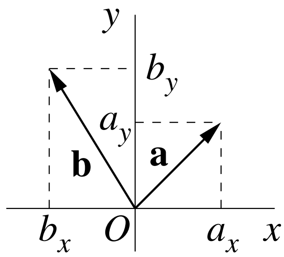
To find the components of a vector, say $\mathbf{a}$, consider point A such that $\overrightarrow{OA} = \mathbf{a}$, where O is the origin. Construct a rectangular prism with the edges parallel to the x, y and z axes , so that $\mathbf{a}$ is its diagonal.



Let the faces of the prism cut the x, y and z axes at points with coordinates a_x, a_y and a_z, respectively. In this case the three vectors $a_x\mathbf{i}$, $a_y\mathbf{j}$ and $a_z\mathbf{k}$ give the three edges of the prism, and

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}. \tag{6.10}$$

This is the expansion of the vector $\mathbf{a}$ in the basis $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and a_x, a_y and a_z are the x, y and z *components* of the vector.



The picture is even simpler for vectors in two dimensions, e.g., in the x-y plane. Here vector $\mathbf{a} = a_x\mathbf{i}+a_y\mathbf{j}$ has components $a_x > 0$ and $a_y > 0$, and $\mathbf{b} = b_x\mathbf{i}+b_y\mathbf{j}$ has components $b_x < 0$ and $b_y > 0$.

When vectors are given in terms of their components in a given basis, operations on vectors become a matter of algebra. For example, considering the sum of two vectors, vector $\mathbf{a}$ from equation (6.10) and vector

$$\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j} + b_z\mathbf{k},$$

we obtain

$$\mathbf{a} + \mathbf{b} = (a_x + b_x)\mathbf{i} + (a_y + b_y)\mathbf{j} + (a_z + b_z)\mathbf{k},$$

which shows that the components of the sum are obtained by adding the corresponding components of the two vectors.

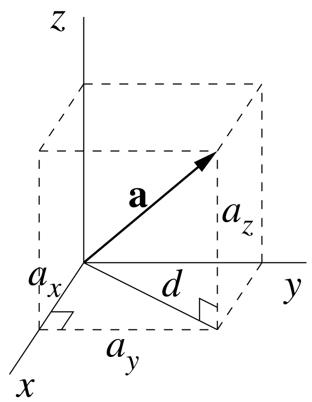
Multiplying vector $\mathbf{a}$ by a number λ, we have

$$\lambda\mathbf{a} = \lambda(a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}) = \lambda a_x\mathbf{i} + \lambda a_y\mathbf{j} + \lambda a_z\mathbf{k},$$

so the components of $\mathbf{a}$ are multiplied by this number. [We use (6.2) here.]

The length of vector $\mathbf{a}$ is given by the three-dimensional analogue of Pythagoras's theorem:

$$a \equiv |\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}. \tag{6.11}$$



Proof. Let d be the diagonal of the base of the rectangular prism containing vector $\mathbf{a}$ (see diagram). Vector $\mathbf{a}$ is then the hypotenuse of the right-angle triangle with short sides d and a_z, so that

$$a^2 = d^2 + a_z^2.$$

In turn, d is the hypotenuse of the right-angle triangle with sides a_x and a_y, so that $d^2 = a_x^2 + a_y^2$. Substituting this in the above, we have

$$a^2 = a_x^2 + a_y^2 + a_z^2,$$

which gives (6.11).

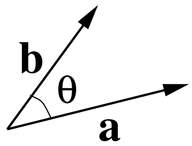
Note that vectors are sometimes shown by listing their components in brackets, as $\mathbf{a} = (a_x, a_y, a_z)$. For example, $\mathbf{a} = (2, -3, 1)$ means $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$.

6.6 Scalar product

Definition. The *scalar product* $\mathbf{a} \cdot \mathbf{b}$ of two vectors $\mathbf{a}$ and $\mathbf{b}$ is a number given by

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta, \tag{6.12}$$

where θ is the angle between the vectors.



Because of its notation, the scalar product is also known as the *dot product.* The right-hand of (6.12) can also be written as $ab\cos\theta$.

The angle θ between two vectors is always in the range

$$0 \le \theta \le \pi.$$

For $0 \le \theta < \frac{\pi}{2}$, $\cos\theta > 0$, so that $\mathbf{a} \cdot \mathbf{b} > 0$ (for nonzero $\mathbf{a}$ and $\mathbf{b}$).

For $\frac{\pi}{2} < \theta \le \pi$, $\cos\theta < 0$, so that $\mathbf{a} \cdot \mathbf{b} < 0$.

Finally, for $\theta = \frac{\pi}{2}$, $\cos\frac{\pi}{2} = 0$, and we have $\mathbf{a} \cdot \mathbf{b} = 0$.

This shows that the scalar product of two *nonzero* vectors can be zero if the vectors are at right angles to each other. The latter property is in fact very useful for checking if two vectors are perpendicular, or for stating this fact.

There is a useful alternative view of the scalar product. Let vector $\mathbf{a}$ be along the x axis. Considering $\mathbf{a} \cdot \mathbf{b} = ab\cos\theta$ in this case, we see that $b\cos\theta = b_x$, which is the x component of vector $\mathbf{b}$ (or *projection* of $\mathbf{b}$ on x). Hence

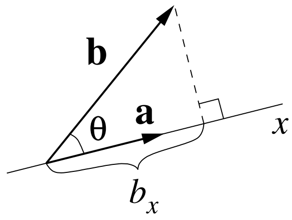
$$\mathbf{a} \cdot \mathbf{b} = ab_x, \tag{6.13}$$

which shows that the scalar product of two vectors equals the length of one vector times the projection of the other vector onto the direction of the first.

Properties of the scalar product.

1. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$, i.e., the scalar product is *commutative.*

   This follows immediately from the definition (6.12).

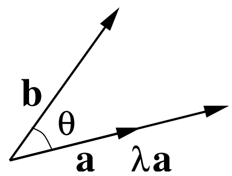


2. $\mathbf{a}\cdot\mathbf{a} \equiv \mathbf{a}^2 = a^2$, i.e., the square of a vector equals its length squared.

   Indeed, from the definition, $\mathbf{a}\cdot\mathbf{a} = |\mathbf{a}||\mathbf{a}|\cos 0 = |\mathbf{a}|^2$.

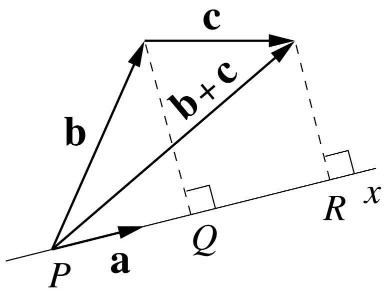
3. $(\lambda\mathbf{a})\cdot\mathbf{b} = \lambda\mathbf{a}\cdot\mathbf{b}$.

   Proof. For $\lambda > 0$, $\lambda\mathbf{a}$ has the same direction as $\mathbf{a}$ and is λ times longer (see diagram). Hence,
   $$(\lambda\mathbf{a})\cdot\mathbf{b} = |\lambda\mathbf{a}||\mathbf{b}|\cos\theta = \lambda|\mathbf{a}||\mathbf{b}|\cos\theta = \lambda\mathbf{a}\cdot\mathbf{b}.$$



   For $\lambda < 0$, $\lambda\mathbf{a}$ is in the opposite direction to $\mathbf{a}$. It makes angle $\pi - \theta$ with $\mathbf{b}$ and its length is $|\lambda||\mathbf{a}|$.[56] So, we have
   $$(\lambda\mathbf{a})\cdot\mathbf{b} = |\lambda\mathbf{a}||\mathbf{b}|\cos(\pi-\theta) = -|\lambda||\mathbf{a}||\mathbf{b}|\cos\theta = \lambda\mathbf{a}\cdot\mathbf{b},$$
   where we used $\cos(\pi-\theta) = -\cos\theta$ and $-|\lambda| = \lambda$ for $\lambda < 0$.

4. $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) = \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{c}$, i.e., the scalar product is *distributive*.



   Proof. Let us choose the x axis in the direction of vector $\mathbf{a}$. According to (6.13), $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) = a|PR|$, where PR is the projection of $\mathbf{b}+\mathbf{c}$ onto the direction of $\mathbf{a}$. From the diagram, $|PR| = |PQ| + |QR|$, so we have
   $$\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) = a(|PQ|+|QR|) = a|PQ| + a|QR| = \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{c},$$

[56] As an exercise, provide a diagram showing vectors $\mathbf{a}$, $\mathbf{b}$, $\lambda\mathbf{a}$ and the relevant angles for $\lambda < 0$.

where we use (6.13) and the fact that $|PQ|$ and $|QR|$ are the projections of $\mathbf{b}$ and $\mathbf{c}$, respectively, onto the direction of $\mathbf{a}$.

Note that properties 1, 3 and 4 look similar to the rules of multiplication of numbers. However, the scalar multiplication of vectors is a new operation, so one cannot simply assume that they hold and it is necessary to verify this.

5. Scalar product of basis vectors:

$$\mathbf{i}\cdot\mathbf{i} = \mathbf{j}\cdot\mathbf{j} = \mathbf{k}\cdot\mathbf{k} = 1, \qquad \mathbf{i}\cdot\mathbf{j} = \mathbf{j}\cdot\mathbf{k} = \mathbf{k}\cdot\mathbf{i} = 0, \tag{6.14}$$

   since $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are unit vectors perpendicular to each other.

6. Scalar product in coordinates.

   For two vectors $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$ and $\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j} + b_z\mathbf{k}$, we have

$$\mathbf{a}\cdot\mathbf{b} = a_xb_x + a_yb_y + a_zb_z. \tag{6.15}$$

   This is derived by expanding the product $\mathbf{a}\cdot\mathbf{b}$ (property 4) and using properties 1 and, crucially, 5.

The scalar product is useful for finding the angle between two vectors, especially when they given in components. From the definition (6.12),

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{ab}, \tag{6.16}$$

and using equations (6.11) and (6.15), we have

$$\cos\theta = \frac{a_xb_x + a_yb_y + a_zb_z}{\sqrt{a_x^2 + a_y^2 + a_z^2}\sqrt{b_x^2 + b_y^2 + b_z^2}}\,. \tag{6.17}$$

In particular, if

$$a_xb_x + a_yb_y + a_zb_z = 0$$

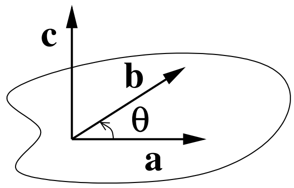
then the angle between the vecors is $\pi/2$ or $90°$, i.e., they are orthogonal.[57]

[57] The scalar product can also be used for finding the angle beteen two vectors in n dimensional space. If the vectors are given in terms of their components, $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, then

$$\cos\theta = \frac{\sum_{i=1}^{n} a_ib_i}{\sqrt{\sum_{i=1}^{n} a_i^2}\sqrt{\sum_{i=1}^{n} b_i^2}}.$$

6.7 Vector product

Definition. The *vector product* of two vectors $\mathbf{a}$ and $\mathbf{b}$ is a vector $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ such that



(1) vector $\mathbf{c}$ is perpendicular to the plane contaning $\mathbf{a}$ and $\mathbf{b}$;

(2) the magnitude of $\mathbf{c}$ is $c = ab\sin\theta$;

(3) the direction of $\mathbf{c}$ is such that, viewed from its "top", the rotation from $\mathbf{a}$ towards $\mathbf{b}$ is anticlockwise (i.e., in the positive direction).

Because of the notation used, the vector product $\mathbf{a}\times\mathbf{b}$ is also known as the *cross product*.

Properties of the vector product.

1. If vectors $\mathbf{a}$ and $\mathbf{b}$ are collinear then $\mathbf{a} \times \mathbf{b} = 0$. In particular, $\mathbf{a} \times \mathbf{a} = 0$.

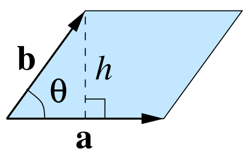
   Indeed, for collinear vectors $\theta = 0$ or $\theta = \pi$, so that $\sin\theta = 0$ and $c = 0$.

   This property shows that the definition consistent. Two collinear vectors do not define a plane, so the direction of $\mathbf{c}$ is not fixed, but then $\mathbf{c} = 0$.

2. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$, i.e., vector multiplication is *anticommutative*.

   The two vectors $\mathbf{a} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{a}$ have the same magnitude and the plane defined by $\mathbf{a}$ and $\mathbf{b}$ is the same. However, for $\mathbf{b} \times \mathbf{a}$ the rotation from $\mathbf{b}$ to $\mathbf{a}$ must be seen as anticlockwise, so the direction of this vector is the opposite of $\mathbf{a} \times \mathbf{b}$.

3. The magnitude of $\mathbf{a} \times \mathbf{b}$ gives the *area of the parallelogram* with sides $\mathbf{a}$ and $\mathbf{b}$.

   We see that $b\sin\theta = h$, the height of the parallelogram, so that

   $$|\mathbf{a} \times \mathbf{b}| = ab\sin\theta = ah = A,$$

   which is the area of the parallelogram.

4. $(\lambda\mathbf{a}) \times \mathbf{b} = \lambda\mathbf{a} \times \mathbf{b}$.

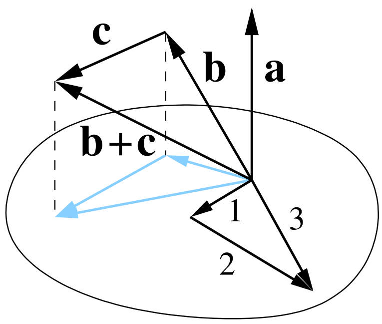
   Proof. The proof for $\lambda > 0$ is trivial. Multiplying $\mathbf{a}$ by $\lambda > 0$ makes this vector λ times longer without changing its direction. This makes the vector product λ times longer too, and in the same direction as $\mathbf{a} \times \mathbf{b}$.

For $\lambda < 0$, $\lambda\mathbf{a}$ is $|\lambda|$ times longer than $\mathbf{a}$ and is in the opposite direction. The angle between $\mathbf{b}$ and $\lambda\mathbf{a}$ is $\pi - \theta$, so we have for the magnitude

$$|(\lambda\mathbf{a}) \times \mathbf{b}| = |\lambda\mathbf{a}||\mathbf{b}| \sin(\pi - \theta) = |\lambda||\mathbf{a}||\mathbf{b}| \sin\theta = |\lambda||\mathbf{a} \times \mathbf{b}|.$$

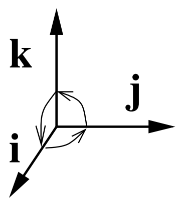
The plane formed by $\lambda\mathbf{a}$ and $\mathbf{b}$ is the same as that of $\mathbf{a}$ and $\mathbf{b}$. However, to see the rotation from $\lambda\mathbf{a}$ (with $\lambda < 0$) towards $\mathbf{b}$ as anticlockwise, one must be looking at the plane from the other side. Hence, the direction of $(\lambda\mathbf{a}) \times \mathbf{b}$ is the opposite to that of $\mathbf{a} \times \mathbf{b}$, which completes the proof.[58]

5. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$, i.e., vector product is *distributive*.



Because of property 4, it is sufficient to prove this for a unit vector $\mathbf{a}$. The proof is then based on the observation that the cross product of vector $\mathbf{a}$ with another vector is given by the projection of this other vector on the plane perpendicular to $\mathbf{a}$, rotated through 90° anticlockwise about $\mathbf{a}$. The relation then holds because the projection of the sum $\mathbf{b} + \mathbf{c}$ onto this plane is equal to the sum of the projections of $\mathbf{b}$ and $\mathbf{c}$. The proof sketched here is illustrated by the diagram in which vectors $\mathbf{a} \times \mathbf{b}$, $\mathbf{a} \times \mathbf{c}$ and $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$ are labelled by numbers 1, 2 and 3, respectively.

6. Cross products of basis vectors.



By property 1, products of identical vectors vanish. Other products are unit vectors (since $\sin\pi/2 = 1$) with appropriate directions, so that

$$\begin{gathered} \mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = 0, \\ \mathbf{i} \times \mathbf{j} = \mathbf{k}, \qquad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \qquad \mathbf{k} \times \mathbf{i} = \mathbf{j}, \\ \mathbf{j} \times \mathbf{i} = -\mathbf{k}, \quad \mathbf{k} \times \mathbf{j} = -\mathbf{i}, \quad \mathbf{i} \times \mathbf{k} = -\mathbf{j}. \end{gathered} \tag{6.18}$$

[58] As an exercise, provide a diagram showing the vectors involved in the proof for $\lambda < 0$.

Note that the relations in the top line and bottom line can be obtained from each other by *circular permutations* $\begin{array}{ccc}\mathbf{i} & \longrightarrow & \mathbf{j}\\ & \nwarrow \mathbf{k} \swarrow & \end{array}$, so it is sufficient to remember one of them, e.g., $\mathbf{i}\times\mathbf{j}=\mathbf{k}$.

7. Vector product in coordinates.

   Multiplying $\mathbf{a}=a_x\mathbf{i}+a_y\mathbf{j}+a_z\mathbf{k}$ and $\mathbf{b}=b_x\mathbf{i}+b_y\mathbf{j}+b_z\mathbf{k}$, we obtain

$$\begin{aligned}\mathbf{a}\times\mathbf{b}&=(a_x\mathbf{i}+a_y\mathbf{j}+a_z\mathbf{k})\times(b_x\mathbf{i}+b_y\mathbf{j}+b_z\mathbf{k})\\&=(a_yb_z-a_zb_y)\mathbf{i}+(a_zb_x-a_xb_z)\mathbf{j}+(a_xb_y-a_yb_x)\mathbf{k},\end{aligned}\tag{6.19}$$

   where we used property 5 to expand the product together with products of unit vectors (6.18). Hence, the components of $\mathbf{c}=\mathbf{a}\times\mathbf{b}$ are

$$\begin{aligned}c_x&=a_yb_z-a_zb_y,\\c_y&=a_zb_x-a_xb_z,\\c_z&=a_xb_y-a_yb_x.\end{aligned}\tag{6.20}$$

   These expressions are highly symmetric. Any component of $\mathbf{c}$ contains only other components of $\mathbf{a}$ and $\mathbf{b}$ (e.g., c_x depends on a_y and b_z). Also, the positive and negative terms on the right-hand sides differ by interchanging the component labels (e.g., y and z for c_x). Equations (6.20) can be obtained from each other by cyclic permutations $\begin{array}{ccc}x & \longrightarrow & y\\ & \nwarrow z \swarrow & \end{array}$.

Why is the vector product useful?

It can be used for finding a vector perpendicular to two given vectors.

Example. Find a unit normal[59] to the plane through three points $A(-1,0,1)$, $B(1,2,3)$ and $C(0,-1,-1)$.

Vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ lie in the plane. Vector $\mathbf{s}=\overrightarrow{AB}\times\overrightarrow{AC}$ is perpendicular to them, so it is perpendicular to the plane. Using

$$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=(\mathbf{i}+2\mathbf{j}+3\mathbf{k})-(-\mathbf{i}+\mathbf{k})=2\mathbf{i}+2\mathbf{j}+2\mathbf{k},$$

and

$$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=(-\mathbf{j}-\mathbf{k})-(-\mathbf{i}+\mathbf{k})=\mathbf{i}-\mathbf{j}-2\mathbf{k},$$

we obtain

[59] A *normal* is a vector perpendicular to a plane or, locally, to any smooth surface.

$$
\begin{aligned}
\mathbf{s} &= (2\mathbf{i}+2\mathbf{j}+2\mathbf{k})\times(\mathbf{i}-\mathbf{j}-2\mathbf{k})\\
&= (-4+2)\mathbf{i}+(2+4)\mathbf{j}+(-2-2)\mathbf{k}\\
&= -2\mathbf{i}+6\mathbf{j}-4\mathbf{k},
\end{aligned}
$$

where we used equations (6.20). (As a check, verify that $\overrightarrow{AB}\ \cdot \mathbf{s} =\overrightarrow{AC}\ \cdot \mathbf{s} = 0$.) The length of $\mathbf{s}$ is

$$|\mathbf{s}| = \sqrt{(-2)^2+6^2+(-4)^2} = \sqrt{56} = 2\sqrt{14}.$$

so the unit normal to the plane is

$$\frac{\mathbf{s}}{|\mathbf{s}|} = \frac{-2\mathbf{i}+6\mathbf{j}-4\mathbf{k}}{2\sqrt{14}} = \frac{-\mathbf{i}+3\mathbf{j}-2\mathbf{k}}{\sqrt{14}}.$$

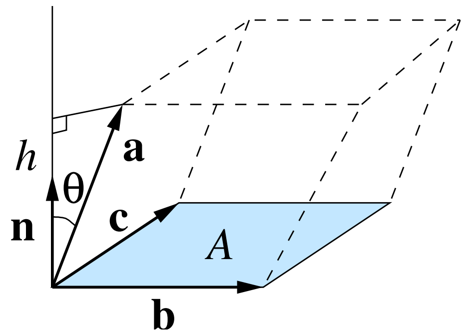
6.8 Triple scalar product

Definition. The triple scalar product of vectors **a**, **a** and **c** is

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}). \tag{6.21}$$

Note that this quantity is a number.

Theorem. Geometrically, the absolute value of the triple scalar product (6.21) gives the volume of the parallelepiped[60] with the edges given by the vectors **a**, **b** and **c**.



Proof. The magnitude of vector $\mathbf{b}\times\mathbf{c}$ gives the area A of the parallelogram at the base of the parallelepiped. Its direction is perpendicular to both **b** and **c**, i.e., perpendicular to the base of the parallelepiped. Let **n** be a unit vector in this direction. We can then write $\mathbf{b}\times\mathbf{c} = A\mathbf{n}$, and the triple scalar product is

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \mathbf{a}\cdot\mathbf{n}A$$

[60] A *parallelepiped* is a "three-dimensional parallelogram". A parallelogram is a shape with two pairs or parallel sides. A parallelepiped is a solid with three pairs of parallel faces which are all parallelograms.

Here $\mathbf{a} \cdot \mathbf{n} = a \cos\theta = h$, where θ is the angle between $\mathbf{a}$ and $\mathbf{n}$ and h is the height of the parallelepiped (perpendicular to the base).

Thus we see that the triple scalar product gives

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = Ah = V,$$

which is the volume of the parallelepiped.

The triple scalar product can be negative, so in general the volume of the parallelepiped is given by the absolute value of $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

Apart from the possible minus sign, the volume of the parallelepiped does not depend on the order in which we list its edges. Swapping two vectors gives

$$\mathbf{a} \cdot (\mathbf{c} \times \mathbf{b}) = -\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}),$$

while *circular* permutations $\begin{array}{ccc}\mathbf{a} & \longrightarrow & \mathbf{b} \\ \nwarrow & \mathbf{c} & \swarrow\end{array}$ leave the triple scalar product unchanged:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}).$$

Triple scalar product in coordinates.

Using expressions (6.15) and (6.20) for the scalar and vector products in coordinates, we obtain

$$\begin{aligned}\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= a_x(\mathbf{b} \times \mathbf{c})_x + a_y(\mathbf{b} \times \mathbf{c})_y + a_z(\mathbf{b} \times \mathbf{c})_z \\ &= a_x(b_y c_z - b_z c_y) + a_y(b_z c_x - b_x c_z) + a_z(b_x c_y - b_y c_x).\end{aligned} \tag{6.22}$$

A complete expansion of the last line produces six terms, three with plus signs and three with minus signs. Each of these terms contains one component from each of the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, and all three component indices x, y and z in some order. The term $a_x b_y c_z$ enters with a plus sign. All terms obtained from it by swapping any of the component indices have minuses, e.g., for $x \rightleftarrows y$ the term in (6.22) is $-a_y b_x c_z$. On the other hand, terms in which two such swaps are performed (which is equivalent to a circular permutation of x, y and z) have plus signs. Finally, note that the six terms correspond to all $6 = 3!$ possible permutations of three objects.

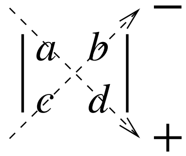
Triple scalar product as a determinant.

The permutation properties of the expression (6.22) for the triple scalar product ara characteristic of the mathematical objects known as *determinants*.

A determinant is a square table of numbers which is evaluated according to certain rules. Below we look at 2×2 and 3×3 determinants.

Definition. The 2×2 determinant is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc. \tag{6.23}$$



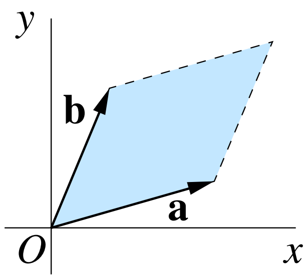
Note how the product of a and d enters with the plus sign and that of b and c with a minus.

Recall that we already met a 2×2 determinant, the Wronskian in (5.52).

Consider the cross product of vectors $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j}$ and $\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j}$ in the x-y plane. Using (6.19), we have

$$\mathbf{a} \times \mathbf{b} = (a_x b_y - a_y b_x)\mathbf{k}, \tag{6.24}$$

which is a vector of length $a_x b_y - a_y b_x$ along the z axis. On the other hand, the magnitude of the vector product $\mathbf{a} \times \mathbf{b}$ is the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$ (Sec. 6.7, property 3).



Comparing (6.24) with (6.23), we see that the determinant

$$\begin{vmatrix} a_x & a_y \\ b_x & b_y \end{vmatrix} = a_x b_y - a_y b_x, \tag{6.25}$$

gives the area of the parallelogram with sides $\mathbf{a} = (a_x, a_y)$ and $\mathbf{b} = (b_x, b_y)$.[61]

Note that if the vectors $\mathbf{a}$ and $\mathbf{b}$ are collinear, i.e., linearly dependent, the area of the parallelogram equals zero and the determinant (6.25) vanishes.

Definition. The 3×3 determinant can be defined by its expansion in the top row

[61] The determinant (6.25) provides the so-called *oriented area*, i.e., it can be positive or negative, depending on the mutual orientation of $\mathbf{a}$ and $\mathbf{b}$.

$$\begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} = a_x \begin{vmatrix} b_y & b_z \\ c_y & c_z \end{vmatrix} - a_y \begin{vmatrix} b_x & b_z \\ c_x & c_z \end{vmatrix} + a_z \begin{vmatrix} b_x & b_y \\ c_x & c_y \end{vmatrix}, \tag{6.26}$$

where each element of the first row is multiplied by the 2×2 determinant which remains after crossing out the first row and the corresponding column.

Note that in the expansion (6.26) the signs on the right-hand side *alternate*, which is an important part of the rule.[62]

Expanding the 2×2 determinants in (6.26) according to (6.23), one obtains

$$\begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} = a_x(b_y c_z - b_z c_y) - a_y(b_x c_z - b_z c_x) + a_z(b_x c_y - b_y c_x).$$

Comparing this with (6.22), we see that

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}. \tag{6.27}$$

This relation means that the 3×3 determinant gives the (oriented) volume of the parallelepiped with edges whose components appear in its rows. It also provides a quick way of calculating the triple scalar product using (6.26).

Theorem. If vector $\mathbf{c}$ is a linear combination of vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{c} = \lambda\mathbf{a} + \mu\mathbf{b}$ (i.e., these three vectors are linearly dependent), the determinant of their components equals zero.

<u>Proof.</u> Of course, this can be verified directly by writing vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ in terms of their components $a_x, \ldots, b_x, \ldots, c_x = \lambda a_x + \mu b_x, \ldots$, and substituting these in (6.26) to show that the result is zero. On the other hand, we know that vector $\mathbf{c} = \lambda\mathbf{a} + \mu\mathbf{b}$ is in the same plane as $\mathbf{a}$ and $\mathbf{b}$. Hence, the parallelepiped with these edges is "flat" and its volume is zero!

This theorem means that we can check that three vectors are linearly independent, by evaluating the corresponding determinant (or, equivalently, the triple scalar product) and verifying that it is nonzero.

The 3×3 determinant (6.26) also offers an alternative way of finding the vector product of two vectors given in terms of their components:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y)\mathbf{i} - (a_x b_z - a_z b_y)\mathbf{i} + (a_x b_y - a_y b_x)\mathbf{k}, \tag{6.28}$$

[62] In a similar way one can define the determinant of any size n by its expansion involving the elements of the first row and determinants of size $(n-1)$ with alternating signs.

which is the same as (6.19). [Note the minus sign in front of the second term above which matches the reverse order of the items in brackets.]

6.9 Triple vector product

Definition. The triple vector product of vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ is

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}). \tag{6.29}$$

The triple vector product (6.29) is a vector. It can be evaluated directly. However, in many cases it easier to expand it using the identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}), \tag{6.30}$$

which is known as the *'bac(k) minus cab'* rule.

To verify (6.30), consider the x component of the vector on the left-hand side:

$$\begin{aligned}
[\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_x &= a_y(\mathbf{b} \times \mathbf{c})_z - a_z(\mathbf{b} \times \mathbf{c})_y \\
&= a_y(b_x c_y - b_y c_x) - a_z(b_z c_x - b_x c_z) \\
&= b_x(a_y c_y + a_z c_z) - c_x(a_y b_y + a_z b_z) \\
&= b_x(a_x c_x + a_y c_y + a_z c_z) - c_x(a_x b_x + a_y b_y + a_z b_z) \\
&= b_x(\mathbf{a} \cdot \mathbf{c}) - c_x(\mathbf{a} \cdot \mathbf{b}),
\end{aligned}$$

which is the x component of the expression on the right-hand side of (6.30). Similarly one can show that the y and z components of the two sides are identical, which completes the proof.

Note that the triple vector product *is not* associative, i.e., in general

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}. \tag{6.31}$$

Indeed, transforming the right-hand side and using (6.30), we have

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = -\mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{c} \times (\mathbf{b} \times \mathbf{a}) = \mathbf{b}(\mathbf{c} \cdot \mathbf{a}) - \mathbf{a}(\mathbf{c} \cdot \mathbf{b}).$$

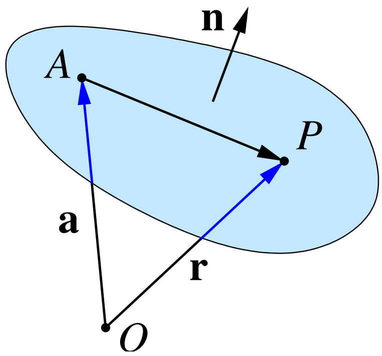
This shows that the vector on right-hand side of (6.31) is a linear combination of vectors $\mathbf{b}$ and $\mathbf{a}$, while the vector on the left-hand side of (6.31) is a linear combination of vectors $\mathbf{b}$ and $\mathbf{c}$, according to (6.30).

6.10 Equation of a plane

In the x-y plane, $ax + by = c$ is the equation of a straigt line (Sec. 2.3). In this section we show that a similar equation $ax + by + cz = d$ describes a plane in three-dimensional space.

We start by deriving the equation of a plane in vector form.

A plane can be defined by providing a point (say A) which it contains and a vector $\mathbf{n}$ *normal* to the plane.



Let point A be given by its position vector $\overrightarrow{OA} = \mathbf{a}$ relative to the origin O. For any other point P in the plane, vector $\overrightarrow{AP}$ lies entirely in the plane, and is hence perpendicular to $\mathbf{n}$. This can be written using the scalar product as

$$\overrightarrow{AP} \cdot \mathbf{n} = 0. \tag{6.32}$$

If the position vector of P is $\overrightarrow{OP} = \mathbf{r}$ then $\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \mathbf{r} - \mathbf{a}$, and (6.32) gives

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0.$$

Thus we have obtained the *vector equation* of the plane

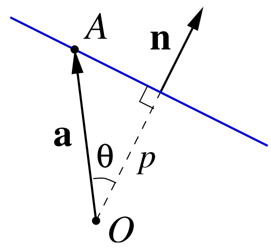
$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}, \tag{6.33}$$

which is satisfied by all points $\mathbf{r}$ in the plane.

If $\mathbf{n}$ is a unit vector, $|\mathbf{n}| = 1$, the right-hand side of (6.33) has a particularly simple geometrical meaning. To find it, let us take the side view of the plane. The diagram shows that $\mathbf{a} \cdot \mathbf{n} = a \cos\theta = p$, where θ is the angle between vectors $\mathbf{a}$ and $\mathbf{n}$ and p is the distance from the the origin to the plane.[63]

We thus see that for the *unit* normal $\mathbf{n}$, the scalar product

[63] The distance is the *shortest* distance measured along the perpendicular.



$$\overrightarrow{OA} \cdot \mathbf{n} \tag{6.34}$$

gives the distance from the origin to the plane.

Note that the scalar product (6.34) can be negative (e.g., if the normal is chosen in the opposite direction), so the distance from O to the plane is given by the absolute value of (6.34). For $\overrightarrow{OA} \cdot \mathbf{n} > 0$ the normal points away from the origin, while for $\overrightarrow{OA} \cdot \mathbf{n} < 0$ the normal points towards it.

Similarly, the distance from any other point B to the plane can be found as $|\overrightarrow{BA} \cdot \mathbf{n}|$, where A is a point in the plane and $\mathbf{n}$ is a unit normal to the plane.

Let us now find the equation of the plane in coordinates. Using the position vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and assuming that the normal $\mathbf{n} = n_x\mathbf{i} + n_y\mathbf{j} + n_z\mathbf{k}$ is a unit vector ($\sqrt{n_x^2 + n_y^2 + n_z^2} = 1$) we obtain from (6.33)

$$n_x x + n_y y + n_z z = p, \tag{6.35}$$

where (the absolute value of) p is the distance from the origin to the plane.

In fact, any equation of the form

$$ax + by + cz = d \tag{6.36}$$

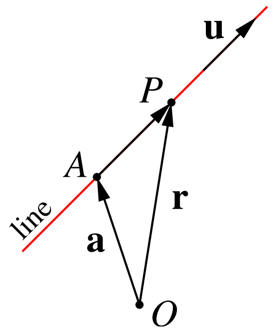
describes a plane with the normal (a, b, c). Dividing (6.36) by $\sqrt{a^2 + b^2 + c^2}$, we obtain

$$\underbrace{\frac{a}{\sqrt{a^2 + b^2 + c^2}}}_{n_x} x + \underbrace{\frac{b}{\sqrt{a^2 + b^2 + c^2}}}_{n_y} y + \underbrace{\frac{c}{\sqrt{a^2 + b^2 + c^2}}}_{n_z} z = \frac{d}{\sqrt{a^2 + b^2 + c^2}}\,.$$

Here the coefficients on the left-hand side are the components of a unit normal and the right-hand side gives the distance from the origin to the plane.

6.11 Equation of a straight line

Let us derive the equation of a straight line through a given point A ($\overrightarrow{OA} = \mathbf{a}$) in the direction of vector $\mathbf{u}$. For any point P on the line, the vector $\overrightarrow{AP}$ is collinear with $\mathbf{u}$, so



that $\overrightarrow{AP} = \lambda\mathbf{u}$. Here λ is a real number which can take values from $-\infty$ to ∞, making point P slide along the line.

Introducing the position vector $\overrightarrow{OP} = \mathbf{r}$, we have from $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$,

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{u} \quad (\lambda \in \mathbb{R}). \tag{6.37}$$

Equation (6.37) is the vector equation of the line.

To find what form it takes in coordinates, we substitute

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}, \quad \mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}, \quad \mathbf{u} = u_x\mathbf{i} + u_y\mathbf{j} + u_z\mathbf{k}$$

into (6.37) and obtain

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k} + \lambda(u_x\mathbf{i} + u_y\mathbf{j} + u_z\mathbf{k}).$$

Comparing the x, y and z components on both sides then gives

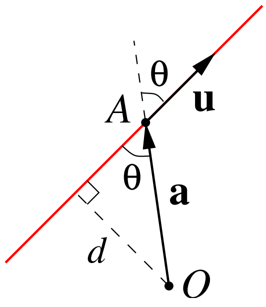
$$x = a_x + \lambda u_x, \quad y = a_y + \lambda u_y, \quad z = a_z + \lambda u_z.$$

If we solve these three equations for λ and equate the right-hand sides, we have

$$\frac{x - a_x}{u_x} = \frac{y - a_y}{u_y} = \frac{z - a_z}{u_z}. \tag{6.38}$$

The solutions (x, y, z) to these equation give the coordinates of all points on the line.

The distance d from the origin to the line is measured along the perpendicular to the line. The diagram shows that $d = |OA|\sin\theta$, where θ is the angle between $\overrightarrow{OA} = \mathbf{a}$ and $\mathbf{u}$. If $\mathbf{u}$ is a *unit* vector, then $|OA|\sin\theta$ is the magnitude of the vector product of the vectors $\overrightarrow{OA}$ and $\mathbf{u}$, see Sec. 6.7. Hence, the distance from the origin O to the line is given by

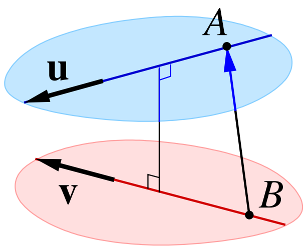


$$d = |\overrightarrow{OA} \times \mathbf{u}|. \tag{6.39}$$

In a similar way we can find the distance from any point to the line, and if $\mathbf{u}$ is not a unit vector, we will divide $\mathbf{u}$ it by its length. Thus, in general, the distance from point B to the line is

$$\frac{|\overrightarrow{BA} \times \mathbf{u}|}{|\mathbf{u}|}, \tag{6.40}$$

where A is any point on the line.



Finally, let us find the distance between skew lines[64] given by the equations

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{u} \quad (\lambda \in \mathbb{R}), \qquad \mathbf{r} = \mathbf{b} + \mu\mathbf{v} \quad (\mu \in \mathbb{R}),$$

one through point $\mathbf{a}$ with direction $\mathbf{u}$, the other through $\mathbf{b}$ with direction $\mathbf{v}$.

This distance must be measured along the line perpendicular to the given lines. Its direction is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, and is given by $\mathbf{u} \times \mathbf{v}$. Hence we find the distance by projecting vector $\overrightarrow{BA} = \mathbf{a} - \mathbf{b}$, which connects two points on the lines, onto the unit vector $(\mathbf{u} \times \mathbf{v})/|\mathbf{u} \times \mathbf{v}|$:

$$\frac{(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{u} \times \mathbf{v})}{|\mathbf{u} \times \mathbf{v}|}.$$

[64] Skew lines are lines that are not parallel and do not intersect. They lie in parallel planes.

7 Matrices

7.1 Matrices and Matrix Operations

Rectangular arrays of real numbers arise in contexts other than augmented matrices for linear systems. In this section we will begin to study matrices as objects in their own right by defining operations of addition, subtraction, and multiplication on them.

7.1.1 Matrix Notation and Terminology

Definition 1 A matrix is a rectangular array of numbers. The numbers in the array are called the entries in the matrix.

A matrix with only one column is called a *column vector* or a column matrix, and a matrix with only one row is called a *row vector* or a row matrix. In Example 7.1, the 2×1 matrix is a column vector, the 1×4 matrix is a row vector, and the 1×1 matrix is both a row vector and a column vector.

Example 7.1 (Examples of Matrices)
Some examples of matrices are:

$$\begin{pmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & e & \sqrt{2} \\ 3 & -1 & x \\ 2 & -2 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 3 \end{pmatrix}.$$

The *size* of a matrix is described in terms of the number of rows (horizontal lines) and columns (vertical lines) it contains. For example, the first matrix in Example 7.1 has three rows and two columns, so its size is 3 by 2 (written 3×2). In a size description, the first number always denotes the number of rows, and the second denotes the number of columns. The remaining matrices in Example 7.1 have sizes $1 \times 4, 3 \times 3, 2 \times 1$, and 1×1, respectively. We will use capital letters to denote matrices and lowercase letters to denote numerical quantities; thus we might write:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & -2 \\ 1 & -1 \end{pmatrix} \quad \text{or} \quad C = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The entry that occurs in row i and column j of a matrix A will be denoted by a_{ij}. Thus a general 3×4 matrix might be written as

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

and a general $m \times n$ matrix as

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

When a compact notation is desired, the preceding matrix can be written as $(a_{ij})_{m\times n}$ or simply (a_{ij}). The entry in row i and column j of a matrix A is also commonly denoted by the symbol $(A)_{ij}$.

A matrix A with n rows and n columns is called a *square matrix* and the entries $a_{11}, a_{22}, \ldots a_{nn}$ form the *main diagonal* of A.

7.1.2 Operations on Matrices

We can use matrices to abbreviate the work in solving systems of linear equations. For other applications, however, it is desirable to develop an "arithmetic of matrices" in which matrices can be added, subtracted, and multiplied in a useful way. The remainder of this section will be devoted to developing this arithmetic.

Definition 2 (Equal Matrices) Two matrices are defined to be *equal* if they have the same size and their corresponding entries are equal.

The equality of two matrices $A = (a_{ij})$ and $B = (b_{ij})$ of the same size can be expressed either by writing

$$(A)_{ij} = (B)_{ij}$$

or by writing $a_{ij} = b_{ij}$ for any i and j. In this case we write $A = B$.

Definition 3 (Sum and Difference of Matrices) If A and B are matrices of the same size, then the *sum* $A + B$ is the matrix obtained by adding the entries of B to the corresponding entries of A, and the difference $A - B$ is the matrix obtained by subtracting the entries of B from the corresponding entries of A. Matrices of different sizes cannot be added or subtracted.

In matrix notation, if $A = (a_{ij})$ and $B = (b_{ij})$ have the same size, then

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad \text{and} \quad (A - B)_{ij} = a_{ij} - b_{ij}$$

Example 7.2 Consider the matrices:

$$A = \begin{pmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 2 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$$

Then

$$A+B=\begin{pmatrix} 1 & -2 & 1 \\ 1 & 3 & 2 \end{pmatrix}, \quad A-B=\begin{pmatrix} 1 & -4 & 3 \\ -1 & -1 & -4 \end{pmatrix},$$

while the expressions $A+C, B+C, A-C$ and $B-C$ are undefined.

Definition 4 (Multiplication by a Scalar)

If A is any matrix and c is any scalar, then the product cA is the matrix obtained by multiplying each entry of the matrix A by c.

In matrix notation, if $A=(a_{ij})$, then $(cA)_{ij}=c(A)_{ij}=ca_{ij}$.

Example 7.3
Let:

$$A=\begin{pmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix}$$

Then

$$2A=\begin{pmatrix} 2 & -6 & 4 \\ 0 & 2 & -2 \end{pmatrix}, \quad (-1)A=-A=\begin{pmatrix} -1 & 3 & -2 \\ 0 & -1 & 1 \end{pmatrix}.$$

Thus far we have defined multiplication of a matrix by a scalar but not the multiplication of two matrices. Since matrices are added by adding corresponding entries and subtracted by subtracting corresponding entries, it would seem natural to define multiplication of matrices by multiplying corresponding entries. However, it turns out that such a definition would not be very useful for most problems. Experience has led mathematicians to the following more useful definition of matrix multiplication.

Definition 5 (Matrix-Matrix Multiplication)

If A is an $m\times r$ matrix and B is an $r\times n$ matrix, then the product AB is the $m\times n$ matrix whose entries are determined as follows: To find the entry in row i and column j of AB, single out row i from the matrix A and column j from the matrix B. Multiply the corresponding entries from the row and column together, and then add up the resulting products.

Example 7.4 (Multiplying Matrices)

Consider the matrices:

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{pmatrix},$$

Since A is a 2×3 matrix and B is a 3×4 matrix, the product AB is a 2×4 matrix. To determine, for example, the entry in row 2 and column 3 of AB, we single out row 2 from A and column 3 from B. Then, as illustrated below, we multiply corresponding entries together and add up these products:

$$\begin{pmatrix} 1 & 2 & 4 \\ \mathbf{2} & \mathbf{6} & \mathbf{0} \end{pmatrix} \begin{pmatrix} 4 & 1 & \mathbf{4} & 3 \\ 0 & -1 & \mathbf{3} & 1 \\ 2 & 7 & \mathbf{5} & 2 \end{pmatrix} = \begin{pmatrix} * & * & * & * \\ * & * & \mathbf{26} & * \end{pmatrix}$$

$$(2 \cdot 4) + (6 \cdot 3) + (0 \cdot 5) = 26$$

Doing all the calculations we get:

$$AB = \begin{pmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{pmatrix}.$$

The definition of matrix multiplication requires that the number of columns of the first factor A be the same as the number of rows of the second factor B in order to form the product AB. If this condition is not satisfied, the product is undefined. A convenient way to determine whether a product of two matrices is defined is to write down the size of the first factor and, to the right of it, write down the size of the second factor. If, as in 3, the inside numbers are the same, then the product is defined. The outside numbers then give the size of the product:

$$\begin{array}{ccccc} A & & B & & AB \\ m \times r & & r \times n & = & m \times n \end{array}$$

In general if A is an $m \times r$ matrix and B is an $r \times n$ matrix, then:

$$AB = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1r} \\ a_{21} & a_{22} & \cdots & a_{2r} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ir} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mr} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{r1} & b_{r2} & \cdots & b_{rj} & \cdots & b_{rn} \end{pmatrix}$$

the entry $(AB)_{ij}$ in row i and column j of AB is given by

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ir}b_{rj} = \sum_{k=1}^{r} a_{ik}b_{kj}$$

7.1.3 Matrix Products as Linear Combinations

The following definition provides yet another way of thinking about matrix multiplication.

Definition 6 If $A_1, A_2, \dots A_r$ are matrices of the same size, and if $c_1, c_2, \dots c_r$ are scalars, then an expression of the form:

$$c_1 A_1 + c_2 A_2 + \dots c_r A_r$$

is called a **linear combination** of $A_1, A_2, \dots A_r$ with **coefficients** $c_1, c_2, \dots c_r$.

To see how matrix products can be viewed as linear combinations, let A be an $m \times n$ matrix and $\mathbf{x}$ an $n \times 1$ column vector, say

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \text{and} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Then

$$A\mathbf{x} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix} = x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \dots x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

This proves the following theorem.

> **Theorem 7.1** *If A is a matrix, and if $\mathbf{x}$ is an column vector, then the product $A\mathbf{x}$ can be expressed as a linear combination of the column vectors of A in which the coefficients are the entries (components) of x.*

7.1.4 Matrix Form of a Linear System

Matrix multiplication has an important application to systems of linear equations. Consider a system of m linear equations in n unknowns:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \quad &\quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

Since two matrices are equal if and only if their corresponding entries are equal, we can replace the m equations in this system by the single matrix equation:

$$\begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

The $m \times 1$ matrix on the left side of this equation can be written as a product to give:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

If we designate these matrices by A, $\mathbf{x}$, and $\mathbf{b}$, respectively, then we can replace the original system of m equations in n unknowns has been replaced by the single matrix equation:

$$A\mathbf{x} = \mathbf{b}$$

The matrix A in this equation is called the **coefficient matrix** of the system.

7.1.5 Transpose of a Matrix

We conclude this section by defining two matrix operations that have no analogs in the arithmetic of real numbers.

Definition 7 If A is any matrix, then the transpose of A, denoted by A^T, is defined to be the matrix that results by interchanging the rows and columns of A; that is, the first column of A^T is the first row of A, the second column of A^T is the second row of A, and so forth.

Example 7.5 The following are some examples of matrices and their transposes.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 \\ -1 & 2 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 3 & 6 \end{pmatrix}$$

$$A^T = \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{pmatrix}, \quad B^T = \begin{pmatrix} 1 & -1 & 0 \\ 4 & 2 & 1 \end{pmatrix}, \quad C^T = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}$$

Observe that not only are the columns of A^T the rows of A, but the rows of A^T are the columns of A. Thus the entry in row i and column j of A^T is the entry in row j and column i of A; that is,

$$(A^T)_{ij} = (A)_{ji}$$

Note the reversal of the subscripts.

In the special case where A is a square matrix, the transpose of A can be obtained by interchanging entries that are symmetrically positioned about the main diagonal. Below we see that A^T can also be obtained by "reflecting" A about its main diagonal.

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 3 & 1 & 2 \\ -1 & 4 & 3 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 1 & 3 & -1 \\ -2 & 1 & 4 \\ 1 & 2 & 3 \end{pmatrix}$$

7.1.6 Trace of a Matrix

Definition 8 (Trace of a Matrix) If A is a square matrix, then the **trace** of A, denoted by $\mathrm{Tr}(A)$, is defined to be the sum of the entries on the main diagonal of A. The trace of A is undefined if A is not a square matrix.

If A is an $n \times n$ matrix then:

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{ii}$$

Example 7.6

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 3 & 1 & 2 \\ -1 & 4 & 3 \end{pmatrix}, \quad \Rightarrow \mathrm{Tr}(A) = 5$$

7.2 Algebraic properties and inverse matrices

In this section we will discuss some of the algebraic properties of matrix operations. We will see that many of the basic rules of arithmetic for real numbers hold for matrices, but we will also see that some do not.

7.2.1 Properties of Matrix Addition and Scalar Multiplication

The following theorem lists the basic algebraic properties of the matrix operations.

Theorem 7.2 (Properties of Matrix Arithmetic)

Assuming that the sizes of the matrices are such that the indicated operations can be performed, the following rules of matrix arithmetic are valid.

1. $A + B = B + A$ *(***Commutative law for addition***)*
2. $A + (B + C) = (A + B) + C$ *(***Associative law for addition***)*
3. $A(BC) = (AB)C$ *(***Associative law for multiplication***)*
4. $A(B + C) = AB + AC$ *(***Left distributive law***)*
5. $(B + C)A = BA + CA$ *(***Right distributive law***)*
6. $a(B + C) = aB + aC$
7. $(a + b)C = aC + bC$
8. $a(bC) = (ab)C$
9. $a(BC) = (aB)C = B(aC)$

To prove any of the equalities in this theorem we must show that the matrix on the left side has the same size as that on the right and that the corresponding entries on the two sides are the same. Most of the proofs follow the same pattern, so we will prove part *4.* as a sample.

Proof of 4. We must show that $A(B + C)$ and $AB + AC$ have the same size and that corresponding entries are equal. To form $A(B+C)$, the matrices B and C must have the same size, say $m \times n$, and the matrix A must then have m columns, so its size must be of the form $r \times m$. This makes $A(B + C)$ an $r \times n$ matrix and, consequently, $A(B + C)$ and $AB + AC$ have the same size.

To show that $A(B + C)$ and $AB + AC$ have the same entries we proceed as follows:

$$\begin{aligned}
(A(B+C))_{ij} &= a_{i1}(b_{1j} + c_{1j}) + a_{i2}(b_{2j} + c_{2j}) + \cdots + a_{im}(b_{mj} + c_{mj}) = \\
&= (a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}) + (a_{i1}c_{1j} + a_{i2}c_{2j} + \cdots + a_{im}c_{mj}) = \\
&= (AB)_{ij} + (AC)_{ij} = (AB + AC)_{ij}
\end{aligned}$$

□

7.2.2 Properties of Matrix Multiplication

Do not let Theorem 7.2 lull you into believing that all laws of real arithmetic carry over to matrix arithmetic. For example, you know that in real arithmetic it is always true that $ab = ba$, which is called the commutative law for multiplication. In matrix arithmetic, however, the equality of AB and BA can fail for three possible reasons:

1. AB may be defined and BA may not (for example, if A is 2×3 and B is 3×4).
2. AB and BA may both be defined, but they may have different sizes (for example, if A is 2×3 and B is 3×2).
3. AB and BA may both be defined and have the same size, but the two matrices may be different (as illustrated in the next example).

Example 7.7 (Order Matters in Matrix Multiplication)

Consider the matrices

$$A = \begin{pmatrix} -1 & 0 \\ 2 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$$

Multiplying gives

$$AB = \begin{pmatrix} -1 & -2 \\ 11 & 4 \end{pmatrix} \quad \text{and} \quad BA = \begin{pmatrix} 3 & 6 \\ -3 & 0 \end{pmatrix}$$

Thus for this example $AB \neq BA$.

It can happen that for two matrices A and B of equal size the products AB and BA are equal. In this case we say that the two matrices A and B **commute**.

7.2.3 Zero Matrices

A matrix whose entries are all zero is called a **zero matrix**. Some examples are

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad (0)$$

We will denote a zero matrix by 0. It should be evident that if A and 0 are matrices of the same size, then

$$A + 0 = 0 + A = A$$

Thus, the zero matrix 0 plays the same role in this matrix equation that the number 0 plays in the numerical equation $a + 0 = 0 + a = a$. The following theorem lists the basic properties of zero matrices. Since the results should be self-evident, we will omit the formal proofs.

Theorem 7.3
If c is a scalar, and if the sizes of the matrices are such that the operations can be performed, then:

1. $A + 0 = 0 + A = A$
2. $A - A = A + (-A) = 0$
3. $0A=0$
4. *If* $cA = 0$, *then* $c = 0$ *or* $A = 0$

The following rules that work in arithmetics with scalars **do not work** in general for matrices:

- If $AB = AC$ and $A \neq 0$ then $B = C$ **FALSE for matrices!**
- If $AB = 0$ then $A = 0$ or $B = 0$ **FALSE for matrices!**

Here are two matrices for which $AB = 0$ but $A \neq 0$ and $B \neq 0$:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 7 \\ 0 & 0 \end{pmatrix}.$$

7.2.4 Identity Matrices

A square matrix with 1's on the main diagonal and zeros elsewhere is called an identity matrix. Some examples are

$$\begin{pmatrix} 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

An identity matrix is denoted by the letter I. If it is important to emphasise the size, we will write I_n for the $n \times n$ identity matrix.

To explain the role of identity matrices in matrix arithmetic, let us consider the effect of multiplying a general 2×3 matrix A on each side by an identity matrix. Multiplying on the right by the 3×3 identity matrix yields

$$AI_3 = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = A$$

and multiplying on the left by the 2×2 identity matrix yields

$$I_2A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} = A$$

The same result holds in general; that is, if A is any $m \times n$ matrix, then

$$AI_n = A \quad \text{and} \quad I_m A = A$$

Thus, the identity matrices play the same role in these matrix equations that the number 1 plays in the numerical equation $a \cdot 1 = 1 \cdot a = a$.

7.2.5 Inverse of a Matrix

In real arithmetic every nonzero number a has a reciprocal $a^{-1} = 1/a$ with the property:

$$a \cdot a^{-1} = a^{-1} \cdot a = 1$$

Our next objective is to develop an analog of this result for matrix arithmetic. For this purpose we make the following definition.

Definition 9 (Inverse of a Matrix)
If A is a square matrix, and if a matrix B of the same size can be found such that $AB = BA = I$, then A is said to be **invertible** (or nonsingular) and B is called an **inverse** of A. If no such matrix B can be found, then A is said to be **singular**.

It follows that if B is the inverse of A then A is the inverse of B.

Example 7.8 (An Invertible Matrix)
Let

$$A = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}.$$

Then

$$\begin{aligned} AB &= \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \\ BA &= \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

Thus, A and B are invertible and each is an inverse of the other.

7.2.6 Properties of Inverses

It is reasonable to ask whether an invertible matrix can have more than one inverse? The next theorem shows that the answer is no: *an invertible matrix has exactly one inverse.*

Theorem 7.4
If B and C are both inverses of the matrix A, then $B = C$.

Proof. Since B is an inverse of A, we have $BA = I$. Multiplying both sides on the right by C gives: $(BA)C = IC = C$. But it is also true that $(BA)C = B(AC) = BI = B$, so $C = B$. $\square$

Thus the inverse of an invertible matrix A is unique and we denote it as A^{-1}. Such that:

$$AA^{-1} = A^{-1}A = I$$

In the next section we will develop a method for computing the inverse of an invertible matrix of any size. For now we give the following theorem that specifies conditions under which a 2×2 matrix is invertible and provides a simple formula for its inverse.

Theorem 7.5 (Inverse of a 2×2 Matrix)

The matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case the inverse is given by the formula

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

We will omit the proof, because we will study a more general version of this theorem later. For now, you should at least confirm the validity of this formula by showing that: $AA^{-1} = A^{-1}A = I$.

The quantity $ad - bc$ in Theorem 7.5 is called the **determinant** of the 2×2 matrix A and is denoted by

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

We have already encountered 2×2 and 3×3 determinants and in Chapter 8 we will study the general theory of determinants and how to calculate larger determinants.

The next theorem is concerned with inverses of matrix products.

Theorem 7.6
If A and B are invertible matrices with the same size, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}$$

Proof. We have:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

and similarly, $(B^{-1}A^{-1})(AB) = I$. $\square$

This result can be extended to three or more factors:

> *A product of any number of invertible matrices is invertible, and the inverse of the product is the product of the inverses in the reverse order.*

7.2.7 Powers of a Matrix

If A is a *square* matrix, then we define the nonnegative integer powers of A to be

$$A^0 = I, \quad \text{and} \quad A^n = \underbrace{AA\cdots A}_{n \text{ factors}}$$

and if A is invertible, then we define the negative integer powers of A to be

$$A^{-n} = \underbrace{A^{-1}A^{-1}\cdots A^{-1}}_{n \text{ factors}}$$

Because these definitions parallel those for real numbers, the usual laws of nonnegative exponents hold; for example,

$$A^rA^s = A^{r+s} \quad \text{and} \quad (A^r)^s = A^{rs}$$

In addition, we have the following properties of negative exponents. If A is invertible and n is a nonnegative integer, then:

1. A^{-1} is invertible and $(A^{-1})^{-1} = A$.
2. A^n is invertible and $(A^n)^{-1} = A^{-n} = (A^{-1})^n$.
3. kA is invertible for any nonzero scalar k, and $(kA)^{-1} = k^{-1}A^{-1}$.

We will prove property 3. and leave 1. and 2. as exercises.

Proof of 3. Using the properties from Theorem 7.2

$$(kA)(k^{-1}A^{-1}) = k^{-1}(kA)A^{-1} = (k^{-1}k)(AA^{-1}) = (1)I = I$$

and similarly, $(k^{-1}A^{-1}) = (kA) = I$. Thus, kA is invertible and $(kA)^{-1} = k^{-1}A^{-1}$. $\square$

7.2.8 Matrix Polynomials

If A is a square matrix, say $n \times n$, and if

$$p(x) = a_0 + a_1 x + a_2 x^2 + \dots a_m x^m$$

is any polynomial of degree m, then we define the $n \times n$ matrix $p(A)$ to be:

$$p(A) = a_0 I + a_1 A + a_2 A^2 + \dots a_m A^m$$

The last expression is called a **matrix polynomial in A**.

Example 7.9 (A Matrix Polynomial)

Find $p(A)$ for

$$p(x) = x^2 - 2x - 3 \quad \text{and} \quad A = \begin{pmatrix} -1 & 2 \\ 1 & 3 \end{pmatrix}$$

We have:

$$\begin{aligned} p(A) &= A^2 - 2A - 3I = \\ &= \begin{pmatrix} -1 & 2 \\ 1 & 3 \end{pmatrix}^2 - 2\begin{pmatrix} -1 & 2 \\ 1 & 3 \end{pmatrix} - 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \\ &= \begin{pmatrix} 3 & 4 \\ 2 & 11 \end{pmatrix} - \begin{pmatrix} -2 & 4 \\ 2 & 6 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \end{aligned}$$

7.2.9 Properties of the Transpose

The following theorem lists the main properties of the transpose.

Theorem 7.7
If the sizes of the matrices are such that the stated operations can be performed, then:

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(kA)^T = kA^T$
4. $(AB)^T = B^T A^T$

If you keep in mind that transposing a matrix interchanges its rows and columns, then you should have little trouble visualising the results in properties 1.,2.,3. For example, property 1. states the obvious fact that interchanging rows and columns twice leaves a matrix unchanged; and part 2. states that adding two matrices and then interchanging the rows and columns produces the same result as interchanging the rows and columns before adding. We will omit the formal proofs. Part 4. is a less obvious, but for brevity we will omit its proof as well. The result in that part can be extended to three or more factors and restated as:

> *The transpose of a product of any number of matrices is the product of the transposes in the reverse order.*

The following theorem establishes a relationship between the inverse of a matrix and the inverse of its transpose.

Theorem 7.8
If A is an invertible matrix, then A^T is also invertible and

$$(A^T)^{-1} = (A^{-1})^T$$

Proof. We can establish the invertibility and obtain the formula at the same time by showing that

$$A^T(A^{-1})^T = (A^{-1})^T A^T = I$$

Using property 4. from Theorem 7.7 and the fact that $I^T = I$ we have:

$$\begin{aligned} A^T(A^{-1})^T &= (A^{-1}A)^T = I^T = I \\ (A^{-1})^T A^T &= (AA^{-1})^T = I^T = I \end{aligned}$$

which completes the proof. □

7.2.10 Solving Linear Systems by Matrix Inversion

The following theorem provides a formula for the solution of a linear system of n equations in n unknowns in the case where the coefficient matrix is invertible.

Theorem 7.9
If A is an invertible $n \times n$ matrix, then for each $n \times 1$ matrix $\mathbf{b}$, the system of equations $A\mathbf{x} = \mathbf{b}$ has exactly one solution, namely:

$$\mathbf{x} = A^{-1}\mathbf{b}$$

Proof. Since $A(A^{-1}\mathbf{b}) = \mathbf{b}$, it follows that $\mathbf{x} = A^{-1}\mathbf{b}$ is indeed a solution of $A\mathbf{x} = \mathbf{b}$. To show that this is the only solution, we will proceed by contradiction and assume that $\mathbf{x}_0$ is another solution therefore satisfying $A\mathbf{x}_0 = \mathbf{b}$ therefore:

$$\begin{aligned}
&A\mathbf{x} = \mathbf{b}, \quad A\mathbf{x}_0 = \mathbf{b} \\
&A\mathbf{x} - A\mathbf{x}_0 = 0 \\
&A(\mathbf{x} - \mathbf{x}_0) = 0 \quad \text{and multiplying from the left by } A^{-1} \\
&\mathbf{x} - \mathbf{x}_0 = 0 \Rightarrow \mathbf{x}_0 = \mathbf{x}
\end{aligned}$$

Therefore the solution is unique. □

Example 7.10 (Solution of a Linear System Using A^{-1})

Consider the system of linear equations

$$\begin{cases} x_1 & +2x_2 & +3x_3 & = 5 \\ 2x_1 & +5x_2 & +3x_3 & = 3 \\ x_1 & & +8x_3 & = 17 \end{cases}$$

In matrix form this system can be written as $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 3 \\ 17 \end{pmatrix}.$$

With the methods developed in the next Chapter, it is possible to show that A is invertible and

$$A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}$$

Using Theorem 7.9, the solution of the system is

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \\ 17 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

or $x_1 = 1, x_2 = -1, x_3 = 2$.

Keep in mind that the method of Theorem 7.9 only applies when the system has as many equations as unknowns and the coefficient matrix is invertible.

7.3 Diagonal, Triangular, and Symmetric Matrices

In this section we will discuss matrices that have various special forms. These matrices arise in a wide variety of applications and will also play an important role in our subsequent work.

7.3.1 Diagonal Matrices

A square matrix in which all the entries off the main diagonal are zero is called a *diagonal matrix*. Here are some examples:

$$\begin{pmatrix} 2 & 0 \\ 0 & -5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

and all identity matrices are obviously diagonal.

A general $n \times n$ diagonal matrix D can be written as

$$D = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & d_n \end{pmatrix}$$

A diagonal matrix is invertible if and only if all of its diagonal entries are nonzero; in this case the inverse of D is (try to verify!)

$$D^{-1} = \begin{pmatrix} 1/d_1 & 0 & \dots & 0 \\ 0 & 1/d_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1/d_n \end{pmatrix}$$

Powers of diagonal matrices are easy to compute; we leave it for you to verify that if D is the diagonal matrix and k is a positive integer, then

$$D^k = \begin{pmatrix} d_1^k & 0 & \dots & 0 \\ 0 & d_2^k & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & d_n^k \end{pmatrix}.$$

7.3.2 Triangular Matrices

A square matrix in which all the entries above the main diagonal are zero is called **lower triangular**, and a square matrix in which all the entries below the main diagonal are zero is called **upper triangular**. A matrix that is either upper triangular or lower triangular is called **triangular**[65].

Example 7.11 (Upper and Lower Triangular Matrices)

$$\begin{array}{cc} \text{Upper triangular} & \text{Lower triangular} \\ \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{pmatrix} & \begin{pmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix} \end{array}$$

7.3.3 Symmetric Matrices

Definition 10 (Symmetric Matrices)
A square matrix A is said to be **symmetric** if $A = A^T$.

It is easy to recognize a symmetric matrix by inspection: The entries on the main diagonal have no restrictions, but mirror images of entries across the main diagonal must be equal:

$$\begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}, \quad \begin{pmatrix} 5 & 2 & -1 \\ 2 & 9 & 3 \\ -1 & 3 & 7 \end{pmatrix}, \quad \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & d_n \end{pmatrix}$$

Clearly all diagonal matrices are symmetric.

It follows that a square matrix A is symmetric if and only if

$$a_{ij} = a_{ji}, \quad \text{for all values of } i \text{ and } j.$$

The following theorem lists the main algebraic properties of symmetric matrices. The proofs are direct consequences of Theorem 7.7 and are omitted.

Theorem 7.10
If A and B are symmetric matrices with the same size, and if k is any scalar, then:

1. *A^T is symmetric.*
2. *$A + B$ and $A - B$ are symmetric.*
3. *kA is symmetric.*

[65] Observe that diagonal matrices are both upper triangular and lower triangular since they have zeros below and above the main diagonal. Observe also that a square matrix in row echelon form is upper triangular since it has zeros below the main diagonal.

It is not true, in general, that the product of symmetric matrices is symmetric. To see why this is so, let A and B be symmetric matrices with the same size. Then it follows from part *4.* of Theorem 7.7 and the symmetry of A and B that

$$(AB)^T = B^T A^T = BA$$

Thus $(AB)^T = AB$ if and only if $AB = BA$, i.e. if and only if A and B commute.

7.3.4 Invertibility of Symmetric Matrices

In general, a symmetric matrix need not be invertible. For example, a diagonal matrix with a zero on the main diagonal is symmetric but not invertible. However, the following theorem shows that if a symmetric matrix happens to be invertible, then its inverse must also be symmetric.

Theorem 7.11
If A is an invertible symmetric matrix, then A^{-1} is symmetric.

Proof. Assume that A is symmetric and invertible. From Theorem 7.8 and the fact that $A = A^T$ we have

$$(A^{-1})^T = (A^T)^{-1} = A^{-1}$$

which proves that A^{-1} is symmetric. □

8 Determinants

In this chapter we will study "determinants" or, more precisely, "determinant functions." Unlike real-valued functions, such as $f(x) = x^2$, that assign a real number to a real variable x, determinant functions assign a real number $f(A)$ to a matrix variable A. Although determinants first arose in the context of solving systems of linear equations, they are no longer used for that purpose in real-world applications. Although they can be useful for solving very small linear systems (say two or three unknowns), our main interest in them stems from the fact that they link together various concepts in linear algebra and provide a useful formula for the inverse of a matrix.

8.1 Determinants by Cofactor Expansion

In this section we will define the notion of a "determinant". This will enable us to give a specific formula for the inverse of an invertible matrix, whereas up to now we have had only a computational procedure for 2×2 matrices. This, in turn, will eventually provide us with a formula for solutions of certain kinds of linear systems.

We have seen that for a 2×2 matrix:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is invertible if and only if $ad - bc \neq 0$ and that the expression $ad - bc$ is called the **determinant** of the matrix A. Recall also that this determinant is denoted by writing

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

and that the inverse of A can be expressed in terms of the determinant as

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

8.1.1 Minors and Cofactors

One of our main goals in this chapter is to obtain an analog of the determinant formula valid for 2×2 matrices that is applicable to square matrices of *all orders*. For this purpose we will find it convenient to use subscripted entries when writing matrices or determinants. Thus, if we denote a 2×2 matrix as

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

the determinant takes the form:

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

We also define the determinant of a 1×1 matrix $A = (a_{11})$ as $\det(A) = a_{11}$.

The following definition will be key to our goal of extending the definition of a determinant to higher order matrices.

Definition 11 (Minors and Cofactors)

If A is a square matrix, then the **minor of entry** a_{ij} is denoted by M_{ij} and is defined to be the determinant of the submatrix that remains after the ith row and jth column are deleted from A. The number $C_{ij} = (-1)^{i+j} M_{ij}$ is called the **cofactor** of entry a_{ij}.

We leave it as an exercise to use the definition of cofactors to verify that $\det(A)$ can be expressed in terms of cofactors in the following four ways:

$$
\begin{aligned}
\det(A) &= a_{11}C_{11} + a_{12}C_{12} = \\
&= a_{21}C_{21} + a_{22}C_{22} = \\
&= a_{11}C_{11} + a_{21}C_{21} = \\
&= a_{12}C_{12} + a_{22}C_{22}
\end{aligned}
$$

Each of last four equations is called a **cofactor** expansion of $\det(A)$. In each cofactor expansion the entries and cofactors all come from the same row or same column of A. For example, in the first equation the entries and cofactors all come from the first row of A, in the second they all come from the second row of A, in the third they all come from the first column of A, and in the fourth they all come from the second column of A.

Example 8.1 (Finding Minors and Cofactors)
Let

$$A = \begin{pmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{pmatrix}$$

The minor of entry a_{11} is

$$M_{11} = \begin{vmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 16$$

The cofactor of a_{11} is thus

$$C_{11} = (-1)^{1+1} M_{11} = 16$$

Similarly, the minor of entry a_{32} is

$$M_{32} = \begin{vmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix} = \begin{vmatrix} 3 & -4 \\ 2 & 6 \end{vmatrix} = 26$$

The cofactor of a_{32} is thus

$$C_{32} = (-1)^{3+2} M_{32} = -26.$$

8.1.2 Definition of a General Determinant

The expression of the determinant for a 2×2 matrix is a special case of the following general result, which we will state without proof.

Theorem 8.1
If A is an $n \times n$ matrix, then regardless of which row or column of A is chosen, the number obtained by multiplying the entries in that row or column by the corresponding cofactors and adding the resulting products is always the same.

This result allows us to make the following definition.

Definition 12 (Cofactor (Laplace) Expansion)
If A is an $n \times n$ matrix, then the number obtained by multiplying the entries in any row or column of A by the corresponding cofactors and adding the resulting products is called the **determinant of A**, and the sums themselves are called **cofactor expansions of A**. That is,

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} \quad \text{[cofactor expansion along the } j\text{th column]}$$

and

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \quad \text{[cofactor expansion along the } i\text{th row]}$$

The cofactor expansion is also called **Laplace expansion**.

Example 8.2 (Cofactor Expansion Along the First Row)
To find the determinant of the matrix

$$A = \begin{pmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{pmatrix}$$

we use the cofactor expansion along the first row:

$$\begin{aligned} \det(A) &= \begin{vmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{vmatrix} = 3 \begin{vmatrix} -4 & 3 \\ 4 & -2 \end{vmatrix} - 1 \begin{vmatrix} -2 & 3 \\ 5 & -2 \end{vmatrix} + 0 \begin{vmatrix} -2 & -4 \\ 5 & 4 \end{vmatrix} = \\ &= 3(-4) - (1)(-11) + 0 = -1 \end{aligned}$$

As an exercise try calculating the same determinant by a cofactor expansion along the first column.

Notice that the best strategy for cofactor expansion is to expand along a row or column with the most zeros.

Example 8.3 (Determinant of a triangular matrix)

The following computation shows that the determinant of a 4×4 lower triangular matrix is the product of its diagonal entries. Each part of the computation uses a cofactor expansion along the first row.

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & 0 & 0 \\ a_{32} & a_{33} & 0 \\ a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11}a_{22} \begin{vmatrix} a_{33} & 0 \\ a_{43} & a_{44} \end{vmatrix} = a_{11}a_{22}a_{33}a_{44}$$

You can verify that this is true for an upper triangular matrix and therefore also for a diagonal matrix.

8.2 Evaluating Determinants by Row Reduction

In this section we will show how to evaluate a determinant by reducing the associated matrix to row echelon form. In general, this method requires less computation than cofactor expansion and hence is the method of choice for large matrices.

8.2.1 Two Preliminary Theorems

We begin with a fundamental theorem that will lead us to an efficient procedure for evaluating the determinant of a square matrix of any size.

Theorem 8.2
Let A be a square matrix. If A has a row of zeros or a column of zeros, then $\det(A) = 0$.

Proof. Since the determinant of A can be found by a cofactor expansion along any row or column, we can use the row or column of zeros. Thus if we let $C_1, C_2, \ldots, C_n$ denote the cofactors of A along that row or column, then it follows

$$\det(A) = 0 \cdot C_1 + 0 \cdot C_2 + \cdots + 0 \cdot C_n = 0$$

as required. □

Theorem 8.3
Let A be a square matrix. Then $\det(A) = \det(A^T)$.

Proof. Since transposing a matrix changes its columns to rows and its rows to columns, the cofactor expansion of A along any row is the same as the cofactor expansion of A^T along the corresponding column. Thus, both have the same determinant. □

8.2.2 Elementary Row Operations

The next theorem shows how an elementary row operation on a square matrix affects the value of its determinant. We will not prove the theorem, but we will later employ it for calculating determinants.

Theorem 8.4 (Elementary operations for determinants)
Let A be an $n \times n$ matrix

1. *If B is the matrix that results when a single row or single column of A is multiplied by a scalar k, then* $\det(B) = k \det(A)$.
2. *If B is the matrix that results when two rows or two columns of A are interchanged, then* $\det(B) = -\det(A)$.
3. *If B is the matrix that results when a multiple of one row of A is added to another row or when a multiple of one column is added to another column, then* $\det(B) = \det(A)$.
4. *If A is a square matrix with two proportional rows or two proportional columns, then* $\det(A) = 0$.
5. *If A is a square matrix with a row (or a column) that can be obtained from a linear combination of the other rows (or columns), then* $\det(A) = 0$.

8.2.3 Evaluating Determinants by Row Reduction

We will now give a method for evaluating determinants that involves substantially less computation than cofactor expansion. The idea of the method is to reduce the given matrix to upper triangular form by elementary row operations, then compute the determinant of the upper triangular matrix (an easy computation), and then relate that determinant to that of the original matrix. Here is an example.

Example 8.4 (Using Row Reduction to Evaluate a Determinant)
Evaluate $\det(A)$ where

$$A = \begin{pmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{pmatrix}$$

We will reduce A to an upper triangular form and then apply Example 8.3

$$\det(A) = \begin{vmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{vmatrix} = -\begin{vmatrix} 3 & -6 & 9 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix}$$

The first and second rows are interchanged

$$= -3\begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix}$$

A common factor 3 from the first row is taken outside the determinant sign

$$= -3\begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{vmatrix}$$

-2 times the first row was added to the third row

$$= -3\begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{vmatrix}$$

-10 times the second row was added to the third row

$$= (-3)(-55)\begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{vmatrix}$$

A common factor of -55 from the last row was taken outside

$$= (-3)(-55)(1) = 165$$

To get the last lined we exploited the fact the last determinant is that of a 3×3 triangular and we just need to multiply its diagonal elements.

8.3 Properties of Determinants; Cramer's Rule

In this section we will develop some fundamental properties of matrices, and we will use these results to derive a formula for the inverse of an invertible matrix and formulas for the solutions of certain kinds of linear systems.

8.3.1 Basic Properties of Determinants

Suppose that A and B are $n \times n$ matrices and k is any scalar. The following properties hold:

1. $\det(kA) = k^n \det(A)$;
2. $\det(AB) = \det(A)\det(B)$;
3. $\det(A + B) \neq \det(A) + \det(B)$ in general;
4. A square matrix A is invertible if and only if $\det(A) \neq 0$.

The first and second properties are very useful and can be employed for calculations and proving theorems involving determinant. You must remember the third property very well, the determinant of the sum is not equal to the sum of determinants. The fourth property provides an important criterion for determining whether a matrix is invertible and gets us a step closer to the general formula for the inverse of a matrix.

To this end we use the following definition:

Definition 13 If A is any $n \times n$ matrix and C_{ij} is the cofactor of a_{ij}, then the matrix

$$\begin{pmatrix} C_{11} & C_{12} & \dots & C_{1n} \\ C_{21} & C_{22} & \dots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \dots & C_{nn} \end{pmatrix}$$

is called the **matrix of cofactors from** A. The transpose of this matrix is called the **adjoint of** A and is denoted $\text{adj}(A)$.

Example 8.5 Let

$$A = \begin{pmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{pmatrix}$$

The cofactors of A are (verify!)

$$\begin{array}{rrr} C_{11} = 12 & C_{12} = 6 & C_{13} = -16 \\ C_{21} = 4 & C_{22} = 2 & C_{23} = 16 \\ C_{31} = 12 & C_{32} = -10 & C_{33} = 16 \end{array}$$

so the matrix of cofactors is

$$\begin{pmatrix} 12 & 6 & -16 \\ 4 & 2 & 16 \\ 12 & -10 & 16 \end{pmatrix}$$

and the adjoint of A is

$$\text{adj}(A) = \begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix}$$

The following theorem is the most important result of this chapter and can be used to find the inverse of an invertible matrix.

Theorem 8.5 (Inverse of a Matrix Using Its Adjoint)
If A is an invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A)$$

Example 8.6 (Using the Adjoint to Find an Inverse Matrix)
We now use Theorem 8.5 to find the inverse of A from Example 8.5 (we leave it to you to check that $\det(A) = 64$):

$$A^{-1} = \frac{1}{64}\begin{pmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{pmatrix} = \begin{pmatrix} \frac{3}{16} & \frac{1}{16} & \frac{3}{16} \\ \frac{3}{32} & \frac{1}{32} & -\frac{5}{32} \\ -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

8.3.2 Cramer's Rule

Our next theorem uses the formula for the inverse of an invertible matrix to produce a formula, called **Cramer's rule**, for the solution of a linear system $A\mathbf{x} = \mathbf{b}$ of n equations in n unknowns in the case where the coefficient matrix A is invertible (or, equivalently, when $\det(A) \neq 0$).

Theorem 8.6 (Cramer's Rule)
If $A\mathbf{x} = \mathbf{b}$ is a system of n linear equations in n unknowns such that $\det(A) \neq 0$, then the system has a unique solution. This solution is

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \quad \ldots, \quad x_n = \frac{\det(A_n)}{\det(A)},$$

where A_j is the matrix obtained by replacing the entries in the jth column of A by the entries in the matrix

$$\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Proof. If $\det(A) \neq 0$, then A is invertible, and by Theorem 7.9 $\mathbf{x} = A^{-1}\mathbf{b}$ is the unique solution of $A\mathbf{x} = \mathbf{b}$. Therefore, by Theorem 8.5 we have

$$\begin{aligned} \mathbf{x} &= A^{-1}\mathbf{b} = \frac{\operatorname{adj}(A)\mathbf{b}}{\det(A)} = \frac{1}{\det(A)}\begin{pmatrix} C_{11} & C_{21} & \ldots & C_{n1} \\ C_{12} & C_{22} & \ldots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \ldots & C_{nn} \end{pmatrix}\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \\ &= \frac{1}{\det(A)}\begin{pmatrix} C_{11}b_1 + C_{21}b_2 + \cdots + C_{n1}b_n \\ C_{12}b_1 + C_{22}b_2 + \cdots + C_{n2}b_n \\ \vdots \\ C_{1n}b_1 + C_{2n}b_2 + \cdots + C_{nn}b_n \end{pmatrix} \end{aligned}$$

The entry in the jth row of x is therefore

$$x_j = \frac{b_1C_{1j} + b_2C_{2j} + \cdots + b_nC_{nj}}{\det(A)}$$

Now let us define

$$A_j = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1j-1} & b_1 & a_{1j+1} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j-1} & b_2 & a_{2j+1} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nj-1} & b_n & a_{nj+1} & \dots & a_{nn} \end{pmatrix}$$

Since A_j differs from A only in the jth column, it follows that the cofactors of entries $b_1, b_2, \dots, b_n$ in A_j are the same as the cofactors of the corresponding entries in the jth column of A. The cofactor expansion of $\det(A_j)$ along the jth column is therefore

$$\det(A_j) = b_1 C_{1j} + b_2 C_{2j} + \dots + b_n C_{nj}$$

Substituting this result in the expression for x_j we finally get:

$$x_j = \frac{\det(A_j)}{\det(A)}$$

□

Example 8.7 (Using Cramer's Rule to Solve a Linear System)
Let us use Cramer's rule to solve

$$\begin{cases} x_1 & & +2x_3 & = 6 \\ -3x_1 & +4x_2 & +6x_3 & = 30 \\ -x_1 & -2x_2 & +3x_3 & = 8 \end{cases}$$

We have:

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{pmatrix}$$
$$A_2 = \begin{pmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & 8 & 3 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{pmatrix}$$

Therefore:

$$\begin{aligned} x_1 &= \frac{\det(A_1)}{\det(A)} = \frac{-40}{44} = -\frac{10}{11}, \\ x_2 &= \frac{\det(A_2)}{\det(A)} = \frac{72}{44} = \frac{18}{11}, \\ x_3 &= \frac{\det(A_3)}{\det(A)} = \frac{152}{44} = \frac{38}{11}. \end{aligned}$$

9 Eigenvalues and Eigenvectors

In this chapter we will define the notions of "eigenvalue" and "eigenvector" and discuss some of their basic properties.

9.1 Definition of Eigenvalue and Eigenvector

We begin with the main definition in this section.

Definition 14
If A is an $n \times n$ matrix, then a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ is called an **eigenvector** of A if $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$, that is:

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar λ. The scalar λ is called an **eigenvalue** of A corresponding to the eigenvector $\mathbf{x}$.

In general, the image of a vector $\mathbf{x}$ under multiplication by a square matrix A differs from $\mathbf{x}$ in both magnitude and direction. However, in the special case where $\mathbf{x}$ is an eigenvector of A, multiplication by A leaves the direction unchanged. For example, in $\mathbb{R}^2$ or $\mathbb{R}^3$ multiplication by A maps each eigenvector $\mathbf{x}$ of A (if any) along the same line through the origin as $\mathbf{x}$. Depending on the sign and magnitude of the eigenvalue corresponding to $\mathbf{x}$, the operation $A\mathbf{x} = \lambda\mathbf{x}$ compresses or stretches $\mathbf{x}$ by a factor of λ, with a reversal of direction in the case where λ is negative:

Example 9.1 (Eigenvector of a 2×2 Matrix) The vector $\mathbf{x}^T = (12)$ is an eigenvector of

$$A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$$

corresponding to the eigenvalue $\lambda = 3$, since

$$A\mathbf{x} \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3\mathbf{x}$$

Geometrically, multiplication by A has stretched the vector $\mathbf{x}$ by a factor 3.

9.2 Computing Eigenvalues and Eigenvectors

Our next objective is to obtain a general procedure for finding eigenvalues and eigenvectors of an $n \times n$ matrix A. We will begin with the problem of finding the eigenvalues of A. Note first that the equation $A\mathbf{x} = \lambda\mathbf{x}$ can be rewritten as $A\mathbf{x} = \lambda I\mathbf{x}$, or equivalently, as

$$(\lambda I - A)\mathbf{x} = 0$$

For λ to be an eigenvalue of A this equation must have a nonzero solution for $\mathbf{x}$. This is so if and only if the coefficient matrix has a zero determinant. Thus, we have the following result.

Theorem 9.1
If A is an $n \times n$ matrix, then λ is an eigenvalue of A if and only if it satisfies the equation

$$\det(\lambda I - A) = 0$$

This is called the **characteristic equation** *of A.*

Example 9.2 (Finding Eigenvalues)
In Example 9.1 we observed that $\lambda = 3$ is an eigenvalue of the matrix

$$A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$$

but we did not explain how we found it. Let us use the characteristic equation to find all eigenvalues of this matrix:

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 3 & 0 \\ 8 & \lambda + 1 \end{vmatrix} = (\lambda - 3)(\lambda + 1) = 0$$

This shows that the eigenvalues of A are $\lambda = 3$ and $\lambda = -1$. Thus, in addition to the eigenvalue $\lambda = 3$ noted in Example 9.1, we have discovered a second eigenvalue.

When the determinant $\det(\lambda I - A)$ is expanded, the result is a polynomial $p(\lambda)$ of degree n that is called the **characteristic polynomial** of A. In Example 9.2 the characteristic polynomial of the matrix A is:

$$p(\lambda) = (\lambda - 3)(\lambda + 1) = \lambda^2 - 2\lambda - 3$$

which is a polynomial of degree 2. In general, the characteristic polynomial of an $n \times n$ matrix has the form

$$p(\lambda) = \lambda^n + c_1\lambda^{n-1} + \cdots + c_n$$

in which the coefficient of λ^n is 1. Since a polynomial of degree n has at most n distinct roots, it follows that the equation

$$\lambda^n + c_1\lambda^{n-1} + \cdots + c_n = 0$$

has at most n distinct solutions and consequently that an $n \times n$ matrix has at most n distinct eigenvalues. Since some of these solutions may be complex numbers, it is possible for a matrix to have complex eigenvalues, even if that matrix itself has real entries. We will normally focus on examples in which the eigenvalues are real numbers.

The following theorem, which we are not going to prove, establishes an important result used in other theorems in linear algebra.

Theorem 9.2 (Caley-Hamilton)
Every matrix A satisfies its own characteristic equation:

$$p(A) = 0$$

Example 9.3
Consider the matrix A of Example 9.2 and let us verify Caley-Hamilton's theorem. The characteristic polynomial is

$$p(\lambda) = \lambda^2 - 2\lambda - 3$$

Therefore we calculate:

$$\begin{aligned} p(A) &= A^2 - 2A - 3I = \begin{pmatrix} 9 & 0 \\ 16 & 1 \end{pmatrix} - 2\begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix} - 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

as expected.

Example 9.4 (Eigenvalues of a Lower Triangular Matrix)
Let us find the eigenvalues of a lower triangular matrix

$$A = \begin{pmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Recalling that the determinant of a triangular matrix is the product of the entries on the main diagonal, we obtain

$$\begin{aligned}\det(\lambda I - A) &= \begin{pmatrix} \lambda - a_{11} & 0 & 0 & 0 \\ a_{21} & \lambda - a_{22} & 0 & 0 \\ a_{31} & a_{32} & \lambda - a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & \lambda - a_{44} \end{pmatrix} = \\ &= (\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44})\end{aligned}$$

Thus the characteristic equation is

$$(\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44}) = 0$$

and the eigenvalues are

$$\lambda = a_{11}, \lambda = a_{22}, \lambda = a_{33}, \lambda = a_{44}$$

which are precisely the diagonal entries of A.

Thus in general:

> *If A is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of A are the entries on the main diagonal of A.*

To find an eigenvector corresponding to a certain eigenvalue we proceed as in the following example.

Example 9.5 (Finding the eigenvector corresponding to an eigenvalue)
For the matrix

$$A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$$

let us find the eigenvector $\mathbf{x}$ corresponding to the eigenvalue $\lambda = 3$. Let us write the vector $\mathbf{x}$ as

$$\mathbf{x} = \begin{pmatrix} a \\ b \end{pmatrix}$$

where a, b are the components of $\mathbf{x}$ that we want to find. The expression $A\mathbf{x} = \lambda\mathbf{x}$ can be explicitly written as

$$\begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 3 \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3a \\ 3b \end{pmatrix}$$

The first equation is trivial while the second gives:

$$8a - b = 3b \Rightarrow a = b/2$$

which leads to the following eigenvector:

$$\mathbf{x} = \begin{pmatrix} t \\ 2t \end{pmatrix}, \quad t \in \mathbb{R}$$

Thus for any real value t the above vector is an eigenvector of matrix A with eigenvalue $\lambda = 3$. Sometimes it is useful to work with a normalised eigenvector obtained by imposing that the length of $\mathbf{x}$ is 1:

$$|\mathbf{x}| = \sqrt{4t^2 + t^2} = \sqrt{5}t = 1 \Leftrightarrow t = \frac{1}{\sqrt{5}}$$

Thus the normalised eigenvector is

$$\tilde{\mathbf{x}} = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$$

Show that $(0 \quad 1)^T$ is the (normalised) eigenvector corresponding to the eigenvalue $\lambda = -1$.